

Joe Das

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CHAPTER 1

Types of Error

1. Inherent Error

Inherent error are those error that are present in the data supplied to the model.(inherent error also known as input error).it contain two components namely,data error & conversion error.

Data error

Data error (also known as empirical error)arises when data for a problem are obtained by some experimental means& are, therefore ,of limited accuracy & precision

Conversion Error

Conversion error are (also known as representation error)arise due to the limitaion of computer to store the exactly,we know that the floating point representation retains only a specified number of digits

2. Roundoff error

Roundoff error are occure when fixed number of digit are used to represent exact number.since number are stored as a every stge of computation,roundoff error are introduced as the end of every arithmetic operation.

3. Truncation error

Truncation error are arise from using an approximatoin in the palce of exact mathematical procedure.we often use some finite number of terms to estimate the sum of an infinite series

$$S = \sum_{i=0}^{\infty} a_i x_i$$

4. Absolute &relative error

Let us suppose that true value of a data item is denoted by x_t & it is approximated value is denoted by x_a then they are as follows.

$$\text{Error} = x_t - x_a$$

$$Rr = \frac{\text{absolute error}}{\text{true value}}$$

$$\text{Percentage error} = Er \times 100\%$$

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Bisection Method: Algorithm

1. Define function $f(x)$ and Error (E)
2. Take two initial point of interval for root as x_1 and x_2
3. Calculate $f(x_1)$ and $f(x_2)$
4. If $f(x_1) * f(x_2) < 0$ then root does not lies between x_1 and x_2 goto step 9
5. If $f(x_1) \geq 0$ then set $a = x_1$ and $b = x_2$
Else Set $a = x_2$ and $b = x_1$
6. Calculate root by $x_n = \frac{a+b}{2}$ and $f(x_n)$
7. If $f(x_n) > 0$ (positive) then
set $b = x_n$ Else set $a = x_n$
8. Repeat step 7 till $(\frac{b-a}{b}) < E$ and print root
9. Stop

Regula False Method : Algorithm

1. Define function $f(x)$ and Error (E)
2. Take two initial point of interval for root as x_1 and x_2
3. Calculate $f(x_1)$ and $f(x_2)$
4. If $f(x_1) * f(x_2) < 0$ then root does not lies between x_1 and x_2 goto step 9
5. If $f(x_1) < 0$ then set $a = x_1$ and $b = x_2$
Else
Set $a = x_2$ and $b = x_1$
6. Calculate root by $x_n = \frac{a f(b) - b f(a)}{f(b) - f(a)}$ and $f(x_n)$
7. If $f(x_n) > 0$ (positive) then set $b = x_n$
Else set $a = x_n$
8. Repeat step 7 till $(\frac{b-a}{b}) < E$ and print root
9. Stop

Note: Algorithm for regula-falsi method is exactly similar to Bisection method only the formula for calculating root is different.

Secant Method: Algorithm

1. Define function $f(x)$ and Error (E)
2. Take two initial point of interval for root as x_1 and x_2
3. Calculate $f(x_1)$ and $f(x_2)$
4. If $f(x_1) * f(x_2) < 0$ then root does not lies between x_1 and x_2 goto step 9
5. If $f(x_1) < 0$ then set $a = x_1$ and $b = x_2$ then
Else Set $a = x_2$ and $b = x_1$
6. Calculate root by $x_n = \frac{a f(b) - b f(a)}{f(b) - f(a)}$ and $f(x_n)$
7. If $(\frac{b-a}{b}) < E$ then print root = x_n
8. stop

Convergence of N-R method [2011 Fall]

Let x_n and x_{n+1} be successive Approximation to the Actual to the Actual root α of $f(x) = 0$ If E_n & E_{n+1} are the Corresponding errors.

We have, $x_n = \alpha + E_n$ and $x_{n+1} = \alpha + E_{n+1}$

Now, From Nrformula We have, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$$\Rightarrow \alpha + E_{n+1} = \alpha + E_n - \frac{f(\alpha + E_n)}{f'(\alpha + E_n)}$$

$$\Rightarrow E_{n+1} = E_n - \frac{f(\alpha + E_n)}{f'(\alpha + E_n)}$$

Expanding by Taylors Series,

$$\Rightarrow E_{n+1} = E_n - \frac{f(\alpha) + E_n f'(\alpha) + \frac{E_n^2}{2!} f''(\alpha) + \dots}{f'(\alpha) + E_n f''(\alpha) + \frac{E_n^2}{2!} f'''(\alpha) + \dots}$$

$$\Rightarrow E_{n+1} = E_n - \frac{E_n f'(\alpha) + \frac{E_n^2}{2!} f''(\alpha)}{f'(\alpha) + E_n f''(\alpha)}$$

[$\therefore f(\alpha) = 0$ and omitting derivative of higher order than two]

$$E_{n+1} = \frac{\frac{E_n^2}{2!} f''(\alpha)}{f'(\alpha) + E_n f''(\alpha)}$$

$$E_{n+1} = \frac{E_n^2 f''(\alpha)}{2! f'(\alpha)} \left(1 + \frac{E_n^2 f''(\alpha)}{f'(\alpha)} \right)^{-1} = \frac{E_n^2 f''(\alpha)}{2! f'(\alpha)} \dots (i)$$

$\therefore \left(1 + \frac{E_n^2 f''(\alpha)}{f'(\alpha)} \right)^{-1}$ is small quantity, we neglect it

Eqn (i) shows that error of current step is directly proportional to the square of error previous step. Hence, N-R method have quadratic convergence or 2nd order convergence

Iteration Method

This method is also known as successive approximation.

Let $f(x) = 0 \dots (i)$ be the given equation whose roots are to be determined. This equation can be written in the form $x = \phi(x) \dots (ii)$

Let x_0 be an initial approximation to the actual root, say, α of equation (i)

Then the first approximation is $x_1 = \phi(x_0)$ & the successive approximation are $x_2 = \phi(x_1)$, $x_3 = \phi(x_2) \dots x_n = \phi(x_{n-1})$

If the sequence of approximate roots $x_1, x_2, x_3 \dots x_n$ converges to α it is taken as the root of the equation (i)

University Question Solutions

1. Bisection Method

Find a root of following equation using bisection method to four decimal places $\log x - \cos x = 0$. [2005 Fall]

Solution: Here function, $f(x) = \log_{10} x - \cos x$

$$f(1) = -0.54 < 0 \text{ \& } f(2) = 1.1092 > 0.$$

Therefore, the root lies between 1 & 2.

$$\therefore a = 1 \text{ \& } b = 2. \text{ Error } (E) = 0.0001$$

$$\text{No of iterations, } n = \frac{\log(b-a) - \log E}{\log 2} = 13.28 \approx 13.$$

$$\text{We have, Bisection formula is } x_n = \frac{a+b}{2} = \frac{1+2}{2} = 1.5$$

$$f(x_n) = 0.3347. \text{ Therefore, root lies between 1.5 \& 1.}$$

Likewise, calculating other roots by tabulation method.

Iteration(n)	a	b	(x_n)	$f(x_n)$
1	1	2	1.5	0.3347
2	1	1.5	1.25	-0.092
3	1.25	1.5	1.3750	0.1239
4	1.25	1.3750	1.3125	0.0164
5	1.25	1.3125	1.28125	-0.0376
6	1.2812	1.3125	1.2968	-0.0106
7	1.2968	1.3125	1.3046	0.0028
8	1.2968	1.3046	1.30074	-0.0038
9	1.30074	1.3046	1.30267	-0.0005
10	1.30267	1.3046	1.30363	0.0015
11	1.30267	1.3036	1.30315	0.0003
12	1.30267	1.30315	1.30315	0.0002
13	1.30267	1.30315	1.30292	END

Hence root is = 1.30292 which is correct up to 3 decimal place.

Calculate the root of given equation using bisection method correct to three decimal points. $f(x) = \tan x + \tanh x$ [2005 Spring]

Solution: Here function, $f(x) = \tan x + \tanh x$

$$f(2.3) = -0.139 < 0 \text{ and } f(2.4) = 0.0676 > 0.$$

Therefore, the root lies between 2.3 & 2.4.

$$\therefore a = 2.3 \text{ \& b = 2.4. Error (E) = 0.0001}$$

$$\text{No of iterations, } n = \frac{\log(b-a) - \log E}{\log 2} = 14.25 \approx 14.$$

$$\text{We have, Bisection formula is } x_n = \frac{a+b}{2} = \frac{2.3+2.4}{2} = 2.35$$

$f(x_n) = -0.030$. Therefore root lies between 2.35 & 2.4.

Likewise, calculating other roots by tabulation method

Iteration(n)	a	b	(x_n)	$f(x_n)$
1	2.3	2.4	2.35	-0.030
2	2.35	2.4	2.375	0.020
3	2.35	2.375	2.3625	-0.005
4	2.3625	2.375	2.36875	0.007
5	2.3625	2.36875	2.36563	0.001
6	2.3625	2.36563	2.36406	-0.002
7	2.36406	2.36563	2.36485	-0.0003
8	2.36485	2.36563	2.36524	0.0004
9	2.36485	2.36524	2.36504	0.00004
10	2.36485	2.36504	2.36459	-0.0001
11	2.36495	2.36504	2.36499	-0.00006
12	2.36499	2.36504	2.36502	-0.000001
13	2.36502	2.36504	2.36503	0.00001
14	2.36502	2.36503	2.36502	END

Hence root is = 2.36502 which is correct up to 4 decimal place

Calculate the root of given equation using Bisection method; correct to three decimal points. $f(x) = \sin x - 2x + 1$. [2009 Fall]

Solution: Here function, $f(x) = \sin x - 2x + 1$.

$$f(1) = -0.1585 < 0 \text{ and } f(0) = 1 > 0.$$

Therefore the root lies between 0 & 1.

$$\therefore a = 1 \text{ \& b = 0. Error (E) = 0.0001}$$

$$\text{No of iterations, } n = \frac{\log(b-a) - \log E}{\log 2} = 13.28 \approx 13.$$

$$\text{We have, Bisection formula is } x_n = \frac{a+b}{2} = \frac{0+1}{2} = 0.5$$

$f(x_n) = 0.479$. Therefore root lies between 0.5 & 1.

Likewise, calculating other roots by tabulation method.

Iteration(n)	a	b	x_n	$f(x_n)$
1	1	0	0.5	0.479
2	1	0.5	0.75	0.1816
3	1	0.75	0.875	0.0175
4	1	0.875	0.9375	-0.06891
5	0.9375	0.875	0.9062	-0.02523
6	0.9062	0.875	0.8906	-0.00375
7	0.8906	0.875	0.8828	0.00692
8	0.8906	0.8828	0.8867	0.00159
9	0.8906	0.8867	0.88865	-0.00107
10	0.8886	0.8867	0.8876	0.00035
11	0.88865	0.8876	0.88816	-0.0004
12	0.88816	0.8876	0.8878	-0.00002
13	0.8878	0.8876	0.8877	END

Hence root is = 0.8877 which is correct up to 3 decimal place.

Find a positive root of the equation $f(x) = \cos x - 1.3x$ by using Bisection method. [2010 Spring]

Solution: Here function, $f(x) = \cos x - 1.3x$

$$f(1) = -0.7596 < 0 \text{ \& } f(0) = 1 > 0.$$

Therefore, the root lies between 1 & 2.

$$\therefore a = 0 \text{ \& b = 1. Error (E) = 0.0001}$$

$$\text{No of iterations, } n = \frac{\log(b-a) - \log E}{\log 2} = 13.28 \approx 13.$$

$$\text{We have, Bisection formula is } x_n = \frac{a+b}{2} = \frac{0+1}{2} = 0.5$$

$f(x_n) = 0.2275$. Therefore, root lies between 0.5 & 1.

Likewise, calculating other roots by tabulation method.

Iteration(n)	a	b	(x_n)	$f(x_n)$
1	0	1	0.5	0.3347
2	0.5	1	0.75	-0.092
3	0.5	0.75	0.625	0.1239
4	0.5	0.625	0.5625	0.0164
5	0.5625	0.625	0.5937	-0.0376
6	0.59375	0.625	0.6093	-0.0106
7	0.60937	0.625	0.6171	0.0028
8	0.6171	0.625	0.6210	-0.0038
9	0.6210	0.625	0.6230	0.0015
10	0.6230	0.625	0.6240	0.0003
11	0.6240	0.625	0.6245	0.0000
12	0.6240	0.6245	0.6242	END

Hence root is = 0.6242 which is correct up to 3 decimal place.

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Find a root of the equation by using Bisection method up to three decimal places. $x^2 - 10 \log_{10} x$. [2011 Spring]

Solution: Here function, $f(x) = x^2 - 10 \log_{10} x$
 $f(3) = -1.98 < 0$ and $f(4) = 2.1370 > 0$.
 Therefore, the root lies between 3 & 4.
 $\therefore a = 3$ & $b = 4$. Error (E) = 0.0001

$$\text{No of Iterations, } n = \frac{\log(b-a) - \log E}{\log 2} = 13.28 \approx 13.$$

$$\text{We have, Bisection formula is } x_n = \frac{a+b}{2} = \frac{3+4}{2} = 3.5$$

$$f(x_n) = -0.2776. \text{ Therefore root lies between 3.5 \& 4.}$$

Likewise, calculating other roots by tabulation method.

Iteration(n)	a	b	x_n	$f(x_n)$
1	3	4	3.5	-0.2776
2	3.5	4	3.75	0.844
3	3.5	3.75	3.625	0.262
4	3.5	3.625	3.5625	-0.013
5	3.5625	3.625	3.5937	0.12
6	3.5625	3.59375	3.5781	0.054
7	3.5625	3.5781	3.5703	0.0206
8	3.5625	3.5703	3.5664	0.0036
9	3.5625	3.5664	3.5644	-0.0047
10	3.5644	3.5664	3.5654	-0.0005
11	3.5654	3.5664	3.5659	0.0015
12	3.5654	3.5659	3.5656	0.0007
13	3.5654	3.5656	3.5655	END

Hence, Root is = 3.5655 which is correct up to 3 decimal place.

Calculate the root of the given equation using Bisection method; correct upto 3 decimal places. $f(x) = \sin x - \frac{1}{x}$ [2012 Fall]

Solution: Here function, $f(x) = \sin x - \frac{1}{x}$

$$f(1) = -0.1585 < 0 \text{ and } f(2) = 0.4092 > 0.$$

Therefore the root lies between 1 & 2.

$$\therefore a = 1 \text{ \& } b = 2.$$

$$\text{We have, Bisection formula is } x_n = \frac{a+b}{2} = \frac{1+2}{2} = 1.5$$

$$f(x_n) = 0.3308. \text{ Therefore root lies between 1 \& 1.5.}$$

Likewise, calculating other roots in tabular form.

Iteration(n)	a	b	x_n	$f(x_n)$
1	1	2	1.5	0.3308
2	1	1.5	1.25	0.1489
3	1	1.25	1.125	0.0133
4	1	1.125	1.0625	-0.0676
5	1.0625	1.125	1.0937	-0.0259
6	1.0937	1.125	1.1093	-0.0060
7	1.1093	1.125	1.1171	END

Hence, root is = 3.5655

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2. Regula Falsi Method

Find a root of following equation using Regula Falsi method to four decimal places $\log x - \cos x = 0$ [2005 Fall]

Solution: Here function, $f(x) = \log_{10} x - \cos x$

$$f(1) = -0.54 < 0, f(2) = 1.1092 > 0;$$

Therefore, root lies between 1 & 2.

$$\therefore a = 1, b = 2$$

$$\begin{aligned} \text{By Regula Falsi formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(1 * 1.1092) - (2 * -0.54)}{1.1092 - (-0.54)} \\ &= 1.32743 \end{aligned}$$

Likewise, calculating the other values of root in tabular form.

Iteration(n)	a	b	f(a)	f(b)	x_n	$f(x_n)$
1	1	2	-0.54	1.1092	1.32743	0.04227
2	1	1.32743	-0.54	0.04227	1.30366	0.00120
3	1	1.30366	-0.54	0.001205	1.30298	0.000028
4	1	1.30298	-0.54	0.000028	1.30296	-0.000007
5	1.30296	1.30298	-0.000007	0.000028	1.30288	END

Hence root = 1.30288 which is correct up to four decimal place.

Find a root of the equation $\sin x - x + 2 = 0$ by False position method correct to three decimal places. [2007 Fall]

Solution: Here function, $f(x) = \sin x - x + 2$

$$f(3) = -0.8588 < 0, f(2) = 0.90930 > 0.$$

Therefore root lies between 3 & 2.

$$\therefore a = 3, b = 2$$

$$\begin{aligned} \text{By Regula Falsi formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(3 * 0.90930) - (2 * -0.8588)}{0.90930 - (-0.8588)} \\ &= 2.5142 \end{aligned}$$

Likewise, calculating the other values of root in tabular form.

Iteration (n)	a	b	f(a)	f(b)	x_n	$f(x_n)$
1	3	2	-0.8588	0.90930	2.5142	0.0728
2	3	2.5142	-0.8588	0.0728	2.5521	0.0038
3	3	2.5521	-0.8588	0.0038	2.5540	0.0004
4	3	2.5540	-0.8588	0.0004	2.5542	END

Hence, root = 2.5542 which is correct up to four decimal place.

Use False- position method to find the root of the following equation, $x^3 - 4x^2 + x + 6 = 0$ correct upto two decimal places: [2009 Spring]

Solution: Here function, $f(x) = x^3 - 4x^2 + x + 6$

$$f(2.5) = -0.875 < 0, f(3.5) = 3.375 > 0.$$

Therefore, root lies between 2.5 & 3.5

$$\therefore a = 2.5, b = 3.5$$

$$\begin{aligned} \text{By Regula Falsi formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(2.5 * 3.375) - (3.5 * -0.875)}{3.375 - (-0.875)} \\ &= 2.7058 \end{aligned}$$

Likewise, calculating the other values of root in tabular form

Iteration (n)	a	b	f(a)	f(b)	x_n	$f(x_n)$
1	2.5	3.5	-0.875	3.375	2.7058	-0.7695
2	2.7058	3.5	-0.7695	3.375	2.8533	-0.48235
3	2.8533	3.5	-0.48235	3.375	2.9342	-0.24184
4	2.9342	3.5	-0.24184	3.375	2.9720	-0.10810
5	2.9720	3.5	-0.10810	3.375	2.9884	-0.04573
6	2.9884	3.5	-0.04573	3.375	2.9952	-0.01908
7	2.9952	3.5	-0.01908	3.375	2.9980	END

Hence, Root = 2.9980 which is correct up to four decimal place.

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Find the root of the equation $x \sin x + \cos x = 0$ with accuracy 0.008, using False Position method. [2011 Fall]

Solution: Here function, $f(x) = x \sin x + \cos x$
 $f(3) = -0.5666 < 0$, $f(2) = 1.4024 > 0$.
 Therefore, Root lies between 3 & 2.
 $\therefore a = 3$, $b = 2$ Error (E) = 0.008

$$\begin{aligned} \text{By Regula Falsi formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(3 * 1.4024) - (2 * -0.5666)}{1.4024 - (-0.5666)} \\ &= 2.7122 \end{aligned}$$

Likewise, calculating the other values of root in tabular form.

Iteration(n)	a	b	f(a)	f(b)	x_n	(x_n)
1	3	2	-0.5666	1.4020	2.7122	0.2199
2	3	2.7122	-0.5666	0.2199	2.7926	0.0152
3	3	2.7926	-0.5666	0.0152	2.7980	0.0010
4	3	2.7980	-0.5666	0.0010	2.7983	0.0002
5	3	2.7983	-0.5666	0.0002	2.7983	END

Hence, root = 2.7983 which is correct up to four decimal place.

Find the root of the equation $f(x) = x^3 + 4x - 1$ with accuracy 0.0001, using False Position method. [2011 Spring]

Solution: Here function, $f(x) = x^3 + 4x - 1$

$f(0) = -1 < 0$, $f(1) = 4 > 0$.
 Therefore, root lies between 0 & 1.
 $\therefore a = 0$, $b = 1$ Error (E) = 0.0001

$$\begin{aligned} \text{By Regula Falsi formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(0 * 4) - (1 * -1)}{4 - (-1)} \\ &= 0.20 \end{aligned}$$

Likewise, calculating the other values of root in tabular form.

Iteration(n)	a	b	f(a)	f(b)	x_n	f(x_n)
1	0	1	-1	4	0.20	-0.1920
2	0	0.20	-1	-0.1920	0.2475	0.0052
3	0.2475	0.20	0.0052	-0.1920	0.2462	-0.0003
4	0.2475	0.2462	0.0052	-0.0003	0.2462	END

Hence, Root = 0.2462 which is correct up to four decimal place.

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3. Secant Method

Find the real root of the function $\tan x + \tanh x = 0$ by Secant method with initial values 2.3 and 2.4 [2005 Spring]

Solution: Here function, $f(x) = \tan x + \tanh x$

$$f(2.3) = -0.13912 < 0, f(2.4) = 0.06766 > 0.$$

Therefore, Root lies between 2.3 & 2.4.

$$\therefore a = 2.3, b = 2.4$$

$$\begin{aligned} \text{By Secant formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(0.06766 \cdot 2.3) - (-0.13912 \cdot 2.4)}{0.06766 - (-0.13912)} \\ &= 2.36728 \end{aligned}$$

Likewise, calculating the other values of root in tabular form.

Iteration (n)	a	b	f(a)	f(b)	x_n	$f(x_n)$
1	2.3	2.4	-0.13912	0.06766	2.36728	0.00451
2	2.4	2.36728	0.06766	0.00451	2.36494	-0.00016
3	2.36728	2.36494	0.00451	-0.00016	2.36502	-0.00000
4	2.36494	2.36502	-0.00016	-0.0000007	2.365020	END

Hence, root = 2.36502 which is correct up to four decimal place

Using secant method, find the root of the equation $x^6 - x^4 - x^3 - 1 = 0$ Correct upto 2-significant digit. [2006 Spring]

Solution: Here function, $f(x) = x^6 - x^4 - x^3 - 1$

$$f(1) = -2 < 0, f(2) = 39 > 0.$$

Therefore root lies between 1 & 2.

$$\therefore a = 1, b = 2$$

$$\begin{aligned} \text{By Secant formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(1 \cdot 39) - (2 \cdot -2)}{39 - (-2)} \\ &= 1.0488 \end{aligned}$$

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Likewise, calculating the other value of root using tabulation method.

Iteration (n)	a	b	f(a)	f(b)	x_n	$f(x_n)$
1	2	-2	39	1.0488	-2.032	
2	1.0488	39	-2.032	-2.02625	1.0959	-2.02625
3	1.0959	-2.02625	-2.02625	24049134	1.0959	-2.026256
4	1.0959	2404913	-2.020256	1.0959	END	

Hence, Root = 1.7321 which is correct upto four decimal place.

Using Secant method, find a real root of $x^3 + x^2 - 3x - 3 = 0$, correct to three decimal places. [2006 Fall, 2008 Fall]

Solution: Here function, $f(x) = x^3 + x^2 - 3x - 3$

$$f(1) = -4 < 0, f(2) = 3 > 0.$$

Therefore root lies between 1 & 2.

$$\therefore a = 1, b = 2$$

$$\begin{aligned} \text{By Secant formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(1 \cdot 3) - (2 \cdot -4)}{3 - (-4)} \\ &= 1.5714 \end{aligned}$$

Likewise, calculating the other value of root in tabular form.

Iteration(n)	a	b	f(a)	f(b)	x_n	$f(x_n)$
1	2	-4	3	1.5714	-1.3646	
2	1.5714	3	-1.3646	-0.5992	1.6659	-0.5992
3	1.6659	-0.5992	0.0746	1.7399	0.0746	
4	1.7399	0.0746	0.1757	1.7504	0.1757	
5	1.7504	0.1757	0.001412	1.7321	0.00046	
6	1.7321	0.00046	0.00046	1.7321	0.00046	

Hence, root = 1.7321 which is correct upto four decimal place.

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Solve $f(x) = 3x + \sin x - e^x$ by secant method up to 5th iteration
[2007 Fall, 2007 Spring, 2010 Spring]

Solution: Here function, $f(x) = 3x + \sin x - e^x$

$$f(0) = -1 < 0, f(1) = 1.123 > 0.$$

Therefore, root lies between 0 & 1.

$$\therefore a = 0, b = 1$$

$$\begin{aligned} \text{By Secant formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(0 \cdot 1.123) - (1 \cdot -1)}{1.123 - (-1)} \\ &= 0.4710 \end{aligned}$$

Likewise, calculating the other value of root in tabular form.

Iteration(n)	a	b	f(a)	f(b)	x_n	f(x_n)
1	0	1	-1	1.123	0.4710	0.2651
2	1	0.4710	1.123	0.2651	0.36753	-0.1347
3	0.4710	0.36753	0.2651	-0.1347	0.40238	0.1033
4	0.36753	0.40238	-0.1347	0.1033	0.3872	0.664
5	0.40238	0.3872	0.1033	0.664	0.3600	END

Hence, root = 0.3600.

Find out at least one real root of $x^3 + x^2 - x - 1 = 0$ using Secant method to three decimal places. [2008 Spring]

Solution: Here function, $f(x) = x^3 + x^2 - x - 1$

$$f(0) = -1 < 0, f(2) = 9 > 0.$$

Therefore, root lies between 0 & 2.

$$\therefore a = 0, b = 2$$

$$\begin{aligned} \text{By Secant formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(0 \cdot 9) - (2 \cdot -1)}{9 - (-1)} \\ &= 1 \end{aligned}$$

Likewise, calculating the other value of root in tabular form.

Iteration(n)	a	b	f(a)	f(b)	x_n	f(x_n)
1	0	2	-1	9	1	0.2000
2	2	1	9	0.20	0.9777	-0.088
3	1	0.9777	0.20	-0.088	0.984	-0.0629
4	0.9777	0.984	-0.088	-0.0629	1.001	0.0040
5	0.984	1.001	-0.0629	0.0040	0.9998	-0.0008
6	1.001	0.9998	0.0040	-0.0008	1	0
7	0.9998	1	-0.0008	0	1	END

Hence, root = 1 which is correct up to three decimal place.

Find a real root of the equation $x^3 - 2x - 5 = 0$ correct to three decimal places using secant method. [2010 Fall]

Solution: Here function, $f(x) = x^3 - 2x - 5$

$$f(2) = -1 < 0, f(3) = 16 > 0.$$

Therefore, root lies between 2 & 3.

$$\therefore a = 2, b = 3$$

$$\begin{aligned} \text{By Secant formula, } x_n &= \frac{a f(b) - b f(a)}{f(b) - f(a)} \\ &= \frac{(2 \cdot 16) - (3 \cdot -1)}{16 - (-1)} \\ &= 2.05882 \end{aligned}$$

Likewise, calculating the other value of root in tabular form.

Iteration(n)	a	b	f(a)	f(b)	x_n	f(x_n)
1	2	3	-1	16	2.05882	-0.39084
2	3	2.05882	16	-0.39084	2.08126	-0.14724
3	2.05882	2.08126	-0.39084	-0.14724	2.09482	0.00300
4	2.08126	2.09482	-0.14724	0.00300	2.09455	END

Hence, root = 2.09455 which is correct up to three decimal place.

Using Secant method find a root of the equation $x - e^x + 2 = 0$ correct to three decimal places. [2011 Fall]

Solution: Here function, $f(x) = x - e^x + 2$

$$f(1) = 0.2817 > 0, f(2) = -3.38905$$

Therefore, root lies between 1 & 2.

$$\therefore a = 1, b = 2$$

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By Secant formula, $x_n = \frac{a f(b) - b f(a)}{f(b) - f(a)}$

$$= \frac{(1 \cdot 3.38905) - (2 \cdot 2.817)}{-3.38905 - 0.2817}$$

$$= 1.07674$$

Likewise, calculating the other value of root in tabular form.

Iteration(n)	a	b	f(a)	f(b)	x_n	$f(x_n)$
1	1	2	0.2817	-3.38905	1.07674	0.14172
2	2	1.07674	-3.38905	0.14172	1.1137	0.0681
3	1.07674	1.1137	0.14172	0.0681	1.1478	-0.0035
4	1.1137	1.1478	0.0681	-0.0035	1.1461	0.0002
5	1.1478	1.1461	-0.0035	0.0002	1.1461	END

Hence, root = 1.1461 which is correct up to three decimal place.

Find the root of the equation $xe^x - \cos x$ using the false position method, correct to 3 decimal places. [2012 Fall]

Solution: Here function, $f(x) = xe^x - \cos x$

$$f(0.25) = 0.6479 > 0, f(0.75) = -0.8561 < 0.$$

Therefore, root lies between 0.25 & 0.75.

$$\therefore a = 0.25, b = 0.75$$

By Secant formula, $x_n = \frac{a f(b) - b f(a)}{f(b) - f(a)}$

$$= \frac{(0.25 \cdot -0.8561) - (0.75 \cdot 0.6479)}{-0.8561 - 0.6479}$$

$$= 0.4654$$

Likewise, calculating the other value of root in tabular form.

Iteration(n)	a	b	f(a)	f(b)	x_n	$f(x_n)$
1	0.25	0.75	0.6479	-0.8561	0.4654	0.1524
2	0.75	0.4654	-0.8561	0.1524	0.5084	0.0282
3	0.4654	0.5084	0.1524	0.0282	0.5182	-0.0013
4	0.5084	0.5182	0.0282	-0.0013	0.5178	-0.00013
5	0.5182	0.5178	-0.0013	-0.00013	0.5177	-0.00001
6	0.5178	0.5177	-0.00013	-0.00001	0.5177	END

Hence, root = 1.1461 which is correct up to three decimal place.

4. Newton-Raphson Method

Find a real root of the equation $x = e^x$ using Newton-Raphson method to six decimal place. [2005 Fall]

Solution: Here function, $f(x) = x - e^x$

$$f'(x) = 1 - e^x \dots\dots\dots (i)$$

Newton-Raphson formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n - e^{-x_n}}{1 + e^{-x_n}} \dots\dots\dots (ii)$$

$$\text{Now, } f(0) = -1 < 0 \text{ and } f(1) = 0.63212 > 0.$$

Therefore, the root of $f(x)$ lies between 0 & 1.

Since, $|f(0)| > |f(1)|$ Let the initial approximation x_0 be 0.8

Putting $n = 0$ in Eqⁿ (ii), the 1st approximation x_1 is given by

$$x_1 = x_0 - \frac{x_0 - e^{-x_0}}{1 + e^{-x_0}} = 0.8 - \frac{0.8 - e^{-0.8}}{1 + e^{-0.8}} = 0.5580$$

Likewise, calculating the other value of root in tabular form..

Iteration (n)	x_n	x_{n+1}
0	0.8	0.55805
1	0.55805	0.56713
2	0.56713	0.56714
3	0.56714	0.56714

Hence, root = 0.56714 which is correct up to four decimal place.

Find the real root of the function $\tan x + \tanh x = 0$ Newton-Raphson method with initial value 2.2. [2005 Spring]

Solution: Here function, $f(x) = \tan x + \tanh x$

$$f'(x) = \sec^2 x + \text{sech}^2 x \dots\dots\dots (i)$$

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Newton- Raphons formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{\tanh x + \tanh x}{\sec^2 x + \operatorname{sech}^2 x} \dots\dots\dots (ii)$$

Now, $f(2) = -1.2210 < 0$ and $f(3) = 0.8525 > 0$.

Therefore, the root of $f(x)$ lies between 2 & 3.

Since, $|f(2)| > |f(3)|$. Let the initial approximation x_0 be 2.2.

Putting $n = 0$ in Eqⁿ (ii), the 1st approximation x_1 is given by

$$x_1 = x_0 - \frac{\tanh x + \tanh x}{\sec^2 x + \operatorname{sech}^2 x} = 0.5580$$

Likewise, calculating the other value of root in tabular form..

Iteration (n)	x_n	x_{n+1}
0	2.2	2.3356
1	2.3356	2.36416
2	2.36416	2.3650200
3	2.3650200	2.3650205
4	2.3650205	2.3650206

Hence, root = 2.3650206 which is correct up to four decimal place.

Using Newton-Raphson method. Solve $f(x) = e^{x-1} - 5x^3$ correct to four decimal places. [2006 Fall, 2011 Spring]

Solution: Here function, $f(x) = e^{x-1} - 5x^3$

$$f'(x) = e^{x-1} - 15x^2 \dots\dots\dots (i)$$

Newton- Raphons formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{e^{x_n-1} - 5x_n^3}{e^{x_n-1} - 15x_n^2} \dots\dots\dots (ii)$$

Now, $f(0) = 0.367 > 0$ and $f(1) = -4 < 0$.

Therefore, the root of $f(x)$ lies between 0 & 1.

Since, $|f(0)| < |f(1)|$ Let the initial approximation x_0 be 0.4.

Putting $n = 0$ in Eqⁿ (ii), the 1st approximation x_1 is given by

$$x_1 = x_0 - \frac{e^{x_0-1} - 5x_0^3}{e^{x_0-1} - 15x_0^2} = 0.4 - \frac{e^{0.4-1} - 5 \times 0.4^3}{e^{0.4-1} - 15 \times 0.4^2} = 0.52360$$

Likewise, calculating the other value of root in tabular form.

Iteration (n)	x_n	x_{n+1}
0	0.4	0.5236
1	0.5236	0.49589
2	0.49589	0.49505
3	0.49405	0.49404
4	0.49404	0.49404

Hence, root = 0.49404 which is correct up to four decimal place.

Find the real root of the equation $x^3 - 4x + 1 = 0$ that lies between 1 and 4, correct to three decimal places by using the Newton-Raphson method. [2006 Spring]

Solution: Here function, $f(x) = x^3 - 4x + 1$

$$f'(x) = 3x^2 - 4 \dots\dots\dots (i)$$

Newton- Raphons formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 4x_n + 1}{3x_n^2 - 4} \dots\dots\dots (ii)$$

Now, $f(1) = -2 > 0$ and $f(4) = 49 < 0$.

Therefore, the root of $f(x)$ lies between 1 & 4.

Since, $|f(1)| < |f(4)|$ Let the initial approximation x_0 be 1.5

Putting $n = 0$ in Eqⁿ (ii), the 1st approximation x_1 is given by

$$x_1 = x_0 - \frac{x_0^3 - 4x_0 + 1}{3x_0^2 - 4} = 1.4897$$

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Likewise, calculating the other value of root in tabular form.

Iteration (n)	x_n	x_{n+1}
0	1.5	2.0909
1	2.0909	1.8953
2	1.8953	1.8618
3	1.8618	1.8608
4	1.8608	1.8608

Hence, root = 1.8608 which is correct up to four decimal place.

Find out all the possible real roots of the equation $x^3 - 2x - 5 = 0$ using Newton Raphson method to three decimal places. [2007 Spring]

Solution: Here function, $f(x) = x^3 - 2x - 5$

$$f'(x) = 3x^2 - 2 \quad \dots\dots\dots (i)$$

Newton-Raphson formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2} \quad \dots\dots\dots (ii)$$

Now, $f(2) = -1 > 0$ and $f(3) = 16 < 0$.

Therefore the root of $f(x)$ lies between 2 & 3.

Since, $|f(2)| < |f(3)|$ Let the initial approximation x_0 be 2.4

Putting $n = 0$ in Eqⁿ (ii), the 1st approximation x_1 is given by

$$x_1 = x_0 - \frac{x_0^3 - 2x_0 - 5}{3x_0^2 - 2} = 2.136649$$

Likewise, calculating the other value of root in tabular form.

Iteration (n)	x_n	x_{n+1}
0	2.4	2.136649
1	2.136649	2.095516
2	2.095516	2.094552
3	2.094552	2.094551
4	2.094551	2.094551

Hence, root = 2.094551 which is correct up to four decimal place.

Find the real root of $x \log x = 1.2$, using Newton-Raphson method. [2008 Fall, 2008 Spring]

Solution: Here function, $f(x) = x \log_{10} x - 1.2$

$$f'(x) = \log_{10} x + x(\log_{10} e)/x$$

$$\left[\therefore \frac{d \log_{10} x}{dx} = \frac{1}{x} \log_{10} e \right] = \log_{10} x + 0.4343 \quad \dots\dots\dots (i)$$

We know that, Newton-Raphson formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = \frac{x_n \log_{10} x_n - 1.2}{\log_{10} x_n + 0.4343} \quad \dots\dots\dots (ii)$$

Now, $f(2.5) = -0.2051 < 0$ and $f(3) = 0.2313 > 0$.

Therefore, the root of $f(x)$ lies between 2.5 & 3.

Since, $|f(2.5)| < |f(3)|$ Let the initial approximation x_0 be 2.7

Putting $n = 0$ in Eqⁿ (ii), the 1st approximation x_1 is given by

$$x_1 = x_0 - \frac{x_0 \log_{10} x_0 - 1.2}{\log_{10} x_0 + 0.4343} = 2.740798$$

Likewise, calculating the other value of root in tabular form.

Iteration (n)	x_n	x_{n+1}
0	2.7	2.740798
1	2.74	2.740646
2	2.740646	2.7406461
3	2.7406461	2.7406461

Hence, root = 2.7406461 which is correct up to four decimal place.

Find the root of equation $x^2 - 3x + 2 = 0$, correct to four decimal digits, using NR method. [2009 Fall]

Solution: Here function, $f(x) = x^2 - 3x + 2$

$$f'(x) = 2x - 3 \quad \dots\dots\dots (i)$$

We know that, Newton-Raphson formula is

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$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^3 - 3x_n + 2}{3x_n^2 - 3} \dots\dots\dots(ii)$$

Now, $f(-1) = 4 > 0$ and $f(-3) = -16 < 0$.

Therefore the root of $f(x)$ lies between -3 & -1.

Since, $|f(-3)| > |f(-1)|$ Let the initial approximation x_0 be -1.1

Putting $n = 0$ in Eqⁿ (ii), the 1st approximation x_1 is given by $x_1 = x_0 - \frac{x_0^3 - 3x_0 + 2}{3x_0^2 - 3} = -7.4$

Likewise, calculating the other value of root in tabular form.

Iteration (n)	x_n	x_{n+1}
0	-1.1	-7.4
1	-7.4	-5.0375
2	-5.0375	-3.52345
3	-3.52345	-2.613156
4	-2.613156	-2.155372
5	-2.155372	-2.013929
6	-2.013929	-2.000127
7	-2.000127	-2.0

Hence, root = -2.0000 which is correct up to four decimal place.

Find the root of the equation $f(x) = x \sin x + \cos x$ using Newton-Raphson method, correct up to three decimal places. [2009 Spring, 2010 Fall]

Solution: Here function, $f(x) = x \sin x + \cos x$.

$$f'(x) = x \cos x \dots\dots\dots(i)$$

We know that, Newton-Raphsons formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n \sin x_n + \cos x_n}{x_n \cos x_n} \dots\dots\dots(ii)$$

Now, $f(3) = -0.566 < 0$ and $f(2) = 1.4024 > 0$.

Therefore the root of $f(x)$ lies between 3 & 2.

Since, $|f(3)| > |f(2)|$. Let the initial approximation x_0 be 2.7

Putting $n = 0$ in Eqⁿ (ii), the 1st approximation x_1 is given by

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$$x_1 = x_0 - \frac{x_0 \sin x_0 + \cos x_0}{x_0 \cos x_0} = 2.8023$$

Likewise, calculating the other value of root in tabular form.

Iteration (n)	x_n	x_{n+1}
0	2.7	2.8023
1	2.8023	2.79839
2	2.79839	2.79839

Hence, root = 2.79839 which is correct up to four decimal place.

Find the root of the equation $f(x) = 3x - \cos x - 1$ using Newton Raphson method, correct upto three decimal places. [2012 Fall]

Solution: Here function, $f(x) = 3x - \cos x - 1$

$$f'(x) = 3 + \sin x \dots\dots\dots(i)$$

We know that, Newton-Raphsons formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = \frac{3x_n - \cos x_n - 1}{3 + \sin x_n} \dots\dots\dots(ii)$$

Now, $f(0) = -2 < 0$ and $f(1) = 1.4596 > 0$.

Therefore the root of $f(x)$ lies between 0 & 1.

Since, $|f(0)| > |f(1)|$ Let the initial approximation x_0 be 0.8

Putting $n = 0$ in Eqⁿ (ii), the 1st approximation x_1 is given by

$$x_1 = x_0 - \frac{3x_0 - \cos x_0 - 1}{3 + \sin x_0} = 0.61080$$

Likewise, calculating the other value of root in tabular form.

Iteration (n)	x_n	x_{n+1}
0	0.8	0.61080
1	0.61080	0.60710
2	0.60710	0.60710

Hence, root = 0.60710 which is correct up to four decimal place.

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5. Fixed Point Iteration Method

Use fixed point iteration method to evaluate root of the equation $x^2 - x - 1 = 0$, correct up to 3-decimal places. [2006 Spring]

Solution: Here function, $f(x) = x^2 - x - 1$ (i)

For fixed point iteration method, arranging Eqⁿ(i) in term of $\phi(x) = x^2 - x - 1 = 0$

$$\text{or, } x^2 - x = 1 \text{ or, } x(x-1) = 1 \therefore \phi(x) = \frac{1}{x-1}$$

By iteration formula, $x_{n+1} = \phi(x_n)$

Let $x_0 = 0$ be the initial approximation to the root. Then

$$x_1 = \phi(x_0) = \frac{1}{0-1} = -1$$

Likewise, calculating the other values of root in tabular form

Iteration(n)	x_n	x_{n+1}
1	0	-1
2	-1	-0.5
3	-0.5	-0.6667
4	-0.6667	-0.6000
5	-0.6000	-0.6250
6	-0.6250	-0.6153
7	-0.6153	-0.6190
8	-0.6190	-0.6176
9	-0.6176	-0.61818
10	-0.61818	-0.61798
11	-0.61798	-0.61806
12	-0.61806	-0.61803
13	-0.61803	-0.61804
14	-0.61804	-0.61803

Hence, root = -0.61803

Find one root of equation $x^2 - 2x - 3 = 0$. Using fixed point iteration method. [2007 Fall]

Solution: Here function, $f(x) = x^2 - 2x - 3$ (i)

Error (E) = 0.0001

For fixed point iteration method, arranging Eqⁿ(i) in term of $\phi(x) = x$

$$x^2 - 2x - 3 = 0$$

$$\text{or, } x^2 - 2x = 3$$

$$\text{or, } x(x-2) = 3 \therefore \phi(x) = \frac{3}{x-2}$$

By iteration formula, $x_{n+1} = \phi(x_n)$

Let $x_0 = 0$ be the initial approximation to the root. Then

$$x_1 = \phi(x_0) = \frac{3}{0-2} = -1.50$$

Likewise, calculating the other values of root in tabular form

Iteration(n)	x_n	x_{n+1}
1	0	-1.50
2	-1.50	-0.857
3	-0.857	-1.05
4	-1.05	-0.9836
5	-0.9836	-1.005
6	-1.005	-0.9981
7	-0.9981	-1.0006
8	-1.0006	-0.9980
9	-0.9980	-1.00007
10	-1.00007	-0.9998
11	-0.9998	-1.00001
12	-1.00001	-1

Hence root = -1

Find one root of equation $f(x) = x^2 - x - 1$. Using fixed point iteration method. [2006 Spring]

Solution: Here function, $f(x) = x^2 - x - 1$ (i)

For fixed point iteration method, arranging Eqⁿ(i) in term of $\phi(x) = x$

$$x^2 - x - 1 = 0 \text{ or, } x^2 - x = 1 \text{ or, } x(x-1) = 1 \therefore \phi(x) = \frac{1}{x-1}$$

By iteration formula, $x_{n+1} = \phi(x_n)$

Let $x_0 = 0$ be the initial approximation to the root.

$$\text{Then, } x_1 = \phi(x_0) = \frac{1}{0-1} = -1$$

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Likewise, calculating the other values of root in tabular form

Iteration(n)	x_n	x_{n+1}
1	0	-1
2	-1	-0.5
3	-0.5	-0.6667
4	-0.6667	-0.6000
5	-0.6000	-0.6250
6	-0.6250	-0.6153
7	-0.6153	-0.6190
8	-0.6190	-0.6176
9	-0.6176	-0.61818
10	-0.61818	-0.61798
11	-0.61798	-0.61806
12	-0.61806	-0.61803
13	-0.61803	-0.61804
14	-0.61804	-0.61803

Hence, root = -0.61803

Find square root of equation. $f(x) = x^2 - 3$, Using fixed point iteration method. [2010 Fall]

Solution: Here function, $f(x) = x^2 - 3$ (i)

Error (E) = 0.0001

For fixed point iteration method, arranging Eqⁿ(i) in term

$$\phi(x) = x$$

$$x^2 - 3 = 0 \quad \text{or, } x + x = \frac{3}{x} + x \quad \text{or, } x = \frac{3+x^2}{2x}$$

$$\therefore \phi(x) = \frac{3+x^2}{2x}$$

By iteration formula, $x_{n+1} = \phi(x_n)$

Let $x_0 = 1$ be the initial approximation to the root.

Then, $x_1 = \phi(x_0) = 2$

Likewise, calculating the other values of root in tabular form

Iteration(n)	x_n	x_{n+1}
1	1	2
2	2	1.75
3	1.75	1.7321
4	1.7321	1.7320
5	1.7320	1.7320

Hence, root = 1.7320

Find the square root of 17. Using fixed point iteration method. [2009 Fall]

Solution: Let, x be the root of 17 then

$$x^2 = 17 \quad \text{i.e. function, } f(x) = x^2 - 17 \quad \text{.....(i)}$$

[Now, done as 2010 fall]

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CHAPTER 2

Interpolation !

The technique of finding the or estimating the value of function for any intermediate value of the independent variable is known as interpolation.

Let $y = f(x)$ be a discrete function having set of value of x

x	x_0	x_1	x_2		x_n
y	y_0	y_1	y_2		y_n

Then the process of finding the value of y at any value of x = between x_0 to x_n is called interpolation

Similarly, the process of computing the value of function out side the given range is called extrapolation

Interpolation method for equispaced data sample

a) Newton forward interpolation

$$y_p = y_0 + p\Delta y_0 + \frac{(p-1)p}{2!}\Delta^2 y_0 + \frac{(p-2)(p-1)p}{3!}\Delta^3 y_0 + \frac{(p-3)(p-2)(p-1)p}{4!}\Delta^4 y_0$$

b) Newton backward interpolation

$$y_p = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!}\nabla^4 y_n$$

Example:

Use table given below gives the value of $\tan x$ for $10^\circ \leq x \leq 30^\circ$. Find the value of $\tan x$ at (i) $x = 12^\circ$, (ii) $x = 26^\circ$.

x	10	15	20	25	30
$\tan x$	0.1003	0.1511	0.2027	0.2553	0.3093

Solution: At $x = 12^\circ$ (Newton's forward interpolation)

Since, the value $x = 12^\circ$ is near the beginning of the difference table. Hence, to get corresponding value, we use Newton forward interpolation formula. The difference table is tabulated below.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.1	0.1003	0.0509			
0.15	0.1511	0.0515	0.0006		
0.2	0.2027	0.0526	0.0011	0.0005	
0.25	0.2553	0.0540	0.0014	0.0003	0.0002
0.3	0.3093				

Since, $x_p = 0.12$ lies between 0.1 and 0.15

therefore $x_0 = 0.1$ and $y = 0.1003$

$$h = 0.05, \quad p = \frac{x_p - x_0}{h} = \frac{0.12 - 0.1}{0.05} = 0.4$$

Let y_p be the value of x when $x = 0.12$

Using Newton's forward interpolation formula,

$$\begin{aligned} y_p &= y_0 + p\Delta y_0 + \frac{(p-1)p}{2!}\Delta^2 y_0 + \frac{(p-2)(p-1)p}{3!}\Delta^3 y_0 + \frac{(p-3)(p-2)(p-1)p}{4!}\Delta^4 y_0 \\ &= 0.1003 + 0.4 * 0.0509 + \frac{0.4 * (-0.6)}{2!} * 0.0006 + \frac{0.4 * (-0.6) * (-1.6)}{3!} * 0.0003 - \frac{0.4 * (-0.6) * (-1.6) * (-2.6)}{4!} * 0.0002 \end{aligned}$$

$\therefore y_{0.12} = 0.1205798$ which is the required value.

At $x = 0.26$ (Newton's backward interpolation)

Since, the value $x = 0.26$ is near the beginning of the difference table.

Hence, to get corresponding value, we use Newton backward interpolation formula. The difference table is tabulated below.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.1	0.1003	0.0509			
0.15	0.1511	0.0515	0.0006		
0.2	0.2027	0.0526	0.0011	0.0005	
0.25	0.2553	0.0540	0.0014	0.0003	0.0002
0.3	0.3093				

Since, $x_p = 0.26$ lies between 0.25 and 0.3 then $x_n = 0.3$ and

$y_n = 0.3093$

$$h = 0.05, \quad p = \frac{x_p - x_n}{h} = -0.8$$

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Using Newton's backward interpolation formula,

$$\begin{aligned}
 y_p &= y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n \\
 &= 0.3093 + (-0.8 * 0.0540) + \frac{-0.8(-0.8+1)}{2!} * 0.0014 + \frac{-0.8(-0.8+1)(-0.8+2)}{3!} * 0.0003 + \frac{-0.8(-0.8+1)(-0.8+2)(-0.8+3)}{4!} * 0.0002 \\
 \therefore y_{0.26} &= 0.26602 \text{ Which is the required value.}
 \end{aligned}$$

Gauss Interpolation

1. Gauss forward interpolation

$$\begin{aligned}
 y_p &= y_0 + p \Delta y_0 + \frac{(p-1)p}{2!} \Delta^2 y_{-1} + \frac{(p+1)(p-1)p}{3!} \Delta^3 y_{-1} \\
 &\quad + \frac{(p+1)(p-1)(p-2)p}{4!} \Delta^4 y_{-2}
 \end{aligned}$$

Example

Find the value of $\log 337.5$ from the following table by using Gauss forward interpolation formula.

x_i	310	320	330	340	350	360
$\log x$	2.49136	2.50515	2.5185	2.53148	2.54407	2.5563

Solution:

Given that, $x_p = 337.5$, $h=10$, $x_0 = 330$, $p = \frac{337.5-330}{10} = 0.75$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
310	2.49136				
320	2.50515	0.01379			
330	2.5185	0.0134	-0.00039		
340	2.53148	0.0126	-0.0003	-0.00011	
350	2.54407	0.01223	-0.00037	-0.00007	-0.00027
360	2.5563				

Now by using Gauss forward interpolation formula,

$$\begin{aligned}
 y_p &= y_0 + p \Delta y_0 + \frac{(p-1)p}{2!} \Delta^2 y_{-1} + \frac{(p+1)(p-1)p}{3!} \Delta^3 y_{-1} + \frac{(p+1)(p-1)(p-2)p}{4!} \Delta^4 y_{-2} \\
 &= 2.5185 + (0.75 * 0.0129) + \frac{(0.75-1)0.75}{2!} * (-0.0005) + \frac{(0.75+1)0.75}{3!} * 0.0002 + \frac{(0.75+1)(0.75-1)0.75}{4!} * 0.00031 \\
 &= 2.5185 + 0.009675 + 0.0000468 + (-0.00001093) + 0.000005297 \\
 &= 2.5282 \text{ which is the required answer.}
 \end{aligned}$$

2. Gauss backward interpolation

$$\begin{aligned}
 y_p &= y_0 + p \Delta y_{-1} + \frac{(p+1)p}{2!} \Delta^2 y_{-1} + \frac{(p+1)(p-1)p}{3!} \Delta^3 y_{-2} + \frac{(p+1)(p-1)(p+2)p}{4!} \Delta^4 y_{-2}
 \end{aligned}$$

Example:

Find the value of $\sin 45^\circ$ from the following table using Gauss backward interpolation formulation.

$X = \theta$	20	30	40	50	60	70	80
$Y = \sin \theta$	0.342	0.502	0.6427	0.7664	0.86603	0.93919	0.98481

Solution:

Given that $x_0 = 40$, $h=10$, $p = \frac{45-40}{10} = 0.5$

Finding the difference values using difference table for Gauss backward interpolation.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
20	0.342				
30	0.502	0.16			
40	0.6427	0.1407	-0.0192		
50	0.7664	0.12334	-0.0173	0.00184	
60	0.866	0.0999	-0.0234	-0.00599	-0.00783
70	0.9391	0.0793	-0.0206	-0.00298	0.00301
80	0.9848	0.04521	-0.0221	-0.00077	

By Gauss backward interpolation formula

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$$\begin{aligned}
 y_p &= y_0 + p\Delta y_{-1} + \frac{(p+1)p}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-2} + \\
 &= \frac{0.6424}{4!} + \frac{0.5}{4!} + \frac{0.1407}{4!} + \frac{0.5(1.5)}{4!} + \frac{(0.01736)}{4!} + \\
 &= \frac{1.5 \times 0.5 \times (-0.5)}{4!} (0.00184) + \frac{2.5 \times (1.5) \times (0.5) \times (-0.5)}{4!} (-0.00783) \\
 &= 0.6424 + 0.07035 - 0.00651 \\
 &= 0.7071 \text{ which is the required answer.}
 \end{aligned}$$

Interpolation with unequally spaced data sample

1. Lagrange interpolation

The lagrange interpolation formula is given by

$$\begin{aligned}
 y = f(x) &= \frac{(x-x_1)(x-x_2) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2) \dots (x_0-x_n)} y_0 \\
 &+ \frac{(x-x_0)(x-x_2) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2) \dots (x_1-x_n)} y_1 + \dots + \frac{(x-x_0)(x-x_1) \dots (x-x_{n-1})}{(x_n-x_0)(x_n-x_1) \dots (x_n-x_{n-1})} y_n
 \end{aligned}$$

Example:

Given that,

X	5	7	11	13	17
Y	150	392	1452	1366	5202

Find the value of f(9).

Solution:

here $x_0 = 5, x_1 = 7, x_2 = 11, x_3 = 13, x_4 = 17$ $y_0 = 150,$
 $y_1 = 392, y_2 = 1452, y_3 = 2366, y_4 = 5202$

By Lagrange's formula,

$$\begin{aligned}
 y &= f(x) \\
 &= \frac{(x-x_1)(x-x_2) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2) \dots (x_0-x_n)} y_0 \\
 &+ \frac{(x-x_0)(x-x_2) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2) \dots (x_1-x_n)} y_1 + \dots \\
 &+ \frac{(x-x_0)(x-x_1) \dots (x-x_{n-1})}{(x_n-x_0)(x_n-x_1) \dots (x_n-x_{n-1})} y_n \\
 &= \frac{(9-7)(9-11)(9-13)(9-17)}{(5-7)(5-11)(5-13)(5-17)} * 150 + \frac{(9-5)(9-11)(9-13)(9-17)}{(7-5)(7-11)(7-13)(7-17)} * 392 + \\
 &\frac{(9-5)(9-7)(9-13)(9-17)}{(11-5)(11-7)(11-13)(11-17)} * 1452 + \frac{(9-5)(9-7)(9-11)(9-17)}{(13-5)(13-7)(13-11)(13-17)} * 2366 + \\
 &\frac{(9-5)(9-7)(9-11)(9-13)}{(17-5)(17-7)(17-11)(17-13)} * 5202
 \end{aligned}$$

$$\begin{aligned}
 &= -16.66 + 209.06 + 1290.66 - 788.666 + 115.6 \\
 &= 809.99
 \end{aligned}$$

which is the required answer.

2. Newton divided difference interpolation

The Newtons divided difference formula is given by

$$\begin{aligned}
 Y &= y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)[x_0, x_1, x_2] \\
 &+ \dots + (x-x_0)(x-x_1) \dots (x-x_n)[x_0, x_1 \dots x_n]
 \end{aligned}$$

Example:

Given that,

X	5	7	11	13	17
Y	150	392	1452	1366	5202

Find the value off f(9). Using Newtons divided difference formula.

Solution:

Here, $x=9$ Calculating the divided difference using the divided difference table as given below

x	y	1 st diff.	2 nd diff	3 rd diff	4 th diff
5	150				
7	392	121			
11	1452	265	24		
13	2366	457	32	1	
17	5202	709	42	1	0

[hint: 1st difference = $\frac{392-150}{7-5} = 121$ & similarly so on]

Now by using formula , then

$$\begin{aligned}
 Y &= y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)[x_0, x_1, x_2] \\
 &+ \dots + (x-x_0)(x-x_1) \dots (x-x_n)[x_0, x_1 \dots x_n] \\
 &= 150 + (9-5)121 + (9-5)(9-7)24 + (9-5)(9-7)(9-11)1 + 0 \\
 &= 634 + 192 - 16 \\
 &= 810 \text{ which is the required answer.}
 \end{aligned}$$

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Using Newton's backward interpolation formula,

$$\begin{aligned}
 y_p &= y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n \\
 &= 0.3093 + (-0.8 \cdot 0.0540) + \frac{-0.8(-0.8+1)}{2!} \cdot 0.0014 + \\
 &\quad \frac{-0.8(-0.8+1)(-0.8+2)}{3!} \cdot 0.0003 + \frac{-0.8(-0.8+1)(-0.8+2)(-0.8+3)}{4!} \cdot 0.0002 \\
 \therefore y_{0.26} &= 0.26602 \text{ Which is the required value.}
 \end{aligned}$$

Gauss Interpolation

1. Gauss forward interpolation

$$\begin{aligned}
 y_p &= y_0 + p \Delta y_0 + \frac{(p-1)p}{2!} \Delta^2 y_{-1} + \frac{(p+1)(p-1)p}{3!} \Delta^3 y_{-1} \\
 &\quad + \frac{(p+1)(p-1)(p-2)p}{4!} \Delta^4 y_{-2}
 \end{aligned}$$

Example

Find the value of $\log 337.5$ from the following table by using Gauss forward interpolation formula.

X	310	320	330	340	350	360
Log x	2.49136	2.50515	2.5185	2.53148	2.54407	2.5563

Solution:

Given that, $x_p = 337.5$, $h=10$, $x_0 = 330$, $p = \frac{337.5-330}{10} = 0.75$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
310	2.49136				
320	2.50515	0.01379			
330	2.5185	0.0134	-0.00039		
340	2.53148	0.0129	-0.0005	-0.00011	
350	2.54407	0.0126	-0.0003	0.0002	0.00031
360	2.5563	0.01223	-0.00037	-0.00007	-0.00027

Now by using Gauss forward interpolation formula,

$$\begin{aligned}
 y_p &= y_0 + p \Delta y_0 + \frac{(p-1)p}{2!} \Delta^2 y_{-1} + \frac{(p+1)(p-1)p}{3!} \Delta^3 y_{-1} + \\
 &\quad \frac{(p+1)(p-1)(p-2)p}{4!} \Delta^4 y_{-2} \\
 &= 2.5185 + (0.75 \cdot 0.0129) + \frac{(0.75-1)0.75}{2!} \cdot (-0.0005) + \frac{(0.75+1)0.75}{3!} \cdot 0.00031 \\
 &\quad + \frac{(0.75+1)(0.75-1)0.75}{4!} \cdot 0.00031 \\
 &= 2.5185 + 0.009675 + 0.0000468 + (-0.00001093) + 0.000005297 \\
 &= 2.5282 \text{ which is the required answer.}
 \end{aligned}$$

2. Gauss backward interpolation

$$\begin{aligned}
 y_p &= y_0 + p \Delta y_{-1} + \frac{(p+1)p}{2!} \Delta^2 y_{-1} + \frac{(p+1)(p-1)p}{3!} \Delta^3 y_{-2} + \\
 &\quad \frac{(p+1)(p-1)(p+2)p}{4!} \Delta^4 y_{-2}
 \end{aligned}$$

Example:

Find the value of $\sin 45^\circ$ from the following table, using Gauss backward interpolation formula.

X	0	20	30	40	50	60	70	80
Y = sin	0	0.342	0.502	0.6427	0.7604	0.86603	0.93919	0.98481

Solution:

Given that $x_0 = 40$, $h=10$, $p = \frac{45-40}{10} = 0.5$

Finding the difference values using difference table for Gauss backward interpolation.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
20	0.342				
30	0.502	0.16			
40	0.6427	0.1407	-0.0192		
50	0.7604	0.12334	-0.0173	0.00184	
60	0.866	0.0999	-0.0234	-0.00599	-0.00783
70	0.9391	0.0793	-0.0206	-0.00298	0.00301
80	0.9848	0.04521	0.03409	-0.00221	0.00077

By Gauss backward interpolation formula

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$$\begin{aligned}
 y_p &= y_0 + p\Delta y_{-1} + \frac{(p+1)p}{2!} \Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!} \Delta^3 y_{-2} + \\
 &\quad \frac{(p+1)p(p-1)(p+2)p}{4!} \Delta^4 y_{-2} \\
 &= 0.6424 + 0.5 * 0.1407 + \frac{0.5(1.5)}{2!} * (0.01736) + \\
 &\quad \frac{1.5 * 0.5 * (-0.5)}{3!} (0.00184) + \frac{2.5 * (1.5) * (0.5) * (-0.5)}{4!} (-0.00783) \\
 &= 0.6424 + 0.07035 - 0.00651 \\
 &= 0.7071 \text{ which is the required answer.}
 \end{aligned}$$

Interpolation with unequally spaced data sample

1. Lagrange interpolation

The lagrange interpolation formula is given by

$$\begin{aligned}
 y &= f(x) = \frac{(x-x_1)(x-x_2) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2) \dots (x_0-x_n)} y_0 \\
 &\quad + \frac{(x-x_0)(x-x_2) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2) \dots (x_1-x_n)} y_1 + \dots + \frac{(x-x_0)(x-x_1) \dots (x-x_{n-1})}{(x_n-x_0)(x_n-x_1) \dots (x_n-x_{n-1})} y_n
 \end{aligned}$$

Example:

Given that,

X	5	7	11	13	17
Y	150	392	1452	1366	5202

Find the value of $f(9)$.

Solution:

here $x_0 = 5, x_1 = 7, x_2 = 11, x_3 = 13, x_4 = 17, y_0 = 150,$

$y_1 = 392, y_2 = 1452, y_3 = 2366, y_4 = 5202$

By Lagrange's formula,

$$\begin{aligned}
 y &= f(x) \\
 &= \frac{(x-x_1)(x-x_2) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2) \dots (x_0-x_n)} y_0 \\
 &\quad + \frac{(x-x_0)(x-x_2) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2) \dots (x_1-x_n)} y_1 + \dots \\
 &\quad + \frac{(x-x_0)(x-x_1) \dots (x-x_{n-1})}{(x_n-x_0)(x_n-x_1) \dots (x_n-x_{n-1})} y_n \\
 &= \frac{(9-7)(9-11)(9-13)(9-17)}{(5-7)(5-11)(5-13)(5-17)} * 150 + \frac{(9-5)(9-11)(9-13)(9-17)}{(7-5)(7-11)(7-13)(7-17)} * 392 + \\
 &\quad \frac{(9-5)(9-7)(9-13)(9-17)}{(11-5)(11-7)(11-13)(11-17)} * 1452 + \frac{(9-5)(9-7)(9-11)(9-17)}{(13-5)(13-7)(13-11)(13-17)} * 2366 + \\
 &\quad \frac{(9-5)(9-7)(9-11)(9-13)}{(17-5)(17-7)(17-11)(17-13)} * 5202
 \end{aligned}$$

$$\begin{aligned}
 &= -16.66 + 209.06 + 1290.66 - 788.666 + 115.6 \\
 &= 809.99
 \end{aligned}$$

Which is the required answer.

2. Newton divided difference interpolation

The Newtons divided difference formula is given by

$$\begin{aligned}
 Y &= y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)[x_0, x_1, x_2] \\
 &\quad + \dots + (x-x_0)(x-x_1) \dots (x-x_n)[x_0, x_1 \dots x_n]
 \end{aligned}$$

Example:

Given that,

X	5	7	11	13	17
Y	150	392	1452	1366	5202

Find the value of $f(9)$. Using Newtons divided difference formula.

Solution:

Here, $x=9$ Calculating the divided difference using the divided difference table as given below

x	y	1 st diff.	2 nd diff.	3 rd diff.	4 th diff.
5	150	<u>121</u>			
7	392	265	<u>24</u>		
11	1452	457	32	<u>1</u>	
13	2366	709	42	1	<u>0</u>
17	5202				

[hint: 1st difference = $\frac{392-150}{7-5} = 121$ & similarly so on]

Now by using formula, then

$$\begin{aligned}
 Y &= y_0 + (x-x_0)[x_0, x_1] + (x-x_0)(x-x_1)[x_0, x_1, x_2] \\
 &\quad + \dots + (x-x_0)(x-x_1) \dots (x-x_n)[x_0, x_1 \dots x_n] \\
 &= 150 + (9-5)121 + (9-5)(9-7)24 + (9-5)(9-7)(9-11)1 + 0 \\
 &= 634 + 192 - 16 \\
 &= 810 \text{ which is the required answer.}
 \end{aligned}$$

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Cubic Spline Interpolation

In previous interpolation, a single polynomial is fitted to the tabulated points. If given sets of points belongs to the polynomials then this method works well otherwise the result are rough approximation only.

If we draw the line through every two closest point, resulting graph will not be smooth. Similarly, we can quadratic curve through every successive points such that slope of two successive curve match at their common points, the resulting graph look better but not quite smooth. So for smooth ness, we draw a cubic curve through A_i, A_{i+1} and another through A_{i+1} and A_{i+2} , such that slope and curvature of two points match at A_{i+1} . Such a curve is called a cubic spline curve.

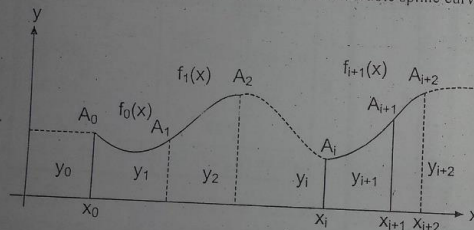


Figure : Curve for cubic spline

University Question Solutions

1. Interpolation

The velocity distribution of the fluid near a flat surface is given below:

X(cm)	0.1	0.3	0.5	0.7	0.9
V(cm/s)	0.72	1.81	2.73	3.47	3.98

X is the distance from the surface using a suitable interpolation formula, obtain the velocity at $x=0.2$ [2006 Spring, 2009 Spring]

Solution: Since the value $x=0.2$ is near the beginning of the difference table. Hence to get corresponding V (cm/s), we use Newton forward interpolation formula. The difference table is tabulated below.

x	y = V	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.1	0.72	1.09			
0.3	1.81	0.92	-0.17		
0.5	2.73	0.74	-0.18	-0.01	
0.7	3.47	0.51	-0.23	-0.05	-0.04
0.9	3.98				

Since $x_p = 0.2$ lies between 0.1 and 0.3 therefore $x_0 = 0.1$ and $V_0 = 0.72$

$$h = 0.3 - 0.1 = 0.2, \quad p = \frac{x_p - x_0}{h} = \frac{0.2 - 0.1}{0.2} = 0.5$$

Let y_p be the value of x when $x = 0.2$

Using Newton's forward interpolation formula,

$$\begin{aligned} y_p &= y_0 + p\Delta y_0 + \frac{(p-1)p}{2!}\Delta^2 y_0 + \frac{(p-2)(p-1)p}{3!}\Delta^3 y_0 + \frac{(p-3)(p-2)(p-1)p}{4!}\Delta^4 y_0 \\ &= 0.72 + 0.5 * 1.09 + \frac{(0.5-1)0.5}{2!} * -0.17 + \frac{(0.5-2)0.5}{3!} * -0.01 + \frac{(0.5-3)(0.5-2)0.5}{4!} * -0.04 \end{aligned}$$

$\therefore y_{0.2} = 1.28$ which is the required value.

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The following data gives the melting point of an alloy of lead and zinc, where t is the temperature and P is the percentage of lead in the alloy. Using Newton's interpolation formula, find the melting point of an alloy containing 75% lead. [2008 Fall]

P	40	50	60	70	80
t	184	204	226	250	276

Solution: Since the value $p = 75\%$ is near the end of the difference table. Hence to get corresponding t , we use Newton backward interpolation formula. The difference table is tabulated below.

$x=p$	$y=t$	∇y	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
40	184				
		20			
50	204		2		
		22		0	
60	226		2		0
		24		0	
70	250		2		
		26			
80	276				

Since,

$x_p = 75\%$ lies between 70 and 80 therefore $x_n = 80$ and $y_n = 276$

$$h = 10, p = \frac{x_p - x_n}{h} = -0.5$$

Using Newton's backward interpolation formula,

$$\begin{aligned}
 y_p &= y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n \\
 &= 276 + (-0.5 * 20) + \frac{-0.5(-0.5+1)}{2!} * 2 + 0 + 0 \\
 \therefore y_{75\%} &= 262.875 \quad \text{Which is the required value.}
 \end{aligned}$$

The following set of data represent the position of a car in a road at specified time. Calculate the position of the car at $T = 1.75$ hours. (You can use any method). [2008 Spring]

Time(hr)	0	0.5	1.0	1.5	2.0
Position(km)	0	0.25	1.0	2.25	4.0

Solution: Since the value $t = 1.75$ is near the end of the difference table. Hence to get corresponding p , we use Newton backward interpolation formula. The difference table is tabulated below.

$x=t$	$y=p$	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
0	0				
		0.25			
0.5	0.25		0.5		
		0.75		0	
1	1		0.5		0
		1.25		0	
1.5	2.25		0.5		
		1.75			
2	4				

Since,

$x_p = 1.75$ lies between 1.5 and 2 therefore $x_n = 2$ and $y_n = 4$

$$h = 0.5, p = \frac{x_p - x_n}{h} = -0.5$$

Using Newton's backward interpolation formula,

$$\begin{aligned}
 y_p &= y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n \\
 &= 4 + (-0.5 * 1.75) + \frac{-0.5(-0.5+1)}{2!} * 0.5 + 0 + 0 \\
 \therefore y_{1.75} &= 3.0625 \text{ km} \quad \text{Which is the required value.}
 \end{aligned}$$

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Find Newton's forward difference interpolating polynomial for the following data. Use the obtained formula to estimate the value of $f(0.24)$. [2010 Fall]

x	0.1	0.2	0.3	0.4	0.5
f(x)	1.40	1.56	1.76	2.00	2.28

Solution: Since the value $x = 0.24$ is near the beginning of the difference table. Hence to get corresponding value of $f(x)$, we use Newton forward interpolation formula. The difference table is tabulated below.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.1	1.40				
0.2	1.56	0.16			
0.3	1.76	0.2	0.04		
0.4	2.0	0.24	0.04	0	
0.5	2.28	0.28	0.04	0	0

Since $x_p = 0.24$ lies between 0.3 and 0.2 therefore $x_0 = 0.2$ and $y_0 = 1.56$

$$h = 0.3 - 0.2 = 0.1, \quad p = \frac{x_p - x_0}{h} = \frac{0.24 - 0.2}{0.1} = 0.4$$

Let y_p be the value of x when $x = 0.24$

Using Newton's forward interpolation formula,

$$y_p = y_0 + p \Delta y_0 + \frac{(p-1)p}{2!} \Delta^2 y_0 + \frac{(p-2)(p-1)p}{3!} \Delta^3 y_0 + \frac{(p-3)(p-2)(p-1)p}{4!} \Delta^4 y_0$$

$$= 1.56 + 0.4 * 0.2 + \frac{(0.4-1)0.4}{2!} * (-0.04) + 0 + 0$$

$\therefore y_{0.24} = 1.6352$ which is the required value.

Estimate the value of $\sin \theta$ at $\theta = 25$ using Newton-Gregory forward difference formula with the help of the following table: [2011 Fall]

θ	10	20	30	40	50
$\sin \theta$	0.1736	0.3420	0.5000	0.6428	0.7660

Solution: Since the value of $\sin \theta$ at $\theta = 25$ is near the beginning of the difference table. Hence to get corresponding value of $\sin 25$, we use Newton forward interpolation formula.

The difference table is tabulated below.

Let $y = \sin \theta$ and $x = \theta$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
10	0.1736				
20	0.3420	0.1684			
30	0.5	0.185	-0.0104		
40	0.6428	0.1428	-0.0152	-0.0048	
50	0.7660	0.1232	-0.0196	-0.0044	0.0004

Since

$x_p = 25$ lies between 20 and 30 therefore $x_0 = 20$ and $y_0 = 0.3420$

$$h = 30 - 20 = 10, \quad p = \frac{x_p - x_0}{h} = \frac{25 - 20}{10} = 0.5$$

Let y_p be the value of x when $x = 25$

Using Newton's forward interpolation formula,

$$y_p = y_0 + p \Delta y_0 + \frac{(p-1)p}{2!} \Delta^2 y_0 + \frac{(p-2)(p-1)p}{3!} \Delta^3 y_0 + \frac{(p-3)(p-2)(p-1)p}{4!} \Delta^4 y_0$$

$$= 0.3420 + 0.5 * 0.158 + \frac{(0.5-1)0.5}{2!} * (-0.0152) + \frac{(0.5-2)(0.5-1)0.5}{3!} * (-0.0044)$$

$\therefore y_{25} = 0.4226$ which is the required value.

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Use appropriate method of interpolation to get $f(0.475)$ from given table. [2010 Spring]

x	0	0.1	0.2	0.3	0.4	0.5
F(x)	1.0	1.1052	1.2214	1.3499	1.4918	1.6487

Solution: Since the value $x = 0.475$ is near the end of the difference table. Hence to get corresponding y , we use Newton backward interpolation formula. The difference table is tabulated below.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	1				
0.1	1.1052	0.1652			
		0.056	-0.1092		
0.2	1.2214		0.0725	0.1817	
		0.1285		0.0591	-0.2408
0.3	1.3499		0.0134		0.0032
		0.1419		0.0016	
0.4	1.4918		0.015		
		0.1569			
0.5	1.6487				

Since,

$x = 0.475$ lies between 0.4 and 0.5 therefore $x_n = 0.5$ and

$$y_n = 1.6487$$

$$h = 0.5 - 0.4 = 0.1, p = \frac{x_p - x_n}{h} = -0.25$$

Using Newton's backward interpolation formula,

$$y_p = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_n$$

$$= 1.6487 + (-0.25 * 0.1569) + \frac{-0.25(-0.25+1)}{2!} * 0.015 + \frac{-0.25(-0.25+1)(-0.25+2)}{3!} * 0.0016 + \frac{-0.25(-0.25+1)(-0.25+2)(-0.25+3)}{4!} * 0.0032$$

$$\therefore y_{0.475} = 1.600981 \text{ which is the required value.}$$

Use appropriate method of interpolation to get $f(0.375)$ from given table. Also calculate the absolute error at this point. [2011 Spring]

x	0.125	0.25	0.375	0.5	0.625	0.75
f(x)	0.7916	0.7733	0.7437	0.7041	0.6536	0.6022

Solution: Since the value $x = 0.375$ is near the beginning of the difference table. Hence to get corresponding y , we use Newton forward interpolation formula. The difference table is tabulated below.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.125	0.7916				
		-0.0183			
0.250	0.7733		-0.05090		
		-0.0692		0.0723	
0.500	0.7041		-0.0478		-0.1
		-0.0541		-0.0063	
0.625	0.6536				
0.750	0.6022				

Since

$x_p = 0.375$ lies between 0.25 and 0.5 therefore $x_0 = 0.25$ and

$$y_0 = 0.7733$$

$$h = 0.125, p = \frac{x_p - x_0}{h} = \frac{0.375 - 0.250}{0.125} = 1$$

Let y_p be the value of x when $x = 0.375$

Using Newton's forward interpolation formula,

$$y_p = y_0 + p \Delta y_0 + \frac{(p-1)p}{2!} \Delta^2 y_0 + \frac{(p-2)(p-1)p}{3!} \Delta^3 y_0 + \frac{(p-3)(p-2)(p-1)p}{4!} \Delta^4 y_0$$

$$= 0.7733 + 1 * (-0.0692) + \frac{(1-1)0.125}{2!} * 0.0214 + 0 + 0$$

$$\therefore y_{0.375} = 0.7041 \text{ which is the required value.}$$

Now, Absolute error = 0.396

$$\text{Relative error} = 0.07562$$

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Or

Write an algorithm for Lagrange's interpolation method to interpolate any functional value of x from given set of data points. Use suitable interpolation formula to find the value of $y(10)$ from the following data: [2007 Spring]

x :	5	6	9	11
y :	12	13	14	16

Solution:

Here, $x_0 = 5, x_1 = 6, x_2 = 9, x_3 = 11, y_0 = 12, y_1 = 13, y_2 = 14, y_3 = 16$.

By Lagrange's formula,

$$y = f(x)$$

$$= \frac{(x-x_1)(x-x_2) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2) \dots (x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2) \dots (x_1-x_n)} y_1 + \dots + \frac{(x-x_0)(x-x_1) \dots (x-x_{n-1})}{(x_n-x_0)(x_n-x_1) \dots (x_n-x_{n-1})} y_n$$

$$= \frac{(x-6)(x-9)(x-11)}{(5-6)(5-9)(5-11)} \cdot 12 + \frac{(x-5)(x-9)(x-11)}{(6-5)(6-9)(6-11)} \cdot 13 + \frac{(x-5)(x-6)(x-11)}{(9-5)(9-6)(9-11)} \cdot 14 + \frac{(x-5)(x-6)(x-9)}{(11-5)(11-6)(11-9)} \cdot 16$$

Putting $x = 10$ in the above Eqⁿ we get

$$y = \frac{(10-6)(10-9)(10-11)}{(5-6)(5-9)(5-11)} \cdot 12 + \frac{(10-5)(10-9)(10-11)}{(6-5)(6-9)(6-11)} \cdot 13 + \frac{(10-5)(10-6)(10-11)}{(9-5)(9-6)(9-11)} \cdot 14 + \frac{(10-5)(10-6)(10-9)}{(11-5)(11-6)(11-9)} \cdot 16$$

$$= 2 \cdot 4.33333 + 11.66666 + 5.33333$$

$$= 14.66666$$

Find the polynomial using Lagrange's method also calculate the absolute and relative error at $x=2.7$. [2009 Fall]

x :	3.2	2.7	1.0	4.8
$f(x)$:	22.0	17.8	14.2	38.3

Solution:

Here $x_0 = 3.2, x_1 = 1.0, x_2 = 4.8, y_0 = 22, y_1 = 14.2, y_2 = 38.3$

By Lagrange's formula,

$$= f(x)$$

$$= \frac{(x-x_1)(x-x_2) \dots (x-x_n)}{(x_0-x_1)(x_0-x_2) \dots (x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2) \dots (x-x_n)}{(x_1-x_0)(x_1-x_2) \dots (x_1-x_n)} y_1 + \dots + \frac{(x-x_0)(x-x_1) \dots (x-x_{n-1})}{(x_n-x_0)(x_n-x_1) \dots (x_n-x_{n-1})} y_n$$

$$= \frac{(x-1.0)(x-4.8)}{(3.2-1.0)(3.2-4.8)} \cdot 22 + \frac{(x-3.2)(x-4.8)}{(1.0-3.2)(1.0-4.8)} \cdot 14.2 + \frac{(x-3.2)(x-1.0)}{(4.8-3.2)(4.8-1.0)} \cdot 38.3$$

Putting $x = 2.7$ in the above Eqⁿ we get

$$y = \frac{(2.7-1.0)(2.7-4.8)}{(3.2-1.0)(3.2-4.8)} \cdot 22 + \frac{(2.7-3.2)(2.7-4.8)}{(1.0-3.2)(1.0-4.8)} \cdot 14.2 + \frac{(2.7-3.2)(2.7-1.0)}{(4.8-3.2)(4.8-1.0)} \cdot 38.3$$

$$= -5.354 + 22.3125 + 1.7834$$

$$= 18.74$$

$$\text{Absolute error} = |x_{abs} - x_{approx}| = 0.9$$

$$\text{Relative error} = \frac{|x_{abs} - x_{approx}|}{x_{abs}} = \left| \frac{0.9}{17.8} \right| = 0.050$$

$$\text{Percentage error} = (\text{relative error} \cdot 100)\% = 5\%$$

The table given below shows the temperatures T (in degree Celsius) and lengths L (in mm) of heated rod. Find the temperature at $L = 800.8$ mm, using Lagrange interpolation method. [2006 Fall]

T	30	40	50	60	70
L	800.4	800.6	800.7	800.9	801.0

Solution:

Here $x_0 = 30, x_1 = 40, x_2 = 50, x_3 = 60, x_4 = 70, y_0 =$

$800.4, y_1 = 800.6, y_2 = 800.7, y_3 = 800.9, y_4 = 801.0$

By Lagrange's inverse interpolation formula,

$$x = f(y)$$

$$= \frac{(y-y_1)(y-y_2) \dots (y-y_n)}{(y_0-y_1)(y_0-y_2) \dots (y_0-y_n)} x_0 + \dots$$

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$$\begin{aligned}
& + \frac{(y-y_0)(y-y_2) \dots (y-y_n)}{(y_1-y_0)(y_1-y_2) \dots (y_1-y_n)} x_1 \dots \\
& + \frac{(y-y_0)(y-y_1) \dots (y-y_{n-1})}{(y_0-y_0)(y_0-y_1) \dots (y_0-y_{n-1})} x_n \\
& = \frac{(800.4-800.6)(800.4-800.7)(800.4-800.9)(800.4-800.1)}{(y-800.4)(y-800.7)(y-800.9)(y-801.0)} * 30 + \\
& \frac{(800.6-800.4)(800.6-800.7)(800.6-800.9)(800.6-801.0)}{(y-800.4)(y-800.6)(y-800.9)(y-801.0)} * 40 + \\
& \frac{(800.7-800.4)(800.7-800.6)(800.7-800.9)(800.7-801.0)}{(y-800.4)(y-800.6)(y-800.7)(y-801.0)} * 50 + \\
& \frac{(800.9-800.4)(800.9-800.6)(800.9-800.7)(800.9-801.0)}{(y-800.4)(y-800.6)(y-800.7)(y-800.9)} * 60 \\
& + \frac{(801-800.4)(801-800.6)(801-800.7)(801-800.9)}{(801-800.4)(801-800.6)(801-800.7)(801-800.9)} * 70
\end{aligned}$$

Putting $y = 800.8$ in the above Eqⁿ we get

$$\begin{aligned}
y & = \frac{(800.8-800.6)(800.8-800.7)(800.8-800.9)(800.8-801.0)}{(800.4-800.6)(800.4-800.7)(800.4-800.9)(800.4-801.0)} * 30 + \\
& \frac{(800.8-800.4)(800.8-800.7)(800.8-800.9)(800.8-801.0)}{(800.6-800.4)(800.6-800.7)(800.6-800.9)(800.6-801.0)} * 40 + \\
& \frac{(800.8-800.4)(800.8-800.6)(800.8-800.9)(800.8-801.0)}{(800.7-800.4)(800.7-800.6)(800.7-800.9)(800.7-801.0)} * 50 + \\
& \frac{(800.8-800.4)(800.8-800.6)(800.8-800.7)(800.8-801.0)}{(800.9-800.4)(800.9-800.6)(800.9-800.7)(800.9-801.0)} * 60 \\
& + \frac{(801-800.4)(801-800.6)(801-800.7)(801-800.9)}{(801-800.4)(801-800.6)(801-800.7)(801-800.9)} * 70
\end{aligned}$$

= 55.94 which is required answer.

2. Least Square Method

Curve fitting

Calculation of $\sum xy, \sum y, \sum x, \sum x^2 y$

step 1. MODE $\Rightarrow$ press

3 $\Rightarrow$ press 3 $\Rightarrow$ set value of x & y respectively then press AC

step 2. SHIFT $\Rightarrow$ STAT(1) $\Rightarrow$ press(4) $\Rightarrow$ then take a value
your requirement
and repeat step 2

Matrix calculation

multiplication:

for matrix A

p 1. MODE $\Rightarrow$ press 6 $\Rightarrow$ press 1(for matrix A) $\Rightarrow$
press 1(for matrix 3*3) then press AC

matrix B

p 2. SHIFT $\Rightarrow$ matrix(4) $\Rightarrow$ press2 $\Rightarrow$ press2(for matrix B)
press1(for 3*3), then AC

p 3. SHIFT $\Rightarrow$ matrix(4) $\Rightarrow$ press3 $\Rightarrow$ press4

verse:

ceed step 1, then

F $\Rightarrow$ matrix(4) $\Rightarrow$ press3 $\Rightarrow$ press x^{-1}

ss seidal

A $\Rightarrow$ press ALPHA $\Rightarrow$ SOLVE $\Rightarrow$ set the equation $\Rightarrow$ ALPHA $\Rightarrow$

ress $\frac{d}{dx}$

2(20-2*A-3*B); B=1/4(10-3*A-4*C); C=1/6(15-2*A+5*C)

AC, then set B=0, C=0, at first iteration then press AC continuously

apter first

the calculation of root interval of the function

p 1. press MODE $\Rightarrow$ press 7= set equation, then press = sign, now start?=
end? = 5, step? = 1 the root lies between -ve sign to +ve sign

from the following data.

x	1	2	3	4	5
y	0.5	2	4.5	8	12.5

a power function model of the form $y = ax^b$. [2006 Fall]

ution:

The given curve is $y = ax^b$ (i)

Taking $\log_{10}$ on both sides we get

$\log_{10} y = \log_{10} a + b \log_{10} x$ Which is of the form

$Y = A + bX$ (ii)

Where, $Y = \log_{10} y$, $A = \log_{10} a$, $X = \log_{10} x$

The normal Equations of (ii) are

$\sum Y = nA + b \sum X$ (iii)

$\sum XY = A \sum X + b \sum X^2$ (iv)

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X	Y	X = log ₁₀ X	Y = log ₁₀ Y	X ²	XY
1	0.5	0	-0.3010	0	0
2	2.0	0.3010	0.3010	0.0906	0.0906
3	4.5	0.4771	0.6532	0.2276	0.3116
4	8.0	0.6021	0.9031	0.3625	0.5438
5	12.5	0.6990	1.0969	0.4886	0.7667
		ΣX=2.0792	ΣY=2.6532	ΣX ² =1.1693	ΣXY=1.71

Putting these value in Eq^a (iii) & (iv),
 we get $2.532 = 2.0792b + 5A$ (*)
 $1.7127 = 1.1693b + 2.0792A$ (**)
 On solving (*) & (**), $b = 2, A = 0.3010$
 $\therefore a = \text{antilog}(-0.3010) = 0.5$

Hence the required curve is $y = 0.5x^2$

Use the data linearization method and determine the exponential fit of $y = Ce^{Ax}$ for the following data: [2006 Spring]

X:	0	1	2	3	4
Y:	1.5	2.5	3.5	5.0	7.5

Solution: The given curve is $y = ce^{Ax}$ (i)

Taking $\log_{10}$ on both sides we get

$$\log_{10} y = \log_{10} c + Ax \log_{10} e$$

which is of the form $Y = V + aX$ (ii)

Where, $Y = \log_{10} y$, $V = \log_{10} c$, $a = A \log_{10} e$

The normal Equations of (ii) are

$$\Sigma Y = nV + a \Sigma X \text{ (iii) \&}$$

$$\Sigma XY = V \Sigma X + a \Sigma X^2 \text{ (iv)}$$

x	y	Y = log ₁₀ y	X ²	XY
0	1.5	0.1760	0	0.2640
1	2.5	0.3979	1	0.9947
2	3.5	0.5440	4	1.9040
3	5.0	0.6989	9	3.4945
4	7.5	0.8750	16	3.5629
ΣX=10		ΣY=2.6918	ΣX ² =30	ΣXY=7.0828

Putting these value in Eq^a (iii) & (iv), we get

$$2.6918 = 5V + 10 \text{ (*) \&}$$

$$7.0824 = 10V + 30 \text{ (**)}$$

On solving (*) & (**),

$$V = 0.1985, a = -0.1699$$

$$A = \frac{a}{\log_{10} e} = -0.3911$$

$$\therefore c = \text{antilog}(0.1985) = 1.5794$$

Hence, the required curve is $y = 1.5794 e^{0.3911x}$

fit the exponential curve of the form $y = ax^b$ for the following data, [2008 Fall]

X	1	2	3	4	5
Y	0.5	2	4.5	8	203

Solution: The given curve is $y = ax^b$ (i)

Taking $\log_{10}$ on both sides we get

$$\log_{10} y = \log_{10} a + b \log_{10} x$$

Which is of the form $Y = A + bX$ (ii)

Where, $Y = \log_{10} y$, $A = \log_{10} a$, $X = \log_{10} x$

The normal Equations of (ii) are

$$\Sigma Y = nA + b \Sigma X \text{ (iii) \&}$$

$$\Sigma XY = A \Sigma X + b \Sigma X^2 \text{ (iv)}$$

x	y	$X = \log_{10}x$	$Y = \log_{10}y$	X^2	XY
1	0.5	0	-0.3010	0	0
2	2	0.3010	0.3010	0.090601	0.090601
3	4.5	0.4771	0.6532	0.2276	0.3116
4	8	0.6020	0.9030	0.3624	0.5436
5	203	0.6989	2.3074	0.48846	1.6126
$\Sigma X = 2.0890$		$\Sigma Y = 3.8636$	$\Sigma X^2 = 1.1752$	$\Sigma XY = 2.56154$	

Putting these value in Eq^a (iii) & (iv), we get

$$3.8636 = 2.0890b + 5A \text{ (*)}$$

$$\& 2.56154 = 1.1752b + 2.0890A \text{ (**)}$$

On solving (*) & (**),

$$b = 3.1325, A = -0.563$$

$$\therefore a = \text{antilog}(-0.563) = 0.2735$$

Hence, the required curve is $y = 0.2735 x^{3.1325}$

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Find the missing value in the following table of a chemical dissolved in water using Least square approximation to straight line fitting. [2005 Fall]

Temperature	10°	15°	20°	25°	30°	35°
Solubility	20	22	23	?	26	27

Solution: The given curve is $y = mx + c$ (i)

The normal Equations of (i) are

$$\sum Y = nC + m \sum X \text{ (ii)}$$

$$\sum XY = C \sum X + m \sum X^2 \text{ (iii)}$$

Now, we have to find missing value in 25

$x = \text{temp}$	$y = \text{soluble}$	X^2	XY
10	20	100	200
15	22	225	330
20	23	400	460
30	26	900	780
35	27	1225	945
$\sum X = 110$	$\sum Y = 118$	$\sum X^2 = 2850$	$\sum XY = 2715$

Putting these value in Eqⁿ (ii) & (iii) we get

$$118 = 110 \cdot m + 5 \cdot C \text{ (*) \&}$$

$$2715 = 2850 \cdot m + 110 \cdot C \text{ (**)}$$

On solving (*) & (**),

$$m = 0.2767, C = 17.5116$$

Hence, the required curve is $y = 0.2767x + 17.5116$

$$\text{At } X = 25,$$

$$Y = 0.2767 \cdot 25 + 17.5116$$

$$= 24.429$$

Use the suitable method to fit a quadratic curve $y = e^{a+bx}$ for the following data: [2010 Spring]

x	-5	-4	-3	-2	-1	0
y	12.96	6.94	4.63	2.11	0.67	0.09

Solution: The given curve is $y = e^{a+bx}$ (i)

Taking $\log_{10}$ on both sides we get

$$y = (a + bx) \log_{10} e$$

$$y = a \log_{10} e + b x \log_{10} e$$

$$\text{Which is of the form } Y = A + BX \text{ (ii)}$$

Where, $Y = \log_{10} y$, $A = \log_{10} e$, $B = b \log_{10} e$

The normal Equations of (ii) are

$$\sum Y = nA + b \sum X \text{ (iii) \&}$$

$$\sum XY = A \sum X + b \sum X^2 \text{ (iv)}$$

x	y	$Y = \log_{10} y$	X^2	XY
-5	12.96	1.1126	25	-5.563
-4	6.94	0.8413	16	-3.3652
-3	4.63	0.6655	9	-1.9965
-2	2.11	0.3242	4	-0.6484
-1	0.67	-0.1739	1	0.17392
0	0.09	-1.0457	0	0
$\sum X = -15$		$\sum Y = 1.7240$	$\sum X^2 = 55$	$\sum XY = -11.3994$

Putting these value in Eqⁿ (iii) & (iv), we get

$$1.7240 = 6 \cdot A - 15 \cdot B \text{ (*) \&}$$

$$-11.3994 = -15 \cdot A + 55 \cdot B \text{ (**)}$$

On solving (*) & (**),

$$A = -0.7254, B = -0.4051$$

$$\therefore a = \frac{A}{\log_{10} e} = -1.67029, b = \frac{B}{\log_{10} e} = -0.9327$$

Hence, the required curve is $y = e^{-1.67029 - 0.9327x}$

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The heat of water H (ft) and the quantity of water Q (ft^3) flowing per second are related by the law $Q = CH^\alpha$. Find out the best fit values for coefficients C and α for the following data: [2009 Fall]

H	5	10	15	20	25	30
Q	20	150	360	800	1500	2200

Solution: The given curve is $Q = CH^\alpha$ (i)

Let $Q = y$ and $C = a$, $H = x$, $\alpha = \beta$

Taking $\log_{10}$ on both sides we get

$\log_{10} y = \log_{10} a + \beta \log_{10} x$ Which is of the form

$Y = A + bX$ (ii)

Where, $Y = \log_{10} y$, $A = \log_{10} a$, $X = \log_{10} x$

The normal Equations of (ii) are

$\sum Y = nA + b \sum X$ (iii) &

$\sum XY = A \sum X + b \sum X^2$ (iv)

x	y	b = log ₁₀ H	Y = log ₁₀ y	x ²	X * Y
5	20	0.699	1.301	0.489	0.909
10	150	1.0	2.176	1	2.176
15	360	1.176	2.556	1.383	3.006
20	800	1.301	2.903	1.693	3.777
25	1500	1.398	3.176	1.954	4.44
30	2200	1.477	3.342	2.182	4.936
		$\sum X = 7.051$	$\sum Y = 15.454$	$\sum X^2 = 8.7$	$\sum XY = 19.244$

Putting these value in Eqⁿ (iii) & (iv), we get

$15.454 = 6A + 7.051b$ (*) &

$19.244 = 7.051A + 8.7b$ (**)

On solving (*) & (**),

$X = 2.617$, $A = -0.499$, $a = \text{antilog}(A) = 0.317$

Hence, the required curve is $y = 0.317x^{2.617}$

From the following data

x	1	2	3	4	5
y	0.5	2	4.5	8	12.5

Fit a power function model of the form $y = ax^b$. [2006 Fall, 2007 Spring]

Solution: The given curve is $y = ax^b$ (i)

Taking $\log_{10}$ on both sides we get

$\log_{10} y = \log_{10} a + b \log_{10} x$ Which is of the form $Y = A + bX$ (ii)

Where, $Y = \log_{10} y$, $A = \log_{10} a$, $X = \log_{10} x$

The normal Equations of (ii) are

$\sum Y = nA + b \sum X$ (iii) &

$\sum XY = A \sum X + b \sum X^2$ (iv)

x	y	X = log ₁₀ x	Y = log ₁₀ y	X ²	XY
1	0.5	0	-0.3010	0	0
2	2	0.3010	0.3010	0.090601	0.090601
3	4.5	0.4771	0.6532	0.2276	0.3116
4	8	0.6020	0.9030	0.3624	0.5436
5	12.5	0.6989	1.0969	0.48846	0.7667
		$\sum X = 2.0890$	$\sum Y = 2.6532$	$\sum X^2 = 1.1752$	$\sum XY = 1.7127$

Putting these value in Eqⁿ (iii) & (iv), we get

$2.6532 = 2.0890b + 5A$ (*)

$1.7127 = 1.1752b + 2.0890A$ (**)

On solving (*) & (**),

$b = 2$, $A = -0.301$

$\therefore a = \text{antilog}(-0.301) = 0.5$

Hence, the required curve is $y = 0.5x^2$

For the following set of data fit a linear curve using Least Square Method and find $f(2)$. [2008 Spring]

x_i	0.5	1.5	4.5	7.5
$f(x_i)$	2.5	3.5	6.5	9.5

Solution: The given curve is $y = ax + b$ (i)

The normal Equations of (i) are

$\sum Y = nb + a \sum X$ (ii) &

$\sum XY = b \sum X + a \sum X^2$ (iii)

Now we have to find missing value in 25

x	y	X ²	XY
0.5	2.5	0.25	1.25
1.5	3.5	2.25	5.25
4.5	6.5	20.25	29.25
7.5	9.5	56.25	71.25
$\sum X = 14$	$\sum Y = 22$	$\sum X^2 = 79$	$\sum XY = 107$

Putting these value in Eq⁽ⁱⁱ⁾ & (iii), we get

$$107 = 79 \cdot a + 14 \cdot b \quad (*)$$

$$22 = 14 \cdot a + 4 \cdot b \quad (**)$$

On solving (*) & (**),

$$a = 1, b = 2$$

Hence, the required curve is $y = x + 2$

Fit an exponential equation of the form $y = ae^{bx}$ using the least square method from the following data. [2010 Fall]

x	1	2	3	4	5	6
y	1.65	2.70	4.50	7.35	12.2	15

Solution: The given curve is $y = ae^{bx}$ (i)

Taking $\log_{10}$ on both sides we get

$$\log_{10} y = \log_{10} a + bx \log_{10} e$$

Which is of the form $Y = A + Bx$ (ii)

Where, $Y = \log_{10} y$, $A = \log_{10} a$, $B = b \log_{10} e$

The normal Equations of (ii) are

$$\sum Y = n \cdot A + B \sum X \quad \text{..... (iii)}$$

$$\sum XY = A \sum X + B \sum X^2 \quad \text{..... (iv)}$$

x	y	$Y = \log_{10} y$	X^2	XY
1	1.65	0.217	1	0.217
2	2.7	0.43	4	0.86
3	4.5	0.65	9	1.95
4	7.35	0.866	16	3.464
5	12.2	1.08	25	5.4
6	15	1.176	36	7.056
$\sum X = 21$		$\sum Y = 4.419$	$\sum X^2 = 91$	$\sum XY = 18.94$

Putting these value in Eq⁽ⁱⁱⁱ⁾ & (iv), we get

$$4.419 = 6 \cdot A + 21 \cdot B \quad (*)$$

$$18.94 = 21 \cdot A + 91 \cdot B \quad (**)$$

On solving (*) & (**),

$$A = 0.0418, B = -0.19848$$

$$\therefore a = \text{antilog}(0.0418) = 1.1010$$

$$b = \frac{B}{\log_{10} e} = 0.4571$$

Hence the required curve is $y = 1.1010e^{0.4571x}$

A Complete Guide to Numerical Methods

Determine the constants a and b by the method of least square such that $y = ae^{bx}$ fits the following data: [2011 Fall]

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

Solution: The given curve is $y = ae^{bx}$ (i)

Taking $\log_{10}$ on both sides we get

$$\log_{10} y = \log_{10} a + bx \log_{10} e$$

Which is of the form

$$Y = A + Bx \quad \text{..... (ii)}$$

Where, $Y = \log_{10} y$, $A = \log_{10} a$, $B = b \log_{10} e$

The normal Equations of (ii) are

$$\sum Y = n \cdot A + B \sum X \quad \text{..... (iii)}$$

$$\sum XY = A \sum X + B \sum X^2 \quad \text{..... (iv)}$$

x	y	$Y = \log_{10} y$	X^2	XY
2	4.077	0.610	4	1.22
4	11.084	1.045	16	4.18
6	30.128	1.479	36	8.874
8	81.897	1.913	64	15.304
10	222.62	2.3475	100	23.475
$\sum X = 30$		$\sum Y = 7.3945$	$\sum X^2 = 220$	$\sum XY = 53.053$

Putting these value in Eq⁽ⁱⁱⁱ⁾ & (iv), we get

$$7.3945 = 5 \cdot A + 30 \cdot B \quad (*)$$

$$53.053 = 30 \cdot A + 220 \cdot B \quad (**)$$

On solving (*) & (**),

$$A = 0.1794, B = 0.21657$$

$$\therefore a = \text{antilog}(A) = 1.5114$$

$$b = \frac{B}{\log_{10} e} = 0.4596x$$

Hence the required curve is $y = 1.5114e^{0.4596x}$

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Fit the following data using least square method to estimate a and b for the relation $y = a + bx$. Also find $f(15)$ from the fitted equation. [2007 Fall]

x	10	12	18	22	30
$f(x)$	0	2	4	8	10

Solution: The given curve is $y = a + bX$ (i)

The normal Equations of (i) are

$$\sum Y = n \cdot a + b \sum X \text{ (ii) \&}$$

$$\sum XY = a \sum X + b \sum X^2 \text{ (iii)}$$

Now, we have to find missing value in 15.

X	y	X^2	XY
10	0	100	0
12	2	144	24
18	4	324	72
22	8	484	176
$\sum X = 62$	$\sum y = 14$	$\sum X^2 = 1052$	$\sum XY = 272$

Putting these value in Eqⁿ(ii) & (iii), we get

$$14 = 4 \cdot a + 62 \cdot b \text{ (*)}$$

$$272 = 62 \cdot a + 1052 \cdot b \text{ (**)}$$

On solving (*) & (**),

$$a = -5.868, b = 0.6043,$$

Hence, the required curve is $y = -5.868 + 0.6043x$

$$\text{at } x = 15,$$

$$Y = -5.868 + 0.6043 \cdot 15$$

$$= 3.196$$

Hence, the required value of y at $x = 15$ is 3.196

Use regression method to fit the geometrical curve $y = ab^x$ to the data given below and obtain the value of y at $x = 5.5$ [2009 Spring]

X	0	1	2	3	4	5	6	7
Y	10	21	35	59	92	200	400	610

Solution: The given curve is $y = ab^x$ (i)

Taking $\log_{10}$ on both sides we get

$$\log_{10} y = \log_{10} a + x \log_{10} b$$

Which is of the form $Y = A + Bx$ (ii)

$$Y = \log_{10} y, A = \log_{10} a, B = \log_{10} b$$

The normal Equations of (ii) are

$$\sum Y = nA + B \sum x \text{ (iii) \&}$$

$$\sum xY = A \sum x + B \sum x^2 \text{ (iv)}$$

Where,

x	y	$Y = \log_{10} y$	X^2	XY
0	10	1	0	0
1	21	1.32	1	1.32
2	35	1.54	4	3.08
3	59	1.77	9	5.31
4	92	1.963	16	7.852
5	200	2.30	25	11.5
6	400	2.60	36	15.6
7	610	2.785	49	19.495
$\sum x = 28$		$\sum Y = 15.27$	$\sum X^2 = 140$	$\sum XY = 64.11$

Putting these value in Eqⁿ(iii) & (iv), we get

$$15.27 = 8 \cdot A + 28 \cdot B \text{ (*)}$$

$$64.11 = 28 \cdot A + 140 \cdot B \text{ (**)}$$

On solving (*) & (**),

$$B = 0.25, A = 1.02$$

$$\therefore a = \text{antilog}(A) = 10.47$$

$$b = \text{antilog}(B) = 1.778$$

$$\text{so, } y = 10.47 \cdot 1.778^x$$

$$\text{Now, at } x = 5.5 \text{ then } y = 10.47 \cdot 1.778^{5.5}$$

$$= 248.068$$

Hence, the required value of y is 248.068

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Given the data table

x_i	0	1	2	3
$F(x_i)$	1.5	4.5	9.0	25.5

Estimate the function value at $x=1.5$ using cubic spline technique. [2011 Spring]

Solution:

We know that the cubic spline interpolation formula

$$y_i(x) = \frac{a_{i-1}}{6h_i}(h_i^2 u_i - u_i^3) + \frac{a_i}{6h_i}(u_i^3 - h_i^2 u_{i-1}) + \frac{1}{h_i}(f_i u_{i-1} - f_{i-1} u_i)$$

and to determine the coefficient a_i we use

$$h_i a_{i-1} + 2(h_i + h_{i+1})a_i + h_{i+1}a_{i+1} = 6\left[\frac{f_{i+1}-f_i}{h_{i+1}} - \frac{f_i-f_{i-1}}{h_i}\right]$$

also $a_0 = a_4 = 0$

now for $i=1$

$$h_1 a_0 + 2(h_1 + h_2)a_1 + h_2 a_2 = 6\left[\frac{f_2-f_1}{h_2} - \frac{f_1-f_0}{h_1}\right]$$

$$h_1=2, h_2=1 \text{ and } h_3=3;$$

$$0 + 2(2+1)a_1 + a_2 = 6\left[\frac{9-4.5}{1} - \frac{4.5-1.5}{2}\right]$$

$$6a_1 + a_2 = 18 \dots \dots \dots (i)$$

For, $i=2$

$$h_2 a_1 + 2(h_2 + h_3)a_2 + h_3 a_3 = 6\left[\frac{f_3-f_2}{h_3} - \frac{f_2-f_1}{h_2}\right]$$

$$a_1 + 8a_2 = 6\left[\frac{25.5-9}{3} - \frac{9-4.5}{1}\right]$$

$$a_1 + 8a_2 = 6 \dots \dots \dots (ii)$$

on solving equation (i) and (ii) we get

$$a_1 = \frac{138}{47}, a_2 = \frac{18}{47}$$

since the interpolating value lies on 1 & 2 so $i=2$;

$$f(1.5) = \frac{a_1}{6h_2}(h_2^2 u_2 - u_2^3) + \frac{a_2}{6h_2}(u_2^3 - h_2^2 u_1) + \frac{1}{h_2}(f_2 u_1 - f_1 u_2)$$

$$\text{where } u_1 = x - x_1 = 1.5 - 1 = 0.5$$

$$u_2 = x - x_2 = 1.5 - 2 = -0.5$$

$$\text{so } f(1.5) = 4.4122$$

which is required answer



CHAPTER 3

Romberg Integration

We can improve the accuracy of trapezoidal rule by the method of Romberg integration.

If integrate a function known as a table of every spaced function value by

$$I(h, \frac{h}{2}) = \frac{1}{3} [4 * I(\frac{h}{2}) - I(h)]$$

i.e.

$$I \approx \frac{1}{3} [4 * I_2 - I_1] \dots \dots \dots (i)$$

So the trapezoidal rule several times halving h & applying (i) to each pair of value as per the following rule illustrate in below table.

$I(h)$	$I(h, \frac{h}{2})$		
$I(\frac{h}{2})$	$I(\frac{h}{2}, \frac{h}{4})$	$I(h, \frac{h}{2}, \frac{h}{4})$	$I(h, \frac{h}{2}, \frac{h}{4}, \frac{h}{8})$
$I(\frac{h}{4})$	$I(\frac{h}{4}, \frac{h}{8})$	$I(\frac{h}{2}, \frac{h}{4}, \frac{h}{8})$	
$I(\frac{h}{8})$			

- The computation can be stoped when two successive values are sufficiently closed to each other.
- This systematic tabulation is called Romberg interation.

Gauss integration (Gauss Legendre) formula

Gauss-Legendre (Gauss-Quadrature) two point ($n=2$) formula is

$$\int_{-1}^1 f(u) du = w_1 * f(u_1) + w_2 * f(u_2)$$

Where,

$$w_1 = 1, w_2 = 1, u_1 = -\frac{1}{\sqrt{3}}, u_2 = \frac{1}{\sqrt{3}}$$

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- (ii) Gauss-Legendre (Gauss-Quadrature) three point (n=3) formula is

$$\int_{-1}^1 f(u) du = w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3)$$

Where,

$$w_1 = 0.555, w_2 = 0.889, w_3 = 0.555, u_1 = -0.77460, u_2 = 0, u_3 = 0.77460$$

- (iii) Gauss-Legendre (Gauss-Quadrature) three point (n=4) formula is

$$\int_{-1}^1 f(u) du = w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3) + w_4 * f(u_4)$$

Where,

$$w_1 = 0.34785, w_2 = 0.65214, w_3 = 0.65214, w_4 = 0.34758, u_1 = -0.86144, u_2 = -0.33998, u_3 = 0.33998, u_4 = 0.86144$$

University Question Solutions

A) TRAPEZOIDAL RULE, SIMPSON'S RULE

Integrate $\int_0^{\pi} (1 + 3 \cos^2 x) dx$ by

- i) Trapezoidal rule
ii) Simpson's 3/8 rule, taking number of intervals (n) = 6 [2005 Spring]

Solution: We have $nh = b - a \therefore h = \frac{b-a}{n} = \frac{\pi-0}{6} = \frac{\pi}{6}$

Let us define a function $f(x_i) = y_i = 1 + 3 \cos^2 x$

x_i	0	$\frac{\pi}{6}$	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$	$\frac{5\pi}{6}$	π
y_i	4	3.9997	3.9989	3.9977	3.9959	3.9937	3.9909

By Trapezoidal rule:

$$\begin{aligned} I &= \frac{h}{2} [y_0 + y_6 + 2(y_1 + y_2 + y_3 + y_4 + y_5)] \\ &= \frac{\pi}{24} [4 + 3.9909 + 2(3.997 + 3.9989 + 3.9977 + 3.9959 + 3.9937)] \\ &\therefore \int_0^{\pi} 1 + 3 \cos^2 x dx = 12.494 \end{aligned}$$

By Simpson's $\frac{3}{8}$ rule:

$$\begin{aligned} I &= \frac{3h}{8} [y_0 + y_6 + 2y_3 + 3(y_1 + y_3 + y_4 + y_5)] \\ &= \frac{3\pi}{48} [4 + 3.9909 + 2 * 3.9977 + 3(3.9997 + 3.9989 + 3.9959)] \\ &\therefore \int_0^{\pi} 1 + 3 \cos^2 x dx = 12.554 \end{aligned}$$

Evaluate the following integral $\int_0^{\frac{\pi}{2}} \sin x dx$ Using

1. Trapezoidal Rule
2. Simpson's $\frac{1}{3}$ rule
3. Simpson's 3/8 Comment on the result. [2006 Spring, 2007 spring, 2010 fall, 2011 fall, 2011 spring]

Solution: We have $nh = b - a \therefore h = \frac{b-a}{n} = \frac{\frac{\pi}{2}-0}{6} = \frac{\pi}{12}$

Let us define a function $f(x) = \sin x$

x_i	0	$\frac{\pi}{12}$	$\frac{2\pi}{12}$	$\frac{3\pi}{12}$	$\frac{4\pi}{12}$	$\frac{5\pi}{12}$	$\frac{\pi}{2}$
y_i	0	0.00456	0.0091	0.0137	0.0182	0.0228	0.0274

(i) By Trapezoidal rule:

$$i = \frac{h}{2} [y_0 + y_6 + 2(y_1 + y_2 + y_3 + y_4 + y_5)]$$

$$= \frac{\frac{\pi}{6}}{2} [0 + 0.0274 + 2(0.00456 + 0.0091 + 0.0137 + 0.0182 + 0.0228)]$$

$$\therefore \int_0^{\frac{\pi}{2}} \sin x \, dx = 0.02148$$

(ii) By Simpson's $\frac{1}{3}$ rule

$$i = \frac{h}{3} [y_0 + y_6 + 2(y_2 + y_4) + 4(y_1 + y_3 + y_5)]$$

$$= \frac{\frac{\pi}{6}}{3} [0 + 0.0274 + 2(0.0091 + 0.0182) + 4(0.00456 + 0.0137 + 0.0228)]$$

$$\therefore \int_0^{\frac{\pi}{2}} \sin x \, dx = 0.02147$$

(iii) By Simpson's $\frac{3}{8}$ rule

$$i = \frac{3h}{8} [y_0 + y_6 + 2y_2 + 3(y_1 + y_3 + y_4 + y_5)]$$

$$= \frac{\frac{\pi}{6}}{8} [0 + 0.0274 + 2 \cdot 0.0137 + 3(0.00456 + 0.0091 + 0.0182 + 0.0228)]$$

$$\therefore \int_0^{\frac{\pi}{2}} \sin x \, dx = 0.023010$$

Exact Value: $\int_0^{\frac{\pi}{2}} \sin x \, dx = 0.02153$

Here, value obtained from Trapezoidal rule and Simpson's $\frac{1}{3}$ rule nearly equal to exact value. But the value obtained from Simpson's $\frac{3}{8}$ rule is largely deviated from exact value.

Integrate the following function by using Simpson's $\frac{3}{8}$ rule.

Take $n = 5$. $\int_1^2 \frac{\sin x}{x} dx$. [2007 Fall]

Solution: We have $nh = b - a \therefore h = \frac{b-a}{n} = \frac{\pi-0}{5} = \frac{\pi}{5}$

Let us define a function $f(x_i) = y_i = \frac{\sin x}{x}$

x_i	0	$\frac{\pi}{10}$	$\frac{2\pi}{10}$	$\frac{3\pi}{10}$	$\frac{4\pi}{10}$	$\frac{\pi}{2}$
y_i	0	0.17453	0.017452	0.017451	0.017451	0.017451

Simpson's $\frac{3}{8}$ rule:

$$i = \frac{3h}{8} [y_0 + y_5 + 2y_1 + 3(y_2 + y_3 + y_4)]$$

$$= \frac{3\pi}{80} [0 + 0.017451 + 2 \cdot 0.017451 + 3(0.017453 + 0.017452 + 0.017451)]$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{\sin x}{x} dx = 0.02467$$

Integrate the given integral

$\int \cos x \, dx$ Using trapezoidal, Simpson's $\frac{1}{3}$ and $\frac{3}{8}$ rule.

[2008 Spring, 2009 Fall]

Solution: We have $nh = b - a \therefore h = \frac{b-a}{n} = \frac{3-1}{5} = \frac{2}{5}$

Let us define a function $f(x_i) = y_i = \cos x$

x_i	1	$\frac{7}{5}$	$\frac{9}{5}$	$\frac{11}{5}$	$\frac{13}{5}$	3
y_i	0.9998	0.9997	0.9995	0.9992	0.9989	0.9986

By Trapezoidal rule:

$$i = \frac{h}{2} [y_0 + y_5 + 2(y_1 + y_2 + y_3 + y_4)]$$

$$= \frac{2}{10} [0.9998 + 0.9986 + 2(0.9997 + 0.9995 + 0.9992 + 0.9989)]$$

$$\therefore \int_1^3 \cos x \, dx = 1.9986$$

By Simpson's $\frac{1}{3}$ rule

$$i = \frac{h}{3} [y_0 + y_5 + 2(y_2 + y_4) + 4(y_1 + y_3)]$$

$$= \frac{2}{15} [0.9998 + 0.9986 + 2(0.9995 + 0.9989) + 4(0.9997 + 0.9992)]$$

$$\therefore \int_1^3 \cos x \, dx = 1.81$$

By Simpson's $\frac{3}{8}$ rule:

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$$i = \frac{3h}{8} [y_0 + y_5 + 2y_3 + 3(y_1 + y_3 + y_4)]$$

$$= \frac{3\pi}{80} [0.9998 + 0.9986 + 2 \cdot 0.9992 + 3(0.9997 + 0.9995 + 0.9989)]$$

$$\therefore \int_1^3 \cos x \, dx = 1.9484$$

Evaluate the following integral using Simpson's 3/8 Rule: $\int_1^2 (x^3 + 1) \, dx$, using $n = 3$. [2009 Spring]

Solution: We have $nh = b - a \therefore h = \frac{b-a}{n} = \frac{2-1}{3} = \frac{1}{3}$
Let us define a function $f(x_i) = y_i = x^3 + 1$

x_i	1	$\frac{4}{3}$	$\frac{5}{3}$	2
y_i	2	3.37	5.62	9

By Simpson's $\frac{3}{8}$ rule:

$$i = \frac{3h}{8} [y_0 + y_3 + 3(y_1 + y_2)]$$

$$= \frac{3}{24} [2 + 9 + 3(3.37 + 5.62)]$$

$$\therefore \int_1^2 x^3 + 1 \, dx = 4.74$$

Integrate the given integral $\int_0^1 \frac{1}{1+x^2} \, dx$ using trapezoidal rule, Simpson's $\frac{1}{3}$ and $\frac{3}{8}$ rule. [2012 fall]

Solution: We have $nh = b - a \therefore h = \frac{b-a}{n} = \frac{1-0}{5} = \frac{1}{5}$

Let us define a function $f(x_i) = y_i = \frac{1}{1+x^2}$

x_i	0	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	1
y_i	1	0.9615	0.8620	0.7352	0.6097	0.5

By Trapezoidal rule:

$$i = \frac{h}{2} [y_0 + y_5 + 2(y_1 + y_2 + y_3 + y_4)]$$

$$= \frac{1}{10} [1 + 0.5 + 2(0.9615 + 0.8620 + 0.7352 + 0.6097)]$$

$$\therefore \int_0^1 \frac{1}{1+x^2} \, dx = 0.78368$$

By Simpson's $\frac{1}{3}$ rule:

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$$i = \frac{h}{3} [y_0 + y_5 + 2(y_2 + y_4) + 4(y_1 + y_3)]$$

$$= \frac{1}{15} [1 + 0.5 + 2(0.8620 + 0.6097) + 4(0.9615 + 0.7352)]$$

$$\therefore \int_0^1 \frac{1}{1+x^2} \, dx = 0.7486$$

By Simpson's $\frac{3}{8}$ rule:

$$i = \frac{3h}{8} [y_0 + y_5 + 2y_3 + 3(y_1 + y_3 + y_4)]$$

$$= \frac{3}{40} [1 + 0.5 + 2 \cdot 0.7352 + 3(0.9615 + 0.8620 + 0.6097)]$$

$$\therefore \int_0^1 \frac{1}{1+x^2} \, dx = 0.77025$$

B) ROMBERG INTEGRATION

Use Romberg integration to evaluate $\int_0^{\pi/2} e^x \sin x \, dx$ [2005 fall]

Solution: We have, $h = \frac{b-a}{n}$ taking $n = 2$, $h = \frac{\frac{\pi}{2} - 0}{2} = \frac{\pi}{4}$

Therefore,

taking $h = \frac{3\pi}{4}, \frac{3\pi}{8}, \frac{3\pi}{16}$, respectively.

Let us calculate the given integral using trapezoidal rule.

$$f(x) = y_i = e^x \sin x$$

a) When $h = \frac{3\pi}{4}$

x_i	0	$\frac{3\pi}{4}$	$\frac{3\pi}{2}$
y_i	0	0.4337	9.1431

$$I_1 = \frac{h}{2} [y_0 + y_2 + 2y_1]$$

$$= \frac{\frac{3\pi}{4}}{2} [0 + 9.1431 + 2 \cdot 0.4337]$$

$$= 11.7927$$

b) When $h = \frac{3\pi}{8}$

x_i	0	$\frac{3\pi}{8}$	$\frac{6\pi}{8}$	$\frac{9\pi}{8}$	$\frac{3\pi}{2}$
y_i	0	0.06678	0.4337	2.1121	9.1431

$$I_2 = \frac{h}{2} [y_0 + y_4 + 2(y_1 + y_2 + y_3)]$$

$$= \frac{\frac{3\pi}{8}}{2} [0 + 9.1431 + 2(0.06678 + 0.4337 + 2.1121)]$$

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$$= 8.463$$

c) When $h = \frac{3\pi}{16}$

x_i	0	$\frac{3\pi}{16}$	$\frac{6\pi}{16}$	$\frac{9\pi}{16}$	$\frac{12\pi}{16}$	$\frac{15\pi}{16}$	$\frac{18\pi}{16}$	$\frac{21\pi}{16}$	$\frac{24\pi}{16}$
y_i	0	0.01853	0.06678	0.18049	0.4337	0.97672	2.1118	4.4392	9.1431

$$I_3 = \frac{h}{2} [y_0 + y_8 + 2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7)]$$

$$= \frac{3\pi}{16} [0 + 9.1431 + 2(0.01853 + 0.06678 + 0.18049 + 0.4337 + 0.97672 + 2.1118 + 4.4392)]$$

$$= 7.578$$

Now, for pair I_1 & I_2 apply Romberg integration formula

$$I_1^* = I_2 + \frac{1}{3}(I_2 - I_1)$$

$$= 8.463 + \frac{1}{3}(8.463 - 11.7927)$$

$$= 7.3531$$

Similarly, for pair I_2 & I_3

$$I_2^* = I_3 + \frac{1}{3}(I_3 - I_2)$$

$$= 7.758 + \frac{1}{3}(7.758 - 8.463)$$

$$= 7.283$$

For pair I_1^* & I_2^*

$$I = I_2^* + \frac{1}{3}(I_2^* - I_1^*)$$

$$= 7.283 + \frac{1}{3}(7.283 - 7.3531)$$

$$= 7.2596$$

$$\therefore \int_0^{2\pi} e^x \sin x dx = 7.2596$$

Use Romberg integration to evaluate $\int_{-0.2}^{1.5} e^{-x^2} dx$ correct to significant digits. [2005 spring]

Solution: We have, $h = \frac{b-a}{n}$ taking $n = 2$, $h = \frac{1.5-(-0.2)}{2} = 0.85$

Therefore, taking $h = 0.65, 0.325, 0.162$, respectively.

Let us calculate the given integral using trapezoidal rule.

$$f(x_i) = y_i = e^{-x^2}$$

a) When $h = 0.65$

x_i	0.2	0.85	1.5
y_i	0.9607	0.4855	0.1053

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$$I_1 = \frac{h}{2} [y_0 + y_2 + 2y_1]$$

$$= \frac{0.65}{2} [0.9607 + 0.1053 + 2 \cdot 0.4855]$$

$$= 0.6617$$

b) When $h = 0.325$

x_i	0.2	0.525	0.850	1.175	1.5
y_i	0.9607	0.759	0.4855	0.251	0.1053

$$I_2 = \frac{h}{2} [y_0 + y_4 + 2(y_1 + y_2 + y_3)]$$

$$= \frac{0.325}{2} [0.9607 + 0.1053 + 2(0.759 + 0.4855 + 0.251)]$$

$$= 0.6589$$

c) When $h = 0.1625$

x_i	0.2	0.363	0.525	0.688	0.850	1.013	1.175	1.338	1.5
y_i	0.9607	0.876	0.759	0.6233	0.4855	0.3591	0.251	0.167	0.1053

$$I_3 = \frac{h}{2} [y_0 + y_8 + 2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7)]$$

$$= \frac{0.1625}{2} [0.9607 + 1.1053 + 2(0.876 + 0.759 + 0.6233 + 0.4855 + 0.3591 + 0.251 + 0.167)]$$

$$= 0.6565$$

Now, for pair I_1 & I_2 apply Romberg integration formula

$$I_1^* = I_2 + \frac{1}{3}(I_2 - I_1)$$

$$= 0.6589 + \frac{1}{3}(0.6589 - 0.6617)$$

$$= 0.657$$

Similarly, for pair I_2 & I_3

$$I_2^* = I_3 + \frac{1}{3}(I_3 - I_2)$$

$$= 0.6565 + \frac{1}{3}(0.6565 - 0.6589)$$

$$= 0.655$$

For pair I_1^* & I_2^*

$$I = I_2^* + \frac{1}{3}(I_2^* - I_1^*) = 0.655 + \frac{1}{3}(0.655 - 0.657)$$

$$= 0.654$$

$$\therefore \int_{-0.2}^{1.5} e^{-x^2} dx = 0.654$$

Apply Romberg's method to evaluate $\int_4^{5.2} \log_{10} x dx$ [2006 fall]

Solution: We have $h = \frac{b-a}{n}$ taking $n = 2$, $h = \frac{5.2-4}{2} = 0.6$ where $b = 5.2$ and $a = 4$.
taking $h = 0.6, 0.3, 0.15$, respectively.

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Let us calculate the given integral using trapezoidal rule.

$$f(x_i) = y_i = \log_{10} x_i$$

a) When $h = 0.6$

x_i :	4	4.6	5.2
y_i :	0.60	0.66	0.71

$$I_1 = \frac{h}{2} [y_0 + y_2 + 2y_1]$$

$$= \frac{0.6}{2} [0.60206 + 0.662758 + 2 \cdot 0.716003]$$

$$= 0.789$$

b) When $h = 0.3$

x_i :	4	4.3	4.6	4.9	5.2
y_i :	0.60	0.63	0.66	0.69	0.71

$$I_2 = \frac{h}{2} [y_0 + y_4 + 2(y_1 + y_2 + y_3)]$$

$$= \frac{0.3}{2} [0.60 + 0.71 + 2(0.63 + 0.66 + 0.69)]$$

$$= 0.7905$$

c) When $h = 0.15$

x_i :	4	4.15	4.3	4.45	4.6	4.75	4.9	5.05	5.2
y_i :	0.60	0.61	0.63	0.64	0.66	0.67	0.69	0.70	0.71

$$I_3 = \frac{h}{2} [y_0 + y_8 + 2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7)]$$

$$= \frac{0.15}{2} [0.60 + 0.71 + 2(0.61 + 0.63 + 0.64 + 0.66 + 0.67 + 0.69 + 0.70)]$$

$$= 0.78825$$

Now, for pair I_1 & I_2 apply Romberg integration formula

$$I_1^* = I_2 + \frac{1}{3}(I_2 - I_1)$$

$$= 0.7905 + \frac{1}{3}(0.7905 - 0.789)$$

$$= 0.791$$

Similarly, for pair I_2 & I_3

$$I_2^* = I_3 + \frac{1}{3}(I_3 - I_2)$$

$$= 0.78825 + \frac{1}{3}(0.78825 - 0.7905)$$

$$= 0.7875$$

For pair I_1^* & I_2^*

$$I = I_2^* + \frac{1}{3}(I_1^* - I_2^*)$$

$$= 0.7875 + \frac{1}{3}(0.7875 - 0.791)$$

$$= 0.7863$$

$$\therefore \int_4^{5.2} \log_{10} x \, dx = 0.7863$$

Find the integral value of $I = e^{-x^3}$, correct to three decimal place using Romberg integration within the limit of (0.2, 1.5).
[2008 fall]

Solution: We have, $h = \frac{b-a}{n}$ taking $n = 2$, $h = \frac{1.5-0.2}{2} = 0.65$.

Therefore, taking $h = 0.65, 0.325, 0.162$, respectively.

Let us calculate the given integral using trapezoidal rule.

$$f(x_i) = y_i = e^{-x^3}$$

a) When $h = 0.65$

x_i :	0.2	0.85	1.5
y_i :	0.992	0.5411	0.03422

$$I_1 = \frac{h}{2} [y_0 + y_2 + 2y_1]$$

$$= \frac{0.65}{2} [0.992 + 0.03422 + 2 \cdot 0.5411]$$

$$= 0.68525$$

b) When $h = 0.325$

x_i :	0.2	0.525	0.850	1.175	1.5
y_i :	0.9920	0.8652	0.5411	0.1974	0.03422

$$I_2 = \frac{h}{2} [y_0 + y_4 + 2(y_1 + y_2 + y_3)]$$

$$= \frac{0.325}{2} [0.992 + 0.03422 + 2(0.8652 + 0.5411 + 0.1974)]$$

$$= 0.688$$

c) When $h = 0.1625$

x_i :	0.2	0.363	0.525	0.688	0.850	1.013	1.175	1.338	1.5
y_i :	0.992	0.953	0.865	0.722	0.5411	0.353	0.197	0.091	0.03422

$$I_3 = \frac{h}{2} [y_0 + y_8 + 2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7)]$$

$$= \frac{0.1625}{2} [0.992 + 0.03422 + 2(0.953 + 0.865 + 0.722 + 0.5411 + 0.353 + 0.197 + 0.091)]$$

$$= 0.688$$

Since, I_2 & I_3 are the same, we conclude that the value of the integral

$$\therefore \int_{0.2}^{1.5} e^{-x^3} \, dx = 0.688$$

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(C) GAUSS-LEGENDRE

Evaluate $\int_3^5 \frac{1}{x^6+5} dx$ by using Gauss-Legendre two and three point formula. [2006 fall]

Solution: By changing the limit (3,5) to (-1,1) by transformation,

$$x = \frac{1}{2}(b-a)U + \frac{1}{2}(b+a) \\ = \frac{1}{2}(5-3)U + \frac{1}{2}(5+3) \\ = U+4$$

$$\therefore dx = du$$

$$\text{Now, } I = \int_3^5 \frac{1}{x^6+5} dx \\ = \int_{-1}^1 \frac{1}{(U+4)^6+5} dU$$

$$\therefore f(u) = \frac{1}{(U+4)^6+5}$$

Gauss-Legendre (Gauss-Quadrature) two point (n=2) formula is

$$\int_{-1}^1 f(u) du = w_1 * f(u_1) + w_2 * f(u_2)$$

$$\text{Where, } w_1 = 1, w_2 = 1, u_1 = -\frac{1}{\sqrt{3}}, u_2 = \frac{1}{\sqrt{3}}$$

$$\therefore \int_3^5 \frac{1}{x^6+5} dx = w_1 * f(u_1) + w_2 * f(u_2) \\ = (1 * 6.2012 * 10^{-4}) + (1 * 1.086 * 10^{-4}) \\ = 7.70612 * 10^{-4}$$

Similarly, Gauss-Legendre (Gauss-Quadrature) three point (n=3) formula is

$$\int_{-1}^1 f(u) du = w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3)$$

$$\text{Where, } w_1 = 0.555, w_2 = 0.889, w_3 = 0.555, \\ u_1 = -0.77460, u_2 = 0, u_3 = 0.77460$$

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$$\int_3^5 \frac{1}{x^6+5} dx = w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3) \\ = (0.555 * 8.84 * 10^{-4}) + (0.889 * 2.438 * 10^{-4}) + (0.555 * 8.4371 * 10^{-4}) \\ = 7.64 * 10^{-4}$$

Using Gauss-quadrature formula for n = 3, evaluate $\int_0^1 \frac{dx}{1+x^2}$

[2007 spring]

Solution: By changing the limit (0,1) to (-1,1) by transformation,

$$x = \frac{1}{2}(b-a)U + \frac{1}{2}(b+a) \\ = \frac{1}{2}(1-0)U + \frac{1}{2}(1+0) \\ = \frac{U}{2} + \frac{1}{2} \\ = \frac{U+1}{2}$$

$$\therefore dx = \frac{1}{2} du$$

$$\text{Now, } I = \int_0^1 \frac{1}{1+x^2} dx \\ = \int_{-1}^1 \frac{1}{1+(\frac{U+1}{2})^2} \frac{1}{2} du$$

$$\therefore f(u) = \frac{2}{U^2 + 2U + 5}$$

Gauss-Legendre (Gauss-Quadrature) three point (n=3) formula is

$$\int_{-1}^1 f(u) du = w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3)$$

$$\text{Where, } w_1 = 0.555, w_2 = 0.889, w_3 = 0.555, \\ u_1 = -0.77460, u_2 = 0, u_3 = 0.77460$$

$$\therefore I = \int_0^1 \frac{1}{1+x^2} dx = w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3) \\ = (0.555 * 0.4937) + (0.889 * 0.4) + (0.555 * 0.2797) \\ = 0.7842$$

Using two and three point Gauss Legendre formula. Evaluate.

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$$\int_{0.5}^{1.5} e^{x^2} dx \quad [\text{2008 spring, 2009 fall}]$$

Solution: By changing the limit (0.5, 1.5) to (-1, 1) by transformation,

$$\begin{aligned} x &= \frac{1}{2}(b-a)U + \frac{1}{2}(b+a) \\ &= \frac{1}{2}(1.5 - 0.5)U + \frac{1}{2}(1.5 + 0.5) \\ &= \frac{1}{2}U + 1 \end{aligned}$$

$$\therefore dx = \frac{1}{2} du$$

$$\begin{aligned} \text{Now, } I &= \int_{0.5}^{1.5} e^{x^2} dx \\ &= \int_{-1}^1 e^{\left(\frac{1}{2}U+1\right)^2} \frac{1}{2} du \end{aligned}$$

$$\therefore f(u) = e^{\left(\frac{1}{2}u+1\right)^2} \frac{1}{2}$$

Gauss-Legendre (Gauss-Quadrature) two point (n=2) formula is

$$\int_{-1}^1 f(u) du = w_1 * f(u_1) + w_2 * f(u_2)$$

$$\text{Where, } w_1 = 1, w_2 = 1, u_1 = -\frac{1}{\sqrt{3}}, u_2 = \frac{1}{\sqrt{3}}$$

$$\therefore \int_{0.5}^{1.5} e^{x^2} dx = w_1 * f(u_1) + w_2 * f(u_2)$$

$$= (1 * 0.8293) + (1 * 2.6310)$$

$$= 3.4603$$

Similarly, Gauss-Legendre (Gauss-Quadrature) three point (n=3) formula is

$$\int_{-1}^1 f(u) du = w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3)$$

$$\text{Where, } w_1 = 0.555, w_2 = 0.889, w_3 = 0.555,$$

$$u_1 = -0.77460, u_2 = 0, u_3 = 0.77460$$

$$\begin{aligned} \therefore \int_{0.5}^{1.5} e^{x^2} dx &= w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3) \\ &= (0.555 * 0.7277) + (0.889 * 1.3591) + (0.555 * 3.426) \\ &= 3.5134 \end{aligned}$$

$$\text{Evaluate the integral } I = \int_0^{10} \exp\left(\frac{-1}{1+x^2}\right) dx, \text{ using Gauss quadrature formula with } n=2 \text{ and } n=3. [\text{2010 fall}]$$

Solution: By changing the limit (0, 10) to (-1, 1) by transformation,

$$\begin{aligned} x &= \frac{1}{2}(b-a)U + \frac{1}{2}(b+a) \\ &= \frac{1}{2}(10 - 0)U + \frac{1}{2}(10 + 0) \\ &= 5U + 5 \end{aligned}$$

$$\therefore dx = 5 du$$

$$\begin{aligned} \text{Now, } I &= \int_0^{10} \exp\left(\frac{-1}{1+x^2}\right) dx \\ &= \int_{-1}^1 \exp\left(\frac{-1}{1+(5U+5)^2}\right) * 5 du \end{aligned}$$

$$\therefore f(u) = \exp\left(\frac{-1}{1+(5U+5)^2}\right) * 5$$

Gauss-Legendre (Gauss-Quadrature) two point (n=2) formula is

$$\int_{-1}^1 f(u) du = w_1 * f(u_1) + w_2 * f(u_2)$$

$$\text{Where } w_1 = 1, w_2 = 1, u_1 = -\frac{1}{\sqrt{3}}, u_2 = \frac{1}{\sqrt{3}}$$

$$\therefore \int_0^{10} \exp\left(\frac{-1}{1+x^2}\right) dx = w_1 * f(u_1) + w_2 * f(u_2)$$

$$= (1 * 4.1640) + (1 * 4.9251)$$

$$= 9.0851$$

Similarly, Gauss-Legendre (Gauss-Quadrature) three point (n=3) formula is

$$\int_{-1}^1 f(u) du = w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3)$$

$$\text{Where, } w_1 = 0.555, w_2 = 0.889, w_3 = 0.555,$$

$$u_1 = -0.77460, u_2 = 0, u_3 = 0.77460$$

$$\therefore \int_0^{10} \exp\left(\frac{-1}{1+x^2}\right) dx$$

$$= w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3)$$

$$= (0.555 * 3.32185) + (0.889 * 4.8113) + (0.555 * 4.9376)$$

$$= 8.98616$$

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Evaluate the integral $I = \int_0^{\frac{\pi}{2}} (1 - (0.25 \sin^2 x))^{\frac{1}{2}} dx$ using Gaussian quadrature formula. Assume $n=3$. [2011 spring]

Solution: By changing the limit $(0, \frac{\pi}{2})$ to $(-1, 1)$ by transformation,

$$\begin{aligned} x &= \frac{1}{2}(b-a)U + \frac{1}{2}(b+a) \\ &= \frac{1}{2}\left(\frac{\pi}{2} - 0\right)U + \frac{1}{2}\left(\frac{\pi}{2} + 0\right) = \frac{\pi}{4}(U+1) \\ \therefore dx &= \frac{\pi}{4} du \end{aligned}$$

$$\begin{aligned} \text{Now, } I &= \int_0^{\frac{\pi}{2}} (1 - (0.25 \sin^2 x))^{\frac{1}{2}} dx \\ &= \int_{-1}^1 \left(1 - 0.25 \sin^2\left(\frac{\pi}{4}(U+1)\right)\right)^{\frac{1}{2}} \frac{\pi}{4} dU \\ \therefore f(u) &= \left(1 - 0.25 \sin^2\left(\frac{\pi}{4}(U+1)\right)\right)^{\frac{1}{2}} \end{aligned}$$

Gauss-Legendre (Gauss-Quadrature) three point ($n=3$) formula is

$$\begin{aligned} \int_{-1}^1 f(u) du &= w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3) \\ \text{Where, } w_1 &= 0.555, w_2 = 0.889, w_3 = 0.555, \\ u_1 &= -0.77460, u_2 = 0, u_3 = 0.77460 \\ \therefore \int_0^{\frac{\pi}{2}} (1 - (0.25 \sin^2 x))^{\frac{1}{2}} dx &= w_1 * f(u_1) + w_2 * f(u_2) + w_3 * f(u_3) \\ &= (0.555 * 0.78234) + (0.889 * 0.73467) + (0.555 * 0.68368) \\ &= 1.467 \end{aligned}$$

Evaluate the integral $\int_0^1 e^x dx$ using two-point Gauss Legendre method. (2009 spring)

Solution: $f(x) = e^x$

Gauss-Legendre (Gauss-Quadrature) two point ($n=2$) formula is

$$\int_{-1}^1 f(x) du = w_1 * f(x_1) + w_2 * f(x_2)$$

Where $w_1 = 1, w_2 = 1, x_1 = -\frac{1}{\sqrt{3}}, x_2 = \frac{1}{\sqrt{3}}$

$$\begin{aligned} \therefore \int_0^1 e^{x^2} dx &= w_1 * f(x_1) + w_2 * f(x_2) \\ &= (1 * 0.56138) + (1 * 1.7813) \\ &= 2.3426 \text{ which is the required answer} \end{aligned}$$



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CHAPTER 4

solving equation of any system we are interested to identify the values the variables that satisfy all equation in the system simultaneously.

In a given arbitrary system of equation it is difficult to say whether the system has a solution or not.

There are four possibilities

- i) System has a unique solution
- ii) System has no solution
- iii) System has infinite solution but unique
- iv) System is ill conditioned

LL Conditioned System

There may be situation where the system has a solution but it is very difficult to find exact point at which the lines intersects. Such system is known as ill conditioned system.

Example:

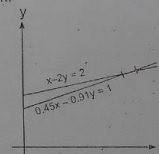


Figure : Ill condition system

In other word, a linear equation is said to be ill conditioned if small change in the coefficients of the equation result in the large change in the value of unknown.

Methods :

The various methods for finding solution of simultaneous algebraic equations are

- (A) Direct method
 - method of determinant (Cramer's rule)
 - matrix inversion method
 - Gauss elimination method

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- > Gauss Jordan method
- > factorization method (LU decomposition method)
- > cholesky factorization method

(B) Iterative methods

- > Jacobi iteration method
- > Gauss-Seidal iteration method

Cramer's rule

Consider the system of equation,

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}$$

Its determinant of coefficient is

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$\text{Then, } \Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}; \Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}; \Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

$$X = \frac{\Delta_1}{\Delta}; Y = \frac{\Delta_2}{\Delta}; Z = \frac{\Delta_3}{\Delta}$$

Matrix Inversion Method

Consider the system of equation,

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}$$

If,

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}; X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; \& B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

Then above equation becomes,

$$AX = B, \text{ so } X = A^{-1}B$$

$$\text{Then, } \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{|A|} \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} \text{ where } A, B, C \text{ are cofactor of } a, b \& c \text{ respectively.}$$

Gauss Elimination Method

In this method, the unknown are eliminated successively and the system is reduced to an upper triangular matrix from which unknown are found by back substitution.

Let the system of equation be

$$a_1x + b_1y + c_1z = d_1 \dots\dots\dots(i)$$

$$a_2x + b_2y + c_2z = d_2 \dots\dots\dots(ii)$$

$$a_3x + b_3y + c_3z = d_3 \dots\dots\dots(iii)$$

Step I: To eliminate x from 2nd and 3rd equation by subtracting $\left(\frac{a_2}{a_1}\right)$ times the first equation from 2nd equation and eliminating x from 3rd equation by $\left(\frac{a_3}{a_1}\right)$ times 1st equation thus, we get

$$a_1x + b_1y + c_1z = d_1 \dots\dots\dots(iv)$$

$$b_2'y + c_2'z = d_2' \dots\dots\dots(v)$$

$$b_3'y + c_3'z = d_3' \dots\dots\dots(vi)$$

Here, the first equation is called pivotal equation and a_1 is called first pivot.

Step II: To eliminate y from (vi) equation in (ii). Assuming $b_2' = 0$, we eliminate y from (vi) equation of (ii) by subtracting $\left(\frac{b_3'}{b_2'}\right)$ times the fifth equation from the sixth equation, we get

$$a_1x + b_1y + c_1z = d_1 \dots\dots\dots(vii)$$

$$b_2'y + c_2'z = d_2' \dots\dots\dots(viii)$$

$$c_3''z = d_3'' \dots\dots\dots(xi)$$

Here the (vii) equation is called pivotal equation and b_2' is called the new pivot.

Step III: To evaluate the unknowns the value of x, y, & z are found from the reduced system (iii) by backward substitution.

Gauss Jordan Method

It is the modification over Gauss elimination method. In this elimination method, unknowns is performed only in equation

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below but in the equation above also ultimately reducing the system to a diagonal matrix i.e

Each equation involving only one unknowns from this the unknowns x, y & z can be obtained radially.

Backward substitution for finding unknowns are solved in this method.

Factorization Method

Based on the fact that every square matrix A can be represented the product of a lower triangular matrix and upper triangular matrix provided that all the principle minors of A are non-singular i.e $A = [a_{ij}]$, then $a_{11} \neq 0$,

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \neq 0$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \neq 0 \text{ etc}$$

Also, such a factorization if exists, is unique.

Now, consider the system of equation,

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Which can be written as $AX = B$ (i)

Where,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}; X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}; \& B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Let,

$$A = LU \text{ (ii)}$$

$$\text{Where } L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}, \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

To compute the matrices L & U, we write (2) as,

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

By multiplying,

$$\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Then,

$$u_{11} = a_{11}, u_{12} = a_{12}$$

$$u_{13} = a_{13}$$

$$l_{21}u_{11} = a_{21} \Rightarrow l_{21} = \frac{a_{21}}{u_{11}}$$

$$l_{21}u_{12} + u_{22} = a_{22}$$

$$l_{21}u_{13} + u_{23} = a_{23}$$

$$l_{31}u_{11} = a_{31}$$

$$l_{31}u_{12} + l_{32}u_{22} = a_{32}$$

$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = a_{33}$$

Then equation (i) becomes

$$LUX = B \text{ (iii)}$$

Let $UX = V$ (iv)

Then equation (iii) becomes

$$LV = B$$

Which is equivalent to,

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\text{ie } v_1 = b_1 \text{ (1)}$$

$$l_{21}v_1 + v_2 = b_2 \text{ (2)}$$

$$l_{31}v_1 + l_{32}v_2 + v_3 = b_3 \text{ (3)}$$

Solving these three equation we get value for v_1, v_2 & v_3 . Then equation (iv) becomes

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$$\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

Which is equivalent to,

$$u_{11}x_1 + u_{12}x_2 + u_{13}x_3 = v_1 \dots\dots\dots(4)$$

$$u_{22}x_2 + u_{23}x_3 = v_2 \dots\dots\dots(5)$$

$$u_{33}x_3 = v_3 \dots\dots\dots(6)$$

From these three equation, we can find the value of x_1, x_2 & x_3 backward substitution.

Cholesky Method

If the matrix of equation is symmetric i.e matrix A is symmetric then LU decomposition can be modified, so that upper triangular matrix for factorization is transpose of lower on or vice versa. That we can write A as

$$A = L L^T \dots\dots\dots(i)$$

$$\text{Or } A = U^T U \dots\dots\dots(ii)$$

As similar to do-little algorithm multiplying r.h.s. of equation (ii) and comparing with elements of l.h.s. of A, we get

$$u_{ij} = \sqrt{a_{ij} - \sum_{k=1}^{i-1} u_{ki}^2} \quad \text{for } i=1 \text{ to } n$$

$$u_{ij} = \frac{1}{u_{ii}} [a_{ij} - \sum_{k=1}^{i-1} u_{ki} \cdot u_{kj}] \quad \text{for } j>i$$

This decomposition is called cholesky factorization method.

Jacobi's Iteration Method

Let the given system of equation is,

$$a_1x + b_1y + c_1z = d_1 \dots\dots\dots(i)$$

$$a_2x + b_2y + c_2z = d_2 \dots\dots\dots(ii)$$

$$a_3x + b_3y + c_3z = d_3 \dots\dots\dots(iii)$$

Where

$$x = \frac{1}{a_1} (d_1 - b_1y - c_1z) \dots\dots\dots(*)$$

$$y = \frac{1}{b_2} (d_2 - a_2x - c_2z) \dots\dots\dots(**)$$

$$z = \frac{1}{c_3} (d_3 - a_3x - b_3y) \dots\dots\dots(***)$$

Let, $x_0 = y_0 = z_0 = 0$

And so on for further iteration.

Gauss-Seidel Iteration Method

Let the given system of equation is,

$$a_1x + b_1y + c_1z = d_1 \dots\dots\dots(i)$$

$$a_2x + b_2y + c_2z = d_2 \dots\dots\dots(ii)$$

$$a_3x + b_3y + c_3z = d_3 \dots\dots\dots(iii)$$

Where

$$x_1 = \frac{1}{a_1} (d_1 - b_1y_0 - c_1z_0)$$

$$y_1 = \frac{1}{b_2} (d_2 - a_2x_1 - c_2z_0)$$

$$z_1 = \frac{1}{c_3} (d_3 - a_3x_1 - b_3y_1)$$

Let, $x_0 = y_0 = z_0 = 0$

And so on for further steps.

Eigen Value and Eigen Vectors

If A is any square matrix of order 'n' with elements a_{ij} , we can find a column matrix X and a constant λ such that,

$$AX = \lambda X$$

$$AX - \lambda I X = 0$$

$$[A - \lambda I]X = 0$$

This matrix equation represent n homogeneous linear equation.

Power Method

It is an iterative method which is used to determine largest or smallest eigen value & corresponding eigen vector. It is well suited for machine computation.

For largest eigen value, the formula for power method is given by.

$$AX = \lambda X$$

$$\text{i.e } AX^{n-1} = \lambda X^n$$

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This process repeats until $[X^n - X^{n-1}]$ becomes negligible. Then λ be the smallest eigen values and X^n be the corresponding eigenvectors.

Similarly for smallest eigen values rewriting,

$$AX = \lambda X \text{ as}$$

$$XA^{-1} = \frac{1}{\lambda} X$$

Where $\frac{1}{\lambda}$ is the smallest eigen value and X is the corresponding eigen vector.

Solve the following system of equations using Gauss elimination method with partial pivoting. [2005 Spring]

$$x_1 + x_2 - x_3 = 2$$

$$2x_1 + 3x_2 + 5x_3 = -3$$

$$3x_1 + 2x_2 - 3x_3 = 6$$

Solution:

Here, the given equation are

$$x_1 + x_2 - x_3 = 2$$

$$2x_1 + 3x_2 + 5x_3 = -3$$

$$3x_1 + 2x_2 - 3x_3 = 6$$

The given system of equation can be arrange as

$$2x_1 + 3x_2 + 5x_3 = -3 \quad \text{.....(i)}$$

$$x_1 + x_2 - x_3 = 2 \quad \text{.....(ii)}$$

$$3x_1 + 2x_2 - 3x_3 = 6 \quad \text{.....(iii)}$$

Now, eliminate x_1 from (ii) and (iii) by applying

$$(ii) \leftarrow (i) - 2^*(ii) \text{ and } (iii) \leftarrow 3^*(i) - 2^*(iii)$$

$$2x_1 + 3x_2 + 5x_3 = -3 \quad \text{.....(iv)}$$

$$x_2 + 7x_3 = -7 \quad \text{.....(v)}$$

$$5x_2 + 21x_3 = -21 \quad \text{.....(vi)}$$

eliminate x_2 from (vi) by applying

$$(v) \leftarrow 5^*(v) - (iv)$$

$$2x_1 + 3x_2 + 5x_3 = -3 \quad \text{.....(vii)}$$

$$x_2 + 7x_3 = -7 \quad \text{.....(viii)}$$

$$14x_3 = -14 \quad \text{.....(ix)}$$

By backward substitution,

$$x_3 = -1$$

$$x_2 = 0$$

$$x_1 = 1$$

Which is the required answer

What are the advantages of LU - Decomposition over Gauss elimination method? Solve the following set of linear equation using Gauss Jordan with partial pivoting. [2006 Fall]

$$x + 3y + 4z = 4$$

$$x + 4y + 3z = -2$$

$$4x + 3y + 4z = 1$$

Solution:

Here, given equation are

$$x + 3y + 4z = 4 \quad \text{.....(i)}$$

$$x + 4y + 3z = -2 \quad \text{.....(ii)}$$

$$4x + 3y + 4z = 1 \quad \text{.....(iii)}$$

Now eliminating, x from (ii) and (iii)

$$\text{Applying } (ii) \leftarrow (i) - (ii) \text{ and } (iii) \leftarrow 4^*(i) - (iii)$$

$$x + 3y + 4z = 4 \quad \text{.....(iv)}$$

$$-y + z = 6 \quad \text{.....(v)}$$

$$9y + 12z = 15 \quad \text{.....(vi)}$$

Now, eliminate y from (iv) and (vi)

By applying

$$(iv) \leftarrow 3^*(v) + (iv) \text{ and } (vi) \leftarrow 9(v) + (vi)$$

$$x + 0 + 7z = 22 \quad \text{.....(vii)}$$

$$-y + z = 6 \quad \text{.....(viii)}$$

$$21z = 69 \quad \text{.....(ix)}$$

eliminate z from (vii) and (viii)

By applying

$$(vii) \leftarrow 3^*(viii) - (ix) \text{ and } (viii) \leftarrow 3^*(viii) - (ix)$$

$$3x + 0 + 0 = 3 \quad \text{.....(x)}$$

$$-21y=57 \dots\dots\dots (xi)$$

$$21z=69 \dots\dots\dots (xii)$$

$$x=-1$$

$$y=-2.71$$

$$z=3.28$$

Which is the required answer

Use Gauss Jordan method to solve the following system of linear equation. [2007 Fall]

$$4x_1 + 3x_2 + 0 - 4x_4 = 7$$

$$3x_2 - 4x_3 + 3x_4 = 5$$

$$x_1 + x_2 - 2x_3 + 3x_4 = -2$$

$$x_1 - 3x_3 + x_4 = -5$$

Solution: Here the equations

$$4x_1 + 3x_2 + 0 - 4x_4 = 7 \dots\dots\dots (i)$$

$$3x_2 - 4x_3 + 3x_4 = 5 \dots\dots\dots (ii)$$

$$x_1 + x_2 - 2x_3 + 3x_4 = -2 \dots\dots\dots (iii)$$

$$x_1 - 3x_3 + x_4 = -5 \dots\dots\dots (iv)$$

We have, $AX=B$

$$\begin{bmatrix} 4 & 3 & 0 & -4 & : & 7 \\ 0 & 3 & -4 & 3 & : & 5 \\ 1 & 1 & -2 & 3 & : & -2 \\ 1 & 0 & -3 & 1 & : & -5 \end{bmatrix}$$

By pivoting,

$$\begin{bmatrix} 1 & 0 & -3 & 1 & : & -5 \\ 0 & 3 & -4 & 3 & : & 5 \\ 1 & 1 & -2 & 3 & : & -2 \\ 4 & 3 & 0 & -4 & : & 7 \end{bmatrix}$$

Applying $R_3 \leftarrow R_3 - R_1$; $R_4 \leftarrow R_4 - 4R_1$ and $R_2 \leftarrow \frac{R_2}{3}$ then the matrix becomes

$$\begin{bmatrix} 1 & 0 & -3 & 1 & : & -5 \\ 0 & 1 & -\frac{4}{3} & 1 & : & \frac{5}{3} \\ 0 & 1 & 1 & 2 & : & 3 \\ 0 & 3 & 0.75 & -4.25 & : & 8.25 \end{bmatrix}$$

Again applying $R_3 \leftarrow R_3 - R_2$; $R_4 \leftarrow R_4 - 3R_2$, then the matrix becomes

$$\begin{bmatrix} 1 & 0 & -3 & 1 & : & -5 \\ 0 & 1 & -\frac{4}{3} & 1 & : & \frac{5}{3} \\ 0 & 0 & \frac{3}{3} & 1 & : & \frac{1.333}{2.333} \\ 0 & 0 & 4.75 & -7.25 & : & 3.25 \end{bmatrix}$$

applying $R_1 \leftarrow R_1 + 3R_3$; $R_2 \leftarrow R_2 + \frac{4}{3}R_3$, $R_4 \leftarrow R_4 - 4.75R_3$ then the matrix becomes

$$\begin{bmatrix} 1 & 0 & 0 & 2.2859 & : & -3.2857 \\ 0 & 1 & 0 & 1.5714 & : & 2.4285 \\ 0 & 0 & 1 & 0.4285 & : & 0.5714 \\ 0 & 0 & 0 & 1 & : & -0.0576 \end{bmatrix}$$

applying $R_1 \leftarrow R_1 - 2.2859R_4$; $R_2 \leftarrow R_2 - 1.57R_4$, $R_3 \leftarrow R_3 - 0.4285R_4$ then the matrix becomes

$$\begin{bmatrix} 1 & 0 & 0 & 0 & : & -3.4175 \\ 0 & 1 & 0 & 0 & : & 2.3378 \\ 0 & 0 & 1 & 0 & : & 0.5466 \\ 0 & 0 & 0 & 1 & : & -0.0576 \end{bmatrix}$$

Hence by solving,

$$\begin{aligned} x_1 &= -3.153828 \\ x_2 &= 2.519154 \\ x_3 &= 0.596144 \\ x_4 &= -0.05768 \end{aligned}$$

Which is the required answer.

Solve the equation by Gauss-Jordan method:

[2008 Fall]

$$x+2y+z=8$$

$$2x+3y+4z=20$$

$$4x+3y+2z=16$$

Solution:

Arranging the above system of equation

$$4x+3y+2z=16 \dots\dots\dots (i)$$

$$2x+3y+4z=20 \dots\dots\dots (ii)$$

$$x+2y+z=8 \dots\dots\dots (iii)$$

Now eliminating, x from (ii) and (iii)

$$\text{Applying } (ii) \leftarrow 2*(i) - (i) \text{ and } (iii) \leftarrow 4*(i) - (i)$$

$$4x+3y+2z=16 \dots\dots\dots (iv)$$

$$\begin{aligned} 0 + 3y + 6z &= 24 \dots\dots\dots(v) \\ 0 + 5y + 2z &= 16 \dots\dots\dots(vi) \end{aligned}$$

Now, eliminate y from (iv) and (vi)

By applying

$$(iv) \leftarrow (v) - (iv) \text{ and } (vi) \leftarrow 3(vi) - 5*(v)$$

$$4x + 0 + 4z = 8 \dots\dots\dots(vii)$$

$$3y + 6z = 24 \dots\dots\dots(viii)$$

$$-24z = -72 \dots\dots\dots(ix)$$

eliminate z from (vii) and (viii)

By applying

$$(vii) \leftarrow 6*(vii) + (ix) \text{ and } (viii) \leftarrow 4*(viii) + (ix)$$

$$24x + 0 + 0 = 4 \dots\dots\dots(x)$$

$$-12y = 24 \dots\dots\dots(xi)$$

$$-24z = -72 \dots\dots\dots(xii)$$

$$x = 1$$

$$y = 2$$

$$z = 3$$

Which is the required answer

Solve the following system of linear equations using Gauss Elimination Method. Use partial pivoting whenever necessary.

[2010 Fall]

$$b + 3c + 2d = 19$$

$$3b + 2c + 2d = 20$$

$$a + 4b + 2d = 17$$

$$-2a + 2b + c + d = 9$$

Solution:

The given system of equation can be arrange as,

$$-2a + 2b + c + d = 9 \dots\dots\dots(i)$$

$$a + 4b + 2d = 17 \dots\dots\dots(ii)$$

$$3b + 2c + 2d = 20 \dots\dots\dots(iii)$$

$$b + 3c + 2d = 19 \dots\dots\dots(iv)$$

Now, eliminate a from (ii)

By applying

$$(ii) \leftarrow (i) + 2*(ii)$$

$$\begin{aligned} -2a + 2b + c + d &= 9 \dots\dots\dots(v) \\ 0 + 10b + c + 5d &= 43 \dots\dots\dots(vi) \end{aligned}$$

$$\begin{aligned} 0 + 3b + 2c + 2d &= 20 \dots\dots\dots(vii) \\ 0 + b + 3c + 2d &= 19 \dots\dots\dots(viii) \end{aligned}$$

Eliminate b from (vii) and (viii)

By applying

$$(vii) \leftarrow 10*(vii) - 3*(vi) \text{ and } (viii) \leftarrow 10*(viii) - (vi)$$

$$-2a + 2b + c + d = 9 \dots\dots\dots(ix)$$

$$0 + 10b + c + 5d = 43 \dots\dots\dots(x)$$

$$17c + 5d = 71 \dots\dots\dots(xi)$$

$$29c + 15d = 147 \dots\dots\dots(xii)$$

Eliminating c from (xii),

By applying (xii) $\leftarrow 17*(xii) - 29*(xi)$

$$-2a + 2b + c + d = 9 \dots\dots\dots(xiii)$$

$$0 + 10b + c + 5d = 43 \dots\dots\dots(xiv)$$

$$17c + 5d = 71 \dots\dots\dots(xv)$$

$$110d = 440 \dots\dots\dots(xvi)$$

By backward substitution,

$$d = 4, c = 3, b = 2, a = 1$$

Which is the required answer

Find the determinant of given matrix using Gauss elimination

$$\begin{bmatrix} 1.2 & -2.1 & 3.2 \\ -1.4 & -2.6 & 3.0 \\ 1.1 & 3.6 & 5.0 \end{bmatrix}$$

(2010 Spring)

Solution:

Here let A be the matrix then,

$$A = \begin{bmatrix} 1.2 & -2.1 & 3.2 \\ -1.4 & -2.6 & 3.0 \\ 1.1 & 3.6 & 5.0 \end{bmatrix}$$

Now, eliminate -1.4 and 1.1 from (ii) and (iii) by applying

$$R_2 \leftarrow R_2 - \frac{-1.4}{1.2} R_1 \text{ and } R_3 \leftarrow R_3 - \frac{1.1}{1.2} R_1$$

Then the matrix becomes

$$A = \begin{bmatrix} 1.2 & -2.1 & 3.2 \\ 0 & -5.05 & 6.733 \\ 0 & 5.525 & 2.066 \end{bmatrix}$$

Again eliminate 5.525 from R_3 by applying

$$R_3 \leftarrow R_3 + \frac{5.525}{-5.05} R_2 \text{ then our matrix becomes}$$

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$$A = \begin{bmatrix} 1.2 & -2.1 & 3.2 \\ 0 & -5.05 & 6.7333 \\ 0 & 0 & 9.4333 \end{bmatrix}$$

Now the matrix determinant is
Determinant of A $|A| = 1.2(-5.05 * 9.4336)$
 $= -57.167$

Which is the required answer

Using Gauss Elimination method, solve the following system of near equation. [2009 Fall]

$$\begin{array}{rrcrcl} 10x & - & 7y & + & 3z & + & 5u & = & 6 \\ -6x & + & 8y & - & & & 4u & = & 5 \\ 3x & + & y & + & 4z & + & 11u & = & 2 \\ 5x & - & 9y & - & 2z & + & 4u & = & 7 \end{array}$$

Solution:

The given system of equation can be arrange as,

$$10x - 7y + 3z + 5u = 6 \dots\dots\dots(i)$$

$$-6x + 8y - z - 4u = 5 \dots\dots\dots(ii)$$

$$3x + y + 4z + 11u = 2 \dots\dots\dots(iii)$$

$$5x - 9y - 2z + 4u = 7 \dots\dots\dots(iv)$$

Now, eliminate x from (ii), (iii) and (iv)

By applying

$$(ii) \leftarrow 5(ii) + 3(i), (iii) \leftarrow 10(iii) - 3(i), (iv) \leftarrow 2(iv) - (i)$$

$$10x - 7y + 3z + 5u = 6 \dots\dots\dots(v)$$

$$0 + 19y + 4z - 5u = 43 \dots\dots\dots(vi)$$

$$0 + 31y + 31z + 95u = 2 \dots\dots\dots(vii)$$

$$0 - 11y - 7z + 3u = 8 \dots\dots\dots(viii)$$

Eliminate y from (vii) and (viii)

By applying

$$(vii) \leftarrow 19(vii) - 31(vi) \text{ and } (viii) \leftarrow 19(viii) + 11(vi)$$

$$10x - 7y + 3z + 5u = 6 \dots\dots\dots(ix)$$

$$0 + 19y + 4z - 5u = 43 \dots\dots\dots(x)$$

$$0 + 0 + 465z + 1960u = -1295 \dots\dots\dots(xi)$$

$$0 + 0 + 89z + 2u = 625 \dots\dots\dots(xii)$$

Eliminating z from (xii),

By applying (xii) $\leftarrow 465(xii) - 89(xi)$

$$10x - 7y + 3z + 5u = 6 \dots\dots\dots(ix)$$

$$0 + 19y + 4z - 5u = 43 \dots\dots\dots(x)$$

$$0 + 0 + 465z + 1960u = -1295 \dots\dots\dots(xi)$$

$$0 + 0 + 0 - 173310u = 405880 \dots\dots\dots(xii)$$

By backward substitution,

$$u = -2.3419, z = 7.07, y = 0.1589, x = -0.239$$

Which is the required answer

Using Gauss Elimination method, solve the following system of near equation. [2008 Spring]

$$\begin{array}{rrcrcl} 2w & + & x & + & 3y & - & z & = & 9 \\ -3w & + & x & - & 2y & + & z & = & -4 \\ w & + & 2x & - & y & + & 2z & = & 5 \\ 3w & - & x & - & y & + & 2z & = & 1 \end{array}$$

Solution:

[Similar to 2009 Fall which is done in above]

Determine the smallest eigen value and its corresponding eigenvector for the following matrix using Power method

$$A = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & -1 \\ -1 & 0 & -1 \end{bmatrix} \quad [2005 fall]$$

Solution:

Let x be the unique vector & initial value $x = [1 \ 0 \ 0]^T$

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{now } Ax = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -0.5 \end{bmatrix}$$

by 1st approximation the eigen value and eigen vector is

$$\lambda^1 = 2 \text{ and } X^1 = [1 \ 0 \ -0.5]$$

Again from second approximation,

$$AX^1 = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -0.5 \end{bmatrix} = \begin{bmatrix} 2.5 \\ 2.5 \\ -0.5 \end{bmatrix}$$

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Then,

$$\lambda^2 = 2 \text{ and } X^2 = \begin{bmatrix} 1 & 0.25 & -0.25 \end{bmatrix}$$

$$AX^2 = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0.25 \\ -0.25 \end{bmatrix} = \begin{bmatrix} 2.75 \\ 0.75 \\ -0.75 \end{bmatrix} = 2.75 \begin{bmatrix} 1 \\ 0.375 \\ -0.375 \end{bmatrix}$$

$$\lambda^3 = 2.75 \text{ and } X^3 = \begin{bmatrix} 1 & 0.375 & -0.375 \end{bmatrix}$$

$$AX^3 = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0.375 \\ -0.375 \end{bmatrix} = \begin{bmatrix} 3.125 \\ 1.125 \\ -0.625 \end{bmatrix} = 3.125 \begin{bmatrix} 1 \\ 0.36 \\ -0.2 \end{bmatrix}$$

$$\lambda^4 = 3.125 \text{ and } X^4 = \begin{bmatrix} 1 & 0.36 & -0.2 \end{bmatrix}$$

$$AX^4 = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0.36 \\ -0.2 \end{bmatrix} = \begin{bmatrix} 3.08 \\ 0.52 \\ -1.2 \end{bmatrix} = 3.8 \begin{bmatrix} 1 \\ 0.1688 \\ -0.3896 \end{bmatrix}$$

$$\lambda^5 = 3.8 \text{ and } X^5 = \begin{bmatrix} 1 & 0.1688 & -0.3896 \end{bmatrix}$$

$$AX^5 = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0.1688 \\ -0.3896 \end{bmatrix} = \begin{bmatrix} 2.5064 \\ 0.7272 \\ -0.6104 \end{bmatrix} = 2.5064 \begin{bmatrix} 1 \\ 0.2901 \\ -0.2435 \end{bmatrix}$$

$$\lambda^6 = 2.5064 \text{ and } X^6 = \begin{bmatrix} 1 & 0.2901 & -0.2435 \end{bmatrix}$$

$$AX^6 = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0.2901 \\ -0.2435 \end{bmatrix} = \begin{bmatrix} 2.8703 \\ 0.8237 \\ -0.7565 \end{bmatrix} = 2.8703 \begin{bmatrix} 1 \\ 0.2869 \\ -0.2435 \end{bmatrix}$$

$$\lambda^7 = 2.8703 \text{ and } X^7 = \begin{bmatrix} 1 & 0.2869 & -0.2435 \end{bmatrix}$$

$$AX^7 = \begin{bmatrix} 2 & 3 & 0 \\ 0 & 2 & -1 \\ -1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0.2869 \\ -0.2435 \end{bmatrix} = \begin{bmatrix} 0.8607 \\ 0.8373 \\ -0.736 \end{bmatrix} = 2.8607 \begin{bmatrix} 1 \\ 0.2926 \\ -0.2572 \end{bmatrix}$$

Hence eigen value $\lambda = 2.8607$ & eigen vector

$$x = \begin{bmatrix} 1 & 0.2926 & -0.2572 \end{bmatrix}$$

Hence the largest eigen value is

Determine the highest eigen value and its corresponding eigen vector for the following matrix by using power method. [2006 Fall]

$$A = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix}$$

Solution:

$$\text{Here, } A = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix}$$

Let X be the unique vector and initial value $X = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T$

$$\text{So, } X = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Now, } AX = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ -0.66 \\ 0 \end{bmatrix}$$

By first approximation the eigen value and eigen vector is

$$\lambda^1 = 3 \text{ and } X^1 = \begin{bmatrix} 1 & -0.66 & 0 \end{bmatrix}$$

Again from second approximation,

$$AX^1 = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -0.66 \\ 0 \end{bmatrix} = \begin{bmatrix} 3.666 \\ -4.664 \\ 0.666 \end{bmatrix} = 4.664 \begin{bmatrix} 0.7860 \\ -1 \\ 0.14279 \end{bmatrix}$$

Then,

$$\lambda^2 = 4.664 \text{ and } X^2 = \begin{bmatrix} 0.7860 & -1 & 0.14279 \end{bmatrix}$$

$$AX^2 = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0.78601 \\ -1 \\ 0.14279 \end{bmatrix} = \begin{bmatrix} 3.358 \\ -6 \\ 1.1427 \end{bmatrix} = 6 \begin{bmatrix} 0.5596 \\ -1 \\ 0.1904 \end{bmatrix}$$

$$\lambda^3 = 6 \text{ and } X^3 = \begin{bmatrix} 0.5596 & -1 & 0.1904 \end{bmatrix}$$

$$AX^3 = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0.5596 \\ -1 \\ 0.1904 \end{bmatrix} = \begin{bmatrix} 2.6788 \\ -5.6904 \\ 1.1904 \end{bmatrix} = 5.6904 \begin{bmatrix} 0.407 \\ -1 \\ 0.2091 \end{bmatrix}$$

$$\lambda^4 = 5.4904 \text{ and } X^4 = \begin{bmatrix} 0.4707 & -1 & 0.2091 \end{bmatrix}$$

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$$AX^4 = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0.4707 \\ -1 \\ 0.2091 \end{bmatrix} = \begin{bmatrix} 2.4121 \\ -5.5687 \\ 1.2091 \end{bmatrix} = 5.5687 \begin{bmatrix} 0.4241 \\ -1 \\ 0.2171 \end{bmatrix}$$

$$\lambda^5 = 5.5687 \text{ and } X^5 = [0.4241 \quad -1 \quad 0.2171]$$

$$AX^5 = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0.4241 \\ -1 \\ 0.2171 \end{bmatrix} = \begin{bmatrix} 2.2723 \\ -5.4995 \\ 1.2171 \end{bmatrix} = 5.4995 \begin{bmatrix} 0.4131 \\ -1 \\ 0.2213 \end{bmatrix}$$

$$\lambda^6 = 5.4995 \text{ and } X^6 = [0.4131 \quad -1 \quad 0.2213]$$

$$AX^6 = \begin{bmatrix} 3 & -1 & 0 \\ -2 & 4 & -3 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0.4131 \\ -1 \\ 0.2213 \end{bmatrix} = \begin{bmatrix} 2.2393 \\ -5.490 \\ 1.2213 \end{bmatrix} = 5.490 \begin{bmatrix} 0.4078 \\ -1 \\ 0.2224 \end{bmatrix}$$

Hence eigen value $\lambda = 5.490$ & eigen vector $[0.4078 \quad -1 \quad 0.2224]$

Find the largest Eigen-value and the corresponding Eigen-vector of the following square matrix using Power method.

[2006 Spring, 2007 Spring]

$$\begin{bmatrix} 1 & -1 & 2 \\ 3 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

Solution :

[Similar to 2006 Fall which is done in above]

Find the largest Eigen-value and the corresponding Eigen-vector of the following square matrix using Power method. [2008 Spring]

$$\begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Solution:

$$\text{Here, } A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Let X be the unique vector and initial value $X = [1 \quad 0 \quad 0]'$

$$\text{So, } X = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Now, } AX = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Now by first approximation the eigen value and eigen vector is

$$\lambda^1 = 3 \text{ and } X^1 = [1 \quad 0 \quad 0]$$

Again from second approximation,

$$AX^1 = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Hence largest eigen vector $[1 \quad 0 \quad 0]$ & eigen value is $\lambda = 3$

[Question may be wrong]

Find the largest eigen value and corresponding eigen vector by power method. [2009 Fall, 2010 Spring]

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Solution:

$$\text{Here, } A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Let X be the unique vector and initial value $X = [1 \quad 0 \quad 0]'$

$$\text{So, } X = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Now, } AX = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -0.5 \\ 0 \end{bmatrix}$$

Now by first approximation the eigen value and eigen vector is

$$\lambda^1 = 2 \text{ and } X^1 = [1 \quad -0.5 \quad 0]$$

Again from second approximation,

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$$AX^1 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 2.5 \\ -2 \\ 0.5 \end{bmatrix} = 2.5 \begin{bmatrix} 1 \\ -0.8 \\ 0.2 \end{bmatrix}$$

$$\lambda^2 = 2.5 \text{ and } X^2 = [1 \quad -0.8 \quad 0.2]$$

$$AX^2 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -0.8 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 2.8 \\ -2.8 \\ 1.2 \end{bmatrix} = 2.8 \begin{bmatrix} 1 \\ -1 \\ 0.42 \end{bmatrix}$$

$$\lambda^3 = 2.8 \text{ and } X^3 = [1 \quad -1 \quad 0.42]$$

$$AX^3 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0.42 \end{bmatrix} = \begin{bmatrix} 3 \\ -3.4 \\ 1.84 \end{bmatrix} = 3.4 \begin{bmatrix} 0.88 \\ -1 \\ 0.54 \end{bmatrix}$$

$$\lambda^4 = 3.4 \text{ and } X^4 = [0.88 \quad -1 \quad 0.54]$$

$$AX^4 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.88 \\ -1 \\ 0.54 \end{bmatrix} = \begin{bmatrix} 2.76 \\ -3.42 \\ 2.08 \end{bmatrix} = 3.42 \begin{bmatrix} 0.807 \\ -1 \\ 0.608 \end{bmatrix}$$

$$\lambda^5 = 3.42 \text{ and } X^5 = [0.807 \quad -1 \quad 0.608]$$

$$AX^5 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.807 \\ -1 \\ 0.608 \end{bmatrix} = \begin{bmatrix} 2.061 \\ -3.41 \\ 2.216 \end{bmatrix} = 3.41 \begin{bmatrix} 0.765 \\ -1 \\ 0.650 \end{bmatrix}$$

$$\lambda^6 = 3.41 \text{ and } X^6 = [0.765 \quad -1 \quad 0.650]$$

$$AX^6 = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0.765 \\ -1 \\ 0.650 \end{bmatrix} = \begin{bmatrix} 2.53 \\ -3.41 \\ 2.30 \end{bmatrix} = 3.41 \begin{bmatrix} 0.742 \\ -1 \\ 0.674 \end{bmatrix}$$

Hence the largest eigen value is

$$\lambda = 3.41 \text{ and largest eigen vector is } x = [0.742 \quad -1 \quad 0.674]$$

Find the largest eigen value and corresponding eigen vector by power method. [2011 Spring]

$$\begin{bmatrix} 2 & -2 & 4 \\ 2 & 3 & 2 \\ -1 & 1 & 1 \end{bmatrix}$$

Solution :

[Similar to 2009 Fall which is done in above]

Find the largest eigen value and corresponding eigen vector of the following matrix using power method

$$\begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 3 \\ 0 & 2 & -3 \end{bmatrix}$$

[2008 Fall]

Solution:

$$\text{Here, } A = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 3 \\ 0 & 2 & -3 \end{bmatrix}$$

Let X be the unique vector and initial value $X = [1 \quad 0 \quad 0]^T$

$$\text{So, } X = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Now, } AX = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 3 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = 1 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Now by first approximation the eigen value and eigen vector is

$$\lambda^1 = 1 \text{ and } X^1 = [-1 \quad 1 \quad 0]$$

Again from second approximation,

$$AX^1 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 3 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} = 3 \begin{bmatrix} 0.3 \\ -1 \\ 0.66 \end{bmatrix}$$

$$\lambda^2 = 3 \text{ and } X^2 = [0.3 \quad -1 \quad 0.66]$$

$$AX^2 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 3 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} 0.3 \\ -1 \\ 0.66 \end{bmatrix} = \begin{bmatrix} -0.30 \\ 4.28 \\ -3.98 \end{bmatrix} = 4.28 \begin{bmatrix} -0.070 \\ 1 \\ -0.929 \end{bmatrix}$$

$$\lambda^3 = 4.28 \text{ and } X^3 = [-0.070 \quad 1 \quad -0.929]$$

$$AX^3 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 3 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} -0.070 \\ 1 \\ -0.929 \end{bmatrix} = \begin{bmatrix} 0.07 \\ -4.85 \\ 4.78 \end{bmatrix} = 4.85 \begin{bmatrix} 0.0144 \\ -1 \\ 0.985 \end{bmatrix}$$

$$\lambda^4 = 4.85 \text{ and } X^4 = [0.0144 \quad -1 \quad 0.985]$$

$$AX^4 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 3 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} 0.0144 \\ -1 \\ 0.985 \end{bmatrix} = \begin{bmatrix} -0.0144 \\ 4.969 \\ -4.959 \end{bmatrix} = 4.969 \begin{bmatrix} -0.0028 \\ 1 \\ -0.9959 \end{bmatrix}$$

$$\lambda^5 = 4.969 \text{ and } X^5 = [-0.0028 \quad 1 \quad -0.9979]$$

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$$AX^5 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 3 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} -0.0028 \\ 1 \\ -0.9979 \end{bmatrix} = \begin{bmatrix} 0.0028 \\ -4.996 \\ 4.993 \end{bmatrix} = 4.996 \begin{bmatrix} 0.00056 \\ -1 \\ 0.999 \end{bmatrix}$$

$$\lambda^6 = 4.996 \text{ and } X^6 = [0.00056 - 1 \ 0.999]$$

$$AX^6 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & -2 & 3 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} 0.00056 \\ -1 \\ 0.999 \end{bmatrix} = \begin{bmatrix} -0.00056 \\ 4.9975 \\ -4.997 \end{bmatrix} = 4.9975 \begin{bmatrix} 0.00011 \\ 1 \\ -1 \end{bmatrix}$$

Hence the largest eigen value is

$$\lambda = 4.9975 \text{ and largest eigen vector is } x = [0.00011 \ 1 - 1]$$

Solve the following set of equation using LU factorisation method. [2007 Spring]

$$3x + 2y + z = 10$$

$$2x + 3y + 2z = 14$$

$$x + 2y + 3z = 14$$

Solution:

Here

$$3x + 2y + z = 10 \dots\dots\dots(i)$$

$$2x + 3y + 2z = 14 \dots\dots\dots(ii)$$

$$x + 2y + 3z = 14 \dots\dots\dots(iii)$$

The given system is $AX = B$, where

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ \& } B = \begin{bmatrix} 10 \\ 14 \\ 14 \end{bmatrix}$$

Let $LU = A$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}, \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$\therefore \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

then

$$u_{11} = 3, u_{12} = 2, u_{13} = 1$$

$$l_{21}u_{11} = 2 \Rightarrow l_{21} = \frac{2}{u_{11}} = \frac{2}{3},$$

$$l_{21}u_{12} + u_{22} = 3 \Rightarrow u_{22} = \frac{5}{3}$$

$$l_{21}u_{13} + u_{23} = 2 \Rightarrow u_{23} = \frac{4}{3}$$

$$l_{31}u_{11} = 1 \Rightarrow l_{31} = \frac{1}{3}$$

$$l_{31}u_{12} + l_{32}u_{22} = 2 \Rightarrow l_{32} = \frac{4}{5}$$

$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = 3 \Rightarrow u_{33} = \frac{8}{5}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0.666 & 1 & 0 \\ 0.333 & 0.8 & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1.666 & 1.333 \\ 0 & 0 & 1.6 \end{bmatrix}$$

$$\text{Let } LV = B, \text{ where } V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0.666 & 1 & 0 \\ 0.333 & 0.8 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 14 \\ 14 \end{bmatrix}$$

$$\therefore v_1 = 10; v_2 = 7.33; v_3 = 4.80$$

$$\text{Then } UX = V, \text{ where } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$U = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 1.666 & 1.333 \\ 0 & 0 & 1.6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 7.33 \\ 4.80 \end{bmatrix}$$

From matrix multiplication we get

$$1.6z = 4.80 \dots\dots\dots(*)$$

$$1.66y + 1.33z = 7.33 \dots\dots\dots(**)$$

$$3z + 2y + z = 10 \dots\dots\dots(***)$$

Solving these, we get, by back substitution

$$z = 3, y = 2, x = 1$$

Use L - U factorization method to solve the following system of equations. [2006 Spring]

$$x + y + z = 1$$

$$3x + y - 3z = 5$$

$$x - 2y - 5z = 10$$

Solution:

Here, the given equation of system are

$$x + y + z = 1 \dots\dots\dots(i)$$

$$3x + y - 3z = 5 \dots\dots\dots(ii)$$

$$x - 2y - 5z = 10 \dots\dots\dots(iii)$$

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The given system is $AX = B$, where

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & -3 \\ 1 & -2 & -5 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \& B = \begin{bmatrix} 1 \\ 5 \\ 10 \end{bmatrix}$$

Let $LU = A$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}, \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$\therefore \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 3 & 1 & -3 \\ 1 & -2 & -5 \end{bmatrix}$$

Then,

$$u_{11} = 1, u_{12} = 1, u_{13} = 1$$

$$l_{21}u_{11} = 3 \Rightarrow l_{21} = \frac{3}{u_{11}} = 3$$

$$l_{21}u_{12} + u_{22} = 1 \Rightarrow u_{22} = -2$$

$$l_{21}u_{13} + u_{23} = -3 \Rightarrow u_{23} = -6$$

$$l_{31}u_{11} = 1 \Rightarrow l_{31} = 1$$

$$l_{31}u_{12} + l_{32}u_{22} = -2 \Rightarrow l_{32} = 1.5$$

$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = -5 \Rightarrow u_{33} = 3$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 1.5 & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -6 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\text{Let } LV = B, \text{ where } V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 1.5 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 10 \end{bmatrix}$$

$$\therefore v_1 = 1; v_2 = 2; v_3 = 6$$

$$\text{Then } UX = V, \text{ where } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & -6 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix}$$

From matrix multiplication we get

$$3z = 6 \dots\dots\dots (*)$$

$$-2y - 6z = 2 \dots\dots\dots (**)$$

$$x + y + z = 1 \dots\dots\dots (***)$$

Solving these, we get, by back substitution

$$z = 2, y = -7, x = 6$$

Use LU decomposition method and solve. [2010 Fall]

$$5x_1 - 2x_2 + x_3 = 4$$

$$7x_1 + x_2 - 5x_3 = 8$$

$$3x_1 + 7x_2 + 4x_3 = 10$$

Solution:

Here, the given equation of system are

$$5x_1 - 2x_2 + x_3 = 4 \dots\dots\dots (i)$$

$$7x_1 + x_2 - 5x_3 = 8 \dots\dots\dots (ii)$$

$$3x_1 + 7x_2 + 4x_3 = 10 \dots\dots\dots (iii)$$

The given system is $AX = B$, where

$$A = \begin{bmatrix} 5 & -2 & 1 \\ 7 & 1 & -5 \\ 3 & 7 & 4 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \& B = \begin{bmatrix} 4 \\ 8 \\ 10 \end{bmatrix}$$

Let $LU = A$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}, \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$\therefore \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -2 & 1 \\ 7 & 1 & -5 \\ 3 & 7 & 4 \end{bmatrix}$$

then

$$u_{11} = 5, u_{12} = -2, u_{13} = 1$$

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$$l_{21}u_{11} = 7 \Rightarrow l_{21} = \frac{7}{u_{11}} = 1.4$$

$$l_{21}u_{12} + u_{22} = 1 \Rightarrow u_{22} = 3.8$$

$$l_{21}u_{13} + u_{23} = -5 \Rightarrow u_{23} = -6.4$$

$$l_{31}u_{11} = 3 \Rightarrow l_{31} = 0.6$$

$$l_{31}u_{12} + l_{32}u_{22} = 7 \Rightarrow l_{32} = 2.15$$

$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = 4 \Rightarrow u_{33} = 17.21$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1.4 & 1 & 0 \\ 0.6 & 2.15 & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} 5 & -2 & 1 \\ 0 & 3.8 & -6.4 \\ 0 & 0 & 17.21 \end{bmatrix}$$

$$\text{Let } LV = B, \text{ where } V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1.4 & 1 & 0 \\ 0.6 & 2.15 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \\ 10 \end{bmatrix}$$

$$\therefore v_1 = 4; v_2 = 2.4; v_3 = 2.42$$

$$\text{Then } UX = V, \text{ where } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$U = \begin{bmatrix} 5 & -2 & 1 \\ 0 & 3.8 & -6.4 \\ 0 & 0 & 17.21 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 2.4 \\ 2.42 \end{bmatrix}$$

From matrix multiplication we get

$$17.21z = 2.42 \dots\dots\dots (*)$$

$$3.8y - 6.4z = 2.4 \dots\dots\dots (**)$$

$$5x - 2y + z = 4 \dots\dots\dots (***)$$

Solving these, we get, by back substitution, $z = 0.1406$,
 $y = 0.8685$, $x = 1.1192$

Solve the following set of equations by using LU decomposition method. [2007 Fall]

$$3x + 2y + 7z = 4$$

$$2x + 3y + z = 5$$

$$3x + 4y + z = 7$$

Solution:

Here, the given equation of system are

$$3x + 2y + 7z = 4 \dots\dots\dots (i)$$

$$2x + 3y + z = 5 \dots\dots\dots (ii)$$

$$3x + 4y + z = 7 \dots\dots\dots (iii)$$

The given system is $AX = B$, where

$$A = \begin{bmatrix} 3 & 2 & 7 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \& B = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}$$

Let $LU = A$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}, \text{ and } U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$\therefore \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix} = \begin{bmatrix} 3 & 2 & 7 \\ 2 & 3 & 1 \\ 3 & 4 & 1 \end{bmatrix}$$

Then

$$u_{11} = 3, u_{12} = 2; u_{13} = 7$$

$$l_{21}u_{11} = 2 \Rightarrow l_{21} = \frac{2}{u_{11}} = 0.66$$

$$l_{21}u_{12} + u_{22} = 3 \Rightarrow u_{22} = 1.67$$

$$l_{21}u_{13} + u_{23} = 1 \Rightarrow u_{23} = -3.667$$

$$l_{31}u_{11} = 3 \Rightarrow l_{31} = 1$$

$$l_{31}u_{12} + l_{32}u_{22} = 4 \Rightarrow l_{32} = 1.197$$

$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = 1 \Rightarrow u_{33} = -1.6$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0.66 & 1 & 0 \\ 1 & 1.197 & 1 \end{bmatrix} \text{ and } U = \begin{bmatrix} 3 & 2 & 7 \\ 0 & 1.67 & -3.66 \\ 0 & 0 & -1.6 \end{bmatrix}$$

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Let $LV=B$, where $V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0.66 & 1 & 0 \\ 1 & 1.197 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 7 \end{bmatrix}$$

$$\therefore v_1 = 4; v_2 = 2.33; v_3 = 0.204$$

Then $UX=V$, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$,

$$U = \begin{bmatrix} 3 & 2 & 7 \\ 0 & 1.67 & -3.66 \\ 0 & 0 & -1.6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 2.33 \\ 0.204 \end{bmatrix}$$

From matrix multiplication we get

$$-1.6z = 0.204 \dots \dots \dots (*)$$

$$1.67y - 3.66z = 2.33 \dots \dots \dots (**)$$

$$3x + 2y + 7z = 4 \dots \dots \dots (***)$$

Solving these,

By back substitution,

$$z = -0.127, y = 1.1157, x = 0.8858$$

By an iterative method solve the following equations. [2008 Fall]

$$10x - 2y - z - w = 3$$

$$-x - y + 10z - 2w = 27$$

$$-x - y - 2z + 10w = -9$$

$$-2x + 10y - z - w = 15$$

Solution:

Here,

Arranging the above system of equation

$$10x - 2y - z - w = 3 \dots \dots \dots (i)$$

$$-2x + 10y - z - w = 15 \dots \dots \dots (ii)$$

$$-x - y + 10z - 2w = 27 \dots \dots \dots (iii)$$

$$-x - y - 2z + 10w = -9 \dots \dots \dots (iv)$$

Now we write the equation in the form

$$x = \frac{1}{10} (3 + 2y + z + w) \dots \dots \dots (v)$$

$$y = \frac{1}{10} (15 + 2x + z + w) \dots \dots \dots (vi)$$

$$z = \frac{1}{10} (27 + x + y + 2w) \dots \dots \dots (vii)$$

$$w = \frac{1}{10} (-9 + x + y + 2z) \dots \dots \dots (viii)$$

We start from an approximation $x_0 = y_0 = z_0 = w_0 = 0$

Substituting these on R.H.S of (v) we get,

First approximation as

$$x^{(1)} = \frac{1}{10} (3 + 0 + 0 + 0) = 0.3$$

$$y^{(1)} = \frac{1}{10} (15 + 2 \cdot 0.3 + 0 + 0) = 1.56$$

$$z^{(1)} = \frac{1}{10} (27 + 0.3 + 1.56 + 0) = 4.29$$

$$w^{(1)} = \frac{1}{10} (-9 + 0.3 + 1.56 + 2 \cdot 4.29) = 0.144$$

Second approximation

$$x^{(2)} = \frac{1}{10} (3 + 2 \cdot 1.56 + 4.29 + 0.144) = 1.05$$

$$y^{(2)} = \frac{1}{10} (15 + 2 \cdot 1.05 + 4.29 + 0.144) = 2.15$$

$$z^{(2)} = \frac{1}{10} (27 + 1.05 + 2.15 + 2 \cdot 0.144) = 3.048$$

$$w^{(2)} = \frac{1}{10} (-9 + 1.05 + 2.15 + 2 \cdot 3.048) = 0.0296$$

Successively we get,

$$x^{(3)} = 1.037; y^{(3)} = 2.015; z^{(3)} = 3.011; w^{(3)} = 0.0074$$

$$x^{(4)} = 1.004; y^{(4)} = 2.002; z^{(4)} = 3.001; w^{(4)} = 0.008$$

$$x^{(5)} = 1.0005; y^{(5)} = 2.0002; z^{(5)} = 3.0002; w^{(5)} = 0.0010$$

$$\text{Hence } 1.0005; y = 2.0002; z = 3.0002; w = 0.0010$$

By an iterative method solve the following equations. [2006 Fall]

$$10x - 2y - z - w = 3$$

$$6x_1 - 2x_2 + x_3 = 11$$

$$x_1 + 2x_2 - 5x_3 = -1$$

$$-2x_1 + 7x_2 + 2x_3 = 5$$

Solution:

[Similar to 2008 fall which is done in above]

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Solve the following system of equation by Jacobi-Iteration method. [2009 Spring]

$$3x + 4y + 15z = 54.8$$

$$x + 12y + 3z = 39.66$$

$$10x + y - 2z = 7.74$$

Solution:

Here

Arranging the above system of equation

$$10x + y - 2z = 7.74 \dots\dots\dots(i)$$

$$x + 12y + 3z = 39.66 \dots\dots\dots(ii)$$

$$3x + 4y + 15z = 54.8 \dots\dots\dots(iii)$$

Now we write the equation in the form

$$x = \frac{1}{10}(7.74 - y + 2z) \dots\dots\dots(iv)$$

$$y = \frac{1}{12}(39.66 - x - 3z) \dots\dots\dots(v)$$

$$z = \frac{1}{15}(54.8 - 4y - 3x) \dots\dots\dots(vi)$$

We start from an approximation $x_0 = y_0 = z_0 = 0$

Substituting these on R.H.S of (iv) we get,

First approximation as

$$x^{(1)} = \frac{1}{10}(7.74 + 0 + 0) = 0.774$$

$$y^{(1)} = \frac{1}{12}(39.66 + 0 + 0) = 3.30$$

$$z^{(1)} = \frac{1}{15}(54.8 - 0 - 0) = 3.653$$

Second approximation

$$x^{(2)} = \frac{1}{10}(7.74 - 3.30 + 2 \cdot 3.653) = 1.174$$

$$y^{(2)} = \frac{1}{12}(39.66 - 0.774 + 2 \cdot 3.653) = 3.84$$

$$z^{(2)} = \frac{1}{15}(54.8 - 4 \cdot 3.30 - 3 \cdot 0.774) = 2.042$$

Successively we get,

$$x^{(3)} = 0.798; y^{(3)} = 2.6966; z^{(3)} = 2.2209;$$

$$x^{(4)} = 0.948; y^{(4)} = 2.683; z^{(4)} = 2.774;$$

$$x^{(5)} = 1.060; y^{(5)} = 2.53; z^{(5)} = 2.766;$$

$$x^{(6)} = 1.074; y^{(6)} = 2.53; z^{(6)} = 2.766;$$

$$\text{Hence, } x = 1.074; y = 2.53; z = 2.766$$

Solve the following set of equation using L_u factorisation method.

[2007 Spring]

$$3x + 2y + z = 10$$

$$2x + 3y + 2z = 14$$

$$x + 2y + 3z = 14$$

Solution:

Here

$$3x + 2y + z = 10 \dots\dots\dots(i)$$

$$2x + 3y + 2z = 14 \dots\dots\dots(ii)$$

$$x + 2y + 3z = 14 \dots\dots\dots(iii)$$

$$\text{Solution: Here, } A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 10 \\ 14 \\ 14 \end{bmatrix}$$

Now, by LU-decomposition formula

$$LU = A;$$

$$\text{or, } \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{or, } \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21} * u_{11} & l_{21} * u_{12} + u_{22} & l_{21} * u_{13} + u_{23} \\ l_{31} * u_{11} & l_{31} * u_{12} + l_{32} * u_{22} & l_{31} * u_{13} + l_{32} * u_{23} + u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Then

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$$u_{11} = 3, u_{12} = 2, u_{13} = 1$$

$$l_{21} * u_{11} = 2 \Rightarrow l_{21} = \frac{2}{3}$$

$$l_{21} * u_{12} + u_{22} = 3 \Rightarrow u_{22} = \frac{5}{3}$$

$$l_{21} * u_{13} + u_{23} = 2 \Rightarrow u_{23} = \frac{4}{3}$$

$$l_{31} * u_{11} = 1 \Rightarrow l_{31} = \frac{1}{3}$$

$$l_{31} * u_{12} + l_{32} * u_{22} = 2 \Rightarrow l_{32} = \frac{4}{5}$$

$$l_{31} * u_{13} + l_{32} * u_{23} + u_{33} = 3 \Rightarrow u_{33} = \frac{8}{5}$$

$$\therefore L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{3} & 1 & 0 \\ \frac{1}{3} & \frac{4}{5} & 1 \end{bmatrix} \& U = \begin{bmatrix} 3 & 2 & 1 \\ 0 & \frac{5}{3} & \frac{4}{3} \\ 0 & 0 & \frac{8}{5} \end{bmatrix}$$

Now, $LV=B$;

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{3} & 1 & 0 \\ \frac{1}{3} & \frac{4}{5} & 1 \end{bmatrix} * \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 14 \\ 14 \end{bmatrix}$$

$$\therefore v_1 = 10$$

$$\frac{2}{3}v_1 + v_2 = 14 \Rightarrow v_2 = 7.33$$

$$\frac{1}{3}v_1 + \frac{4}{5}v_2 + v_3 = 14 \Rightarrow v_3 = 4.80$$

Again, writing the system as

$$UX=V;$$

$$\begin{bmatrix} 3 & 2 & 1 \\ 0 & \frac{5}{3} & \frac{4}{3} \\ 0 & 0 & \frac{8}{5} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 7.33 \\ 4.80 \end{bmatrix}$$

By backward substitution

$$z = 3, y = 2, x = 1$$

Solve the following set of equation using gauss seidel iteration method. [2006 Fall]

$$6x_1 - 2x_2 + x_3 = 11$$

$$x_1 + 2x_2 - 5x_3 = -1$$

$$-2x_1 + 7x_2 + 2x_3 = 5$$

Solution: Here the given set of equation is

$$6x_1 - 2x_2 + x_3 = 11$$

$$x_1 + 2x_2 - 5x_3 = -1$$

$$-2x_1 + 7x_2 + 2x_3 = 5$$

Arranging the system of equation as

$$6x_1 - 2x_2 + x_3 = 11$$

$$-2x_1 + 7x_2 + 2x_3 = 5$$

$$x_1 + 2x_2 - 5x_3 = -1$$

Now by gauss seidel iteration method,

$$X_1 = \frac{1}{6}(11 + 2x_2 - x_3) \dots \dots \dots (i)$$

$$X_2 = \frac{1}{7}(5 + 2X_1 - 2x_3) \dots \dots \dots (ii)$$

$$X_3 = \frac{1}{5}(1 + X_1 + 2X_2) \dots \dots \dots (iii)$$

Let, $x_0 = y_0 = z_0 = 0$ then,

in first iteration, Second Iteration, third iteration,

$$X_1 = 1.8333 \quad X_2 = 2.06904 \quad X_3 = 1.9982$$

$$X_1 = 1.23801 \quad X_2 = 1.0204 \quad X_3 = 0.9953$$

$$X_1 = 1.0619 \quad X_2 = 1.01462 \quad X_3 = 0.9977$$

Fourth iteration, fifth iteration, sixth iteration

$$X_1 = 1.9988 \quad X_1 = 1.9988 \quad X_1 = 2.00001$$

$$X_2 = 1.00029 \quad X_2 = 1.00029 \quad X_2 = 0.9999$$

$$X_3 = 0.99989 \quad X_3 = 0.99989 \quad X_3 = 0.9999$$

which is the required values.

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CHAPTER 5

1. Taylors Series Method

Consider the first order equation $\frac{dy}{dx} = f(x, y) \dots \dots \dots (i)$

Now, differentiating eqn(i) successively we can find $y', y'', y''', \dots$

Expanding $y(x)$ about $x = x_0$ by Taylors series,

$$y_1 = y_0 + (x_1 - x_0)y'_0 + \frac{1}{2!}(x_1 - x_0)^2 y''_0 + \frac{1}{3!}(x_1 - x_0)^3 y'''_0 + \dots$$

Expanding $y(x)$ about $x = x_1$ by Taylors series

$$y_2 = y_1 + (x_2 - x_1)y'_1 + \frac{1}{2!}(x_2 - x_1)^2 y''_1 + \frac{1}{3!}(x_2 - x_1)^3 y'''_1 + \dots$$

Proceeding on, we get

$$y_n = y_{n-1} + (x_n - x_{n-1})y'_{n-1} + \frac{1}{2!}(x_n - x_{n-1})^2 y''_{n-1} + \frac{1}{3!}(x_n - x_{n-1})^3 y'''_{n-1} + \dots$$

Picards Method For 1st Order Equations

Consider the first order equation $\frac{dy}{dx} = f(x, y) \dots \dots \dots (i)$

It is required to find that particular solution of (i) which assumes value y_0 , when $x = x_0$ integrating equation (i), between the limits x_0 and x , we get

$$\int_{y_0}^y dy = \int_{x_0}^x f(x, y) dx \Rightarrow y = y_0 + \int_{x_0}^x f(x, y) dx \dots \dots (ii)$$

This is an integral equation equivalent to (i) for it contains unknown y under the integral sign

As the first approximation y_1 to the solution we put $y = y_0$ in (ii) and integrate equation (ii) giving

$$\Rightarrow y_1 = y_0 + \int_{x_0}^x f(x, y_0) dx \dots \dots (iii)$$

As the second approximation y_2 to the solution we put $y = y_1$ in (ii) and integrate equation (ii) giving

$$\Rightarrow y_2 = y_0 + \int_{x_0}^x f(x, y_1) dx \dots \dots (iv)$$

Similarly for n steps

$$\Rightarrow y_n = y_0 + \int_{x_0}^x f(x, y_{n-1}) dx \dots \dots (*)$$

Hence this method gives a sequence of approximation $y_1, y_2, y_3, \dots, y_n$ each giving a better result than previous one.

Note: This method is applicable only to a limited class of equations in which the successive integrations can be performed easily.

Picards Method For 2nd Order Equations

Consider the second order differential equation

$$\frac{d^2y}{dx^2} = f(x, y, \frac{dy}{dx})$$

By putting $\frac{dy}{dx} = z$, it can be reduced to two first order simultaneous differential equations:

$$\frac{dy}{dx} = z = f(x, y, z) \& \frac{dz}{dx} = \phi(x, y, z)$$

for n steps,

$$y_n = y_0 + \int_{x_0}^x f(x, y_{n-1}, z_{n-1}) dx$$

$$z_n = z_0 + \int_{x_0}^x \phi(x, y_{n-1}, z_{n-1}) dx$$

Eulers Method for 1st order Equations

Consider the first order equation $\frac{dy}{dx} = f(x, y) \dots \dots \dots (i)$

where $y(x_0) = y_0$

Suppose that we wish to find successively $y_1, y_2, y_3, \dots, y_m$ where

x_m is the value of x corresponding to $x = x_0$ where $x = x_0 + mh$, $m = 1, 2, \dots, h$ being small, here we use the fact that in a small interval, a curve is nearly a straight line.

The value of y corresponding to $x = x_0 + h$ is

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The value of y corresponding to $x = x_2$ is

$$y_2 = y_1 + hf(x_1, y_1)$$

Similarly, in general it can be shown that

$$y_{m+1} = y_m + hf(x_m, y_m) \text{ at } x = x_m \text{ where } m = 1, 2, 3, \dots$$

Eulers method for 2nd order Equations

Consider the second order differential equation

$$\frac{d^2y}{dx^2} = f(x, y, \frac{dy}{dx})$$

By putting $\frac{dy}{dx} = z$, it can be reduced to two first order simultaneous differential equations:

$$\frac{dy}{dx} = z = f(x, y, z) \quad \& \quad \frac{dz}{dx} = \phi(x, y, z)$$

$$y_{m+1} = y_m + hf(x_m, y_m, z_m) \text{ at } x = x_m \text{ where } m = 1, 2, 3, \dots$$

$$z_{m+1} = z_m + h\phi(x_m, y_m, z_m) \text{ at } x = x_m \text{ where } m = 1, 2, 3, \dots$$

Improved Eulers or Heauns Method for 1st order Equations

Consider the first order equation $\frac{dy}{dx} = f(x, y) \dots \dots \dots (i)$

Then

$$y_{m+1} = y_m + \frac{h}{2} [f(x_m, y_m) + f(x_{m+1}, y_{m+1}^e)]$$

where

$$y_{m+1}^e = y_m + hf(x_m, y_m) \text{ at } x = x_m \text{ where } m = 1, 2, 3, \dots$$

Improved Eulers or Heauns Method for 2nd order Equations

Consider the second order differential equation

$$\frac{d^2y}{dx^2} = f(x, y, \frac{dy}{dx})$$

By putting $\frac{dy}{dx} = z$, it can be reduced to two first order simultaneous differential equations:

$$\frac{dy}{dx} = z = f(x, y, z) \quad \& \quad \frac{dz}{dx} = \phi(x, y, z)$$

$$y_{m+1} = y_m + \frac{h}{2} [f(x_m, y_m, z_m) + f(x_{m+1}, y_{m+1}^e, z_{m+1}^e)]$$

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$$z_{m+1} = z_m + \frac{h}{2} [\phi(x_m, y_m, z_m) + \phi(x_{m+1}, y_{m+1}^e, z_{m+1}^e)]$$

where

$$y_{m+1}^e = y_m + hf(x_m, y_m, z_m) \text{ at } x = x_m \text{ where } m = 1, 2, 3, \dots$$

$$z_{m+1}^e = z_m + h\phi(x_m, y_m, z_m)$$

Runge-Kutta 2nd order (RK-2 method):

i) For 1st order differential equation

Consider the first order equation $\frac{dy}{dx} = f(x, y) \dots \dots \dots (i)$

Then

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf(x_0 + h, y_0 + k_1)$$

$$k = \frac{k_1 + k_2}{2} \quad \text{hence } y_x = k + y_0$$

ii) For 2nd order differential equation

Consider the second order differential equation

$$\frac{d^2y}{dx^2} = f(x, y, \frac{dy}{dx})$$

By putting $\frac{dy}{dx} = z$, it can be reduced to two first order simultaneous differential equations:

$$\frac{dy}{dx} = z = f(x, y, z) \quad \& \quad \frac{dz}{dx} = \phi(x, y, z)$$

Then

$$k_1 = hf(x_0, y_0, z_0)$$

$$l_1 = h\phi(x_0, y_0, z_0)$$

$$k_2 = hf(x_0 + h, y_0 + k_1, z_0 + l_1) \quad l_2 = h\phi(x_0 + h, y_0 + k_1, z_0 + l_1)$$

$$k = \frac{k_1 + k_2}{2} \quad \& \quad l = \frac{l_1 + l_2}{2}$$

$$\text{hence } y_x = k + y_0 \quad \& \quad z_x = y'_x = l + z_0$$

Runge-Kutta 4th order (RK-4 method)

(i) For 1st order differential equation

Consider the first order equation $\frac{dy}{dx} = f(x, y) \dots \dots \dots (i)$

Then

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$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2})$$

$$k_3 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2})$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

$$\therefore k = \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}$$

$$\text{Hence, } y_x = k + y_0$$

(ii) For 2nd order differential equation

Consider the second order differential equation

$$\frac{d^2 y}{dx^2} = f(x, y, \frac{dy}{dx})$$

By putting $\frac{dy}{dx} = z$, it can be reduced to two first order simultaneous differential equations:

$$\frac{dy}{dx} = z = f(x, y, z) \text{ \& } \frac{dz}{dx} = \frac{d^2 y}{dx^2} = \phi(x, y, z)$$

Then

$$k_1 = hf(x_0, y_0, z_0)$$

$$l_1 = h\phi(x_0, y_0, z_0)$$

$$k_2 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2})$$

$$l_2 = h\phi(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2})$$

$$k_3 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2})$$

$$l_3 = h\phi(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2})$$

$$k_4 = hf(x_0 + h, y_0 + k_3, z_0 + l_3)$$

$$l_4 = h\phi(x_0 + h, y_0 + k_3, z_0 + l_3)$$

$$\therefore k = \frac{k_1 + 2k_2 + 2k_3 + k_4}{6} \text{ \& } l = \frac{l_1 + 2l_2 + 2l_3 + l_4}{6}$$

$$\text{Hence } y_x = k + y_0 \text{ \& } z_x = y'_x = l + z_0$$

Boundary Value Problem

There are so many type of boundary value problem among which we have to solve the two point linear boundary value problem of type

$$1. \frac{d^2 y}{dx^2} + \lambda(x) \frac{dy}{dx} + \mu(x)y = \gamma(x) \text{ with initial condition } y(x_0) =$$

$$2. \frac{d^4 y}{dx^4} + \lambda(x)y = \mu(x) \text{ with the condition } y(x_0) = y'(x_0) = a$$

$$\text{ \& } y(x_n) = y'(x_n) = b$$

There exists two numerical method for solving such boundary value problem are

1. Finite differential method
2. Shooting method

Finite Differential Method

- It makes use of finite difference equivalent of derivatives.
- If $y(x)$ it's derivatives are single valued continuous function of x then by Taylor's series expansion we have

$$y'_i = \frac{1}{h}(y_{i+1} - y_{i-1}) \dots \dots \dots (i)$$

$$y''_i = \frac{1}{h^2}(y_{i+1} - 2y_i + y_{i-1}) \dots \dots \dots (ii)$$

$$y'''_i = \frac{1}{h^3}(y_{i+2} - 2y_{i+1} + 2y_{i-1} - y_{i-2}) \dots \dots \dots (iii)$$

$$y^{(iv)}_i = \frac{1}{h^4}(y_{i+2} - 4y_{i+1} + 6y_i - 4y_{i-1} + y_{i-2}) \dots \dots \dots (iv)$$

Example:

Solve the equation $y' = x+y$ with boundary condition $y(0) = y(1) = 0$ taking $n = 4$.

Solution:

We divide the interval $(0, 1)$ into four sub interval so that $h = \frac{1}{4}$ and pivot point at $x_0 = 0, x_1 = \frac{1}{4}, x_2 = \frac{1}{2}, x_3 = \frac{3}{4}, x_4 = 1$ then the differential equation is approximated as

$$\frac{1}{h^2}(y_{i+1} - 2y_i + y_{i-1}) = (x_i + y_i) \text{ for } i = 1, 2, 3$$

$$\text{Using } y_0 = y_4 = 0$$

$$\text{At } i = 1,$$

$$16(y_2 - 2y_1 + y_0) = (x_1 + y_1)$$

$$16y_2 - 32y_1 = \frac{1}{4} \dots \dots \dots (i)$$

$$\text{At } i = 2,$$

$$16y_3 - 32y_2 + 16y_1 = \frac{1}{2} \dots \dots \dots (ii)$$

$$\text{At } i = 3,$$

$$-32y_3 + 16y_2 = \frac{3}{4} \dots \dots \dots (iii)$$

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Solving equation (i), (ii) & (iii), we get
 $y_1 = -0.03482$, $y_2 = -0.05632$, $y_3 = -0.05003$

Shooting Method

In this method, the given boundary value problem is first transformed to an initial value problem. This initial value problem is solved by Taylors series or R-K method etc. finally the given boundary value problem is solved.

Consider the given boundary value problem

$$y''(x) = y(x), y(a) = A, y(b) = B, \dots \dots (1)$$

On the condition is $y(a) = m$ which represent the slope m , then find the corresponding value of $y(b)$, using any initial value method.

Let the two guesses be m_0, m_1

So that the corresponding value of $y(b)$ are and $y(m_1, b)$.

Assuming that the values of m and $y(b)$ are linearly related, we obtained a better approximation m_2 for m from relation.

$$\frac{m_2 - m_1}{y(b) - y(m_1, b)} = \frac{m_1 - m_0}{y(m_1, b) - y(m_0, b)}$$

This gives, the value of m_2 .

Now we solve the initial value problem

$y''(x) = y(x), y(a) = A, y'(a) = m_2$ and obtain the solution $y(m_2, b)$.

To obtain the better approximation m_3 for m , we again use the linear relation shown above as $[y(m_1, y(m_1, b))] & [y(m_2, y(m_2, b))]$. This process is repeated until the value of $y(m_i, b)$ agreed with $y(b)$ to desired accuracy.

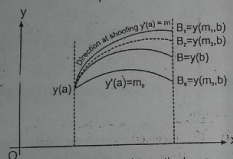


Figure : Shooting method

Solve the following questions by Picards methods and estimate y at $x = 0.25$ and 0.5 $\frac{dy}{dx} = x^2 y - y, y_0 = 1$ [2011 Spring]

Solution:

Here,

$$\frac{dy}{dx} = x^2 y - y, y_0 = 1, x_0 = 0 \text{ we have to find } y_{0.25} \text{ and } y_{0.5}$$

We have from picard's formula,

$$y_n = y_0 + \int_{x_0}^x f(x, y_{n-1}) dx$$

Now first approximation

$$y_1 = y_0 + \int_{x_0}^x f(x, y_0) dx$$

$$= 1 + \int_0^x (x^2 y_0 - y_0) dx$$

Putting $y_0 = 1$, at $x = 0.25$

$$y_1 = 1 + \int_0^{0.25} (x^2 * 1 - 1) dx = 0.7552$$

Similarly,

$$y_2 = 1 + \int_0^{0.25} (x^2 * 0.7552 - 0.7552) dx = 0.8151$$

Successively we get,

$$y_3 = 0.8004, y_4 = 0.8046, y_5 = 0.8031$$

At $x = 0.5$

$$y_1 = 1 + \int_0^{0.5} (x^2 * 1 - 1) dx = 0.54166$$

$$y_2 = 1 + \int_0^{0.5} (x^2 * 0.54166 - 0.54166) dx = 0.7517$$

Successively we get,

$$y_3 = 0.6555, y_4 = 0.6995, y_5 = 0.6793, y_6 = 0.6886,$$

$$y_7 = 0.6843$$

Solve the following equation by Picards method

$$\frac{dy}{dx} = x^2 + y^2, y_0 = 0 \text{ and estimate } y(0.1), y(0.2) \text{ and } y(1) \text{ [2010 Spring]}$$

Solution:

Here,

$$\frac{dy}{dx} = x^2 + y^2, y_0 = 0, x_0 = 0 \text{ we have to find } y \text{ at } 0.1, 0.2 \text{ and } 1$$

We have from picard's formula,

$$y_n = y_0 + \int_{x_0}^x f(x, y_{n-1}) dx$$

Now first approximation

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$$y_1 = y_0 + \int_{x_0}^x f(x, y_0) dx$$

$$= 0 + \int_0^x (x^2 + y_0^2) dx$$

Putting $y_0 = 0$, at $x = 0.1$

$$y_1 = 1 + \int_0^{0.1} (x^2 + 0) dx = 3.33 \times 10^{-4}$$

Similarly,

$$y_2 = 0 + \int_0^{0.1} (x^2 + 3.33 \times 10^{-4}) dx = 3.666 \times 10^{-4}$$

Successively we get,

$$y_3 = 3.7 \times 10^{-4}, y_4 = 3.703 \times 10^{-4}, y_5 = 3.703 \times 10^{-4}$$

At $x = 0.2$

$$y_1 = 0 + \int_0^{0.2} (x^2 + 0) dx = 2.666 \times 10^{-3}$$

$$y_2 = 0 + \int_0^{0.2} (x^2 + 2.666 \times 10^{-3}) dx = 3.2 \times 10^{-3}$$

Successively we get,

$$y_3 = 3.306 \times 10^{-3}, y_4 = 3.328 \times 10^{-3}, y_5 = 3.332 \times 10^{-3},$$

$$y_6 = 3.333 \times 10^{-3}$$

At $x = 1$;

$$y_1 = 0 + \int_0^1 (x^2 + 0) dx = 0.3333$$

$$y_2 = 0 + \int_0^1 (x^2 + 0.3333) dx = 0.6666$$

Successively we get,

$$y_3 = 1, y_4 = 1.333, y_5 = 1.666, y_6 = 2, y_7 = 2.333$$

Solve the differential equation $\frac{dy}{dx} = x^3 + y^2$, $y(0) = 0$ for $y(0.5) = ?$ by

Euler's method. [2008 Fall]

Solution:

Here, $\frac{dy}{dx} = x^3 + y^2$, $y_0 = 0$, $x_0 = 0$ we have to find $y(0.5) = ?$

We have from Euler's formula, $y_{m+1} = y_m + hf(x_m, y_m)$

Taking $h = 0.1$ and $f(x_m, y_m) = x_m^3 + y_m^2$, then

$$y_{m+1} = y_m + 0.1(x_m^3 + y_m^2)$$

Putting $m = 0, 1, 2, 3, \dots$ we get

$$y_{0.1} = y_1 = y_0 + 0.1(x_0^3 + y_0^2) = 0$$

$$y_{0.2} = y_2 = y_1 + 0.1(x_1^3 + y_1^2) = 1 \times 10^{-4}$$

$$y_{0.3} = y_3 = y_2 + 0.1(x_2^3 + y_2^2) = 9 \times 10^{-4}$$

$$y_{0.4} = y_4 = y_3 + 0.1(x_3^3 + y_3^2) = 3.6 \times 10^{-3}$$

$$y_{0.5} = y_5 = y_4 + 0.1(x_4^3 + y_4^2) = 0.010$$

Hence, $y_{0.5} = 0.010$

Solve the differential equation $y' = x + y$ using appropriate method within $0 \leq x \leq 0.2$ with $y(0) = 1$. [2010 Fall]

Solution:

Here, $\frac{dy}{dx} = x + y$ for $0 \leq x \leq 0.2$ with $y(0) = 1$

Let $h = 0.1$, now

We have, from Euler's modified formula, $y_1 = y_0 + \frac{h}{2}(m_1 + m_2)$

Where, $m_1 = f(x_0, y_0) = 1$

$$y_1^e = y_0 + hm_1 = 1 + (0.1 \times 1) = 1.1$$

$$m_2 = f(x_1, y_1^e) = 0.1 + 1.1 = 1.2$$

$$y_1(0.1) = y_0 + \frac{h}{2}(m_1 + m_2) = 1 + \frac{0.1}{2}(1 + 1.2) = 1.11$$

Second iteration

$$m_1 = f(x_1, y_1) = 1.21$$

$$y_1^e = y_1 + hm_1 = 1.11 + (0.1 \times 1.21) = 1.231$$

$$m_2 = f(x_2, y_2^e) = 1.431$$

$$y_2(0.2) = y_1 + \frac{h}{2}(m_1 + m_2) = 1.11 + \frac{0.1}{2}(1.21 + 1.431) = 1.24205$$

Hence $y_2(0.2) = 1.24205$

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Find the value of xy at 2.0 using Heun's Method.
 $y' = x + y^2, y(1) = -8$, step size $h = 0.5$ [2009 Fall]

Solution:

Here, $\frac{dy}{dx} = x + y^2, y_0 = -8, x_0 = 1$

Let $h = 0.5$, now

We have, from Euler's modified formula, $y_1 = y_0 + \frac{h}{2}(m_1 + m_2)$

Where, $m_1 = f(x_0, y_0) = 65$

$y_1^e = y_0 + hm_1 = -8 + (0.5 \cdot 65) = 24.5$

$m_2 = f(x_1, y_1^e) = 601.75$

$y_1(1.5) = y_0 + \frac{h}{2}(m_1 + m_2) = -8 + \frac{0.5}{2}(65 + 601.75) = 158.68$

Second iteration

$m_1 = f(x_1, y_1) = 25183.222$

$y_1^e = y_1 + hm_1 = 158.68 + (0.5 \cdot 25183.22) = 12750.298$

$m_2 = f(x_2, y_2^e) = 162570122.2$

$y_2(2) = y_1 + \frac{h}{2}(m_1 + m_2)$

$= 158.68 + \frac{0.5}{2}(25183.22 + 162570122.2)$

$= 40649126.81$

Hence $y_2(2) = 40649126.81$, [question may be wrong]

Use Heun's method to solve $y'' - xy + 4y = 0; y(0) = 3; y'(0) = 0$ at $x = 0.2$. Take $h = 0.1$ [2066 Spring]

Here, $y'' = xy - 4y, y_0 = 3, x_0 = 0$

Let $h = 0.1$, now

Let

$y' = z \Rightarrow f(x_0, y_0, z_0); z_0 = 0$

Then,

$\frac{dz}{dx} = xy - 4y \Rightarrow \phi(x_0, y_0, z_0)$

We have, from Euler's modified formula,

$y_1 = y_0 + \frac{h}{2}[f(x_0, y_0, z_0) + f(x_1, y_1^e, z_1^e)]$

Where, $y_1^e = y_0 + hf(x_0, y_0, z_0) = 3 + 0.1f(0, 3, 0) = 3$

$z_1^e = z_0 + h\phi(x_0, y_0, z_0) = 0 + 0.1(0, 3, 0) = (0.1 \cdot -12) = -1.2$

$f(x_1, y_1^e, z_1^e) = f(0.1, 3, -1.2) = -1.2$

$y_1(0.1) = 3 + \frac{0.1}{2}[f(0, 3, 0) + f(0.1, 3, -1.2)] = 2.94$

$z_1 = z_0 + \frac{0.1}{2}[\phi(0, 3, 0) + \phi(0.1, 3, -1.2)] = 0.05 \cdot -23.7 = -1.185$

Second iteration

$y_2 = y_1 + \frac{h}{2}[f(x_1, y_1, z_1) + f(x_2, y_2^e, z_2^e)]$

$y_2^e = y_1 + hf(x_1, y_1, z_1) = 2.94 + 0.1f(0.1, 2.94, -1.185)$

$= 2.8215$

$z_2^e = z_1 + h\phi(x_1, y_1, z_1) = -1.185 + 0.1\phi(0.1, 2.94, -1.185)$

$= -2.3316$

$y_2(0.2) = 2.94 + \frac{0.1}{2}[f(0.1, 2.94, -1.185)$

$+ f(0.2, 2.8215, -2.3316)] = 2.76417$

$z_2 = -1.185 + \frac{0.1}{2}[\phi(0.1, 2.94, -1.185)$

$+ \phi(0.2, 2.8215, -2.3316)] = -2.29438$

Using Shooting method solve the following equation.

$\frac{d^2y}{dx^2} = 6x, y(1) = 2 \text{ \& } y(2) = 9$ in the interval 1 & 2.

[2007 Fall]

Solution:

Here by transformation we obtained the following,

$\frac{dy}{dx} = z, y(1) = 2$ and $\frac{dz}{dx} = 6x$

Let us assume that $z(2) = 2(m_1)$ Applying the Heun's method we obtain the solution as follows

Iteration 1

Let $h = 0.5$

$X_0 = 1, y(1) = y_0 = 2, z(1) = z_0 = 2$

$m_1(1) = z_0 = 2$

$m_1(2) = 6x_0 = 6$

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$$m_2(1) = z_0 + h m_1(2) = 2 + 0.5(6) = 5$$

$$m_2(2) = 6(x_0 + h) = 6(1.5) = 9$$

$$m(1) = \frac{m_1(1) + m_2(1)}{2} = \frac{2+5}{2} = 3.5$$

$$m(2) = \frac{m_1(2) + m_2(2)}{2} = \frac{6+9}{2} = 7.5$$

$$y(x_1) = y(1.5) = y(1) + m(1)h = 2 + 3.5 \cdot 0.5 = 3.75$$

$$z(x_1) = z(1.5) = z(1) + m(2)h = 2 + 7.5 \cdot 0.5 = 5.75$$

Iteration 2

$$h = 0.5$$

$$X_1 = 1.5, y(1) = 3.75, z(1) = 5.75$$

$$m_1(1) = z_1 = 5.75$$

$$m_1(2) = 6x_1 = 9$$

$$m_2(1) = z_1 + h m_1(2) = 5.75 + 0.5(9) = 10.25$$

$$m_2(2) = 6(x_1 + h) = 12$$

$$m(1) = \frac{m_1(1) + m_2(1)}{2} = 8$$

$$m(2) = \frac{m_1(2) + m_2(2)}{2} = 10.5$$

$$y(x_2) = y(1) + m(1)h = 3.75 + 8 \cdot 0.5 = 7.75$$

This gives $B_1 = 7.75$ which is less than $B = 9$.

Now, let us assume $z(1) = y(1) = 4$ (m_2) and again estimate $y(2)$.

Iteration 1

$$h = 0.5$$

$$X_0 = 1, y(1) = y_0 = 2, z_0 = 4$$

$$m_1(1) = z_0 = 4$$

$$m_1(2) = 6x_0 = 6$$

$$m_2(1) = z_0 + h m_1(2) = 4 + 0.5(6) = 7$$

$$m_2(2) = 6(x_0 + h) = 6(1.5) = 9$$

$$m(1) = \frac{m_1(1) + m_2(1)}{2} = \frac{4+7}{2} = 5.5$$

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$$m(2) = \frac{m_1(2) + m_2(2)}{2} = \frac{6+9}{2} = 7.5$$

$$y(x_1) = y(1.5) = y(1) + m(1)h = 2 + 5.5 \cdot 0.5 = 4.75$$

$$z(x_1) = z(1.5) = z(1) + m(2)h = 4 + 7.5 \cdot 0.5 = 7.75$$

Iteration 2

$$h = 0.5$$

$$X_1 = 1.5, y(1) = 4.75, z(1) = 7.75$$

$$m_1(1) = z_1 = 7.75$$

$$m_1(2) = 6x_1 = 9$$

$$m_2(1) = z_1 + h m_1(2) = 7.75 + 0.5(9) = 12.25$$

$$m_2(2) = 6(x_1 + h) = 12$$

$$m(1) = \frac{m_1(1) + m_2(1)}{2} = 12$$

$$m(2) = \frac{m_1(2) + m_2(2)}{2} = 10.5$$

This gives $B_2 = 9.75$ which is greater than $B = 9$.

Which is the required answer.

Solve the differential equation for $y(0.25)$ using RK 4th order

method. $10 \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx}\right)^2 + 6x = 0$, with $y(0) = 1$ and $y'(0) = 0$.

(take $h = 0.25$ [2011 Spring, 2011 Fall, 2010 Fall])

Solution:

Here, $10y'' + (y')^2 + 6x = 0$, $y(0) = 1$, $y'(0) = 0$, $h = 0.5$

Let $y' = z = f(x, y, z)$, $z(0) = 0$ then

$$10z' + z^2 + 6x = 0 \Rightarrow z' = -\frac{(6x + z^2)}{10} = \phi(x, y, z)$$

Now using Runge-Kutta method,

$$k_1 = hf(x_0, y_0, z_0) = 0.5 \cdot f(0, 0, 0) = 0$$

$$l_1 = h\phi(x_0, y_0, z_0) = 0.5 \cdot \phi(0, 0, 0) = 0$$

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$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) = 0.5 * f(0.25, 0, 0) = 0$$

$$l_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) = 0.5 * \phi(0.25, 1, 0) = -0.075$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) = 0.5 * f(0.25, 0, -0.0375) = -0.01875$$

$$l_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) = 0.5 * \phi(0.25, 0, -0.0375) = -0.0749$$

$$k_4 = hf(x_0 + h, y_0 + k_3, z_0 + l_3) = 0.5 * f(0.5, -0.01875, -0.0749) = -0.03746$$

$$l_4 = hf(x_0 + h, y_0 + k_3, z_0 + l_3) = 0.5 * \phi(0.5, -0.01875, -0.0749) = -0.299$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = -0.0124$$

$$l = \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4) = -0.0998$$

$$\text{Now, } y(0.5) = y_0 + k = 0.9876$$

$$z(0.25) = z_0 + l = 0.0998$$

Solve the differential equation $\frac{d^2 y}{dx^2} + 2\left(\frac{dy}{dx}\right) + 6y = x$, with $y(0) = 0$ and $y'(0) = 1$. For $y(0.2)$ (take $h = 0.20$) using RK-2 method. [2008 Spring]

Solution:

$$\text{Here, } y'' + 2(y') + 6y = x, y(0) = 0, y'(0) = 1, h = 0.20$$

$$\text{Let } y' = z = f(x, y, z), z(0) = 1 \text{ then}$$

$$z' + 2z + 6y = x \Rightarrow z' = -2z - 6y + x = \phi(x, y, z)$$

Now using Runge-Kutta second method,

$$k_1 = hf(x_0, y_0, z_0) = 0.2 * f(0, 0, 1) = 0.2 * 1 = 0.2$$

$$l_1 = hf(x_0, y_0, z_0) = 0.2 * \phi(0, 0, 1) = -0.4$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) = 0.2 * f(0.1, 0.1, 0.5) = 0.2 * 0.5 = 0.1$$

$$y = 0.2 * 0.6 = 0.12$$

$$l_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) = 0.2 * \phi(0.1, 0.1, 0.5) = 0.44$$

$$\text{Then, } k = \frac{1}{2}(k_1 + k_2) = 0.16, l = \frac{1}{2}(l_1 + l_2) = -0.42$$

$$\text{Now, } y(0.2) = y_0 + k = 0.16$$

$$z(0.2) = z_0 + l = 0.58$$

Use the classical 4th order RK method to estimate $y(0.5)$ of the differential equation with $h = 0.25$, $\frac{dy}{dx} = y + \sin x$, $y(0) = 2$ [2008 Fall]

Solution:

$$\text{Here, } y' = y + \sin x, y(0) = 2, h = 0.25$$

$$x_0 = 0, y_0 = 2, f(x, y) = y + \sin x$$

Now using Runge-Kutta method,

$$k_1 = hf(x_0, y_0) = 0.25 * f(0, 2) = 0.5$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.25 * f(0.125, 2.25) = 0.5630$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.25 * f(0.125, 2.2815) = 0.57092$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = 0.25 * f(0.25, 2.57092) = 0.64382$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.5686$$

$$\text{Now, } y(0.25) = y_0 + k = 2 + 0.5686 = 2.5686$$

$$\text{Again, } x_1 = 0.25, y_1 = 2.5686$$

$$k_1 = hf(x_1, y_1) = 0.25 * f(0.25, 2.5686) = 0.6432$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = 0.25 * f(0.375, 2.8902) = 0.7241$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = 0.25 * f(0.375, 2.9306) = 0.7342$$

$$k_4 = hf(x_1 + h, y_1 + k_3) = 0.25 * f(0.5, 2.9357) = 0.7361$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.7159$$

$$\text{Now, } y(0.5) = y_1 + k = 2.5686 + 0.7159 = 3.2845$$

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Solve the differential equation by using 4th order R-K method for $y(0.6)$ from $\frac{dy}{dx} = x^2 + y$, $y(0) = 3$ and taking $h = 0.3$ [2007 Fall]

Solution:

Here, $y' = x^2 + y$, $y(0) = 3$, $h = 0.3$

$x_0 = 0$, $y_0 = 3$, $f(x, y) = x^2 + y$

Now using Runge-Kutta method,

$$k_1 = hf(x_0, y_0) = 0.3 * f(0, 3) = 0.9$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.3 * f(0.15, 3.45) = 1.04175$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.3 * f(0.15, 3.520) = 1.06301$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = 0.3 * f(0.3, 4.0636) = 1.2459$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 1.0591$$

$$\text{Now, } y(0.3) = y_0 + k = 3 + 1.0591 = 4.0591$$

Again, $x_1 = 0.6$, $y_1 = 4.0591$

$$k_1 = hf(x_1, y_1) = 0.3 * f(0.3, 4.0591) = 1.2447$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = 0.3 * f(0.45, 4.6814) = 1.4651$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = 0.3 * f(0.45, 4.79169) = 1.4982$$

$$k_4 = hf(x_1 + h, y_1 + k_3) = 0.3 * f(0.6, 5.5573) = 1.7751$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 2.506$$

$$\text{Now, } y(0.6) = y_1 + k = 4.0591 + 2.506 = 6.5651$$

Solve $y' = y + e^x$, $y(0) = 0$ for $y(0.2)$ and $y(0.4)$ by RK-4th order method. [2006 Spring]

Solution:

Here, $y' = y + e^x$, $y(0) = 0$, $h = 0.2$

$x_0 = 0$, $y_0 = 0$, $f(x, y) = e^x + y$

Now using Runge-Kutta method,

$$k_1 = hf(x_0, y_0) = 0.2 * f(0, 0) = 0.2$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.2 * f(0.1, 0.1) = 0.241$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.2 * f(0.1, 0.1205) = 0.2451$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = 0.2 * f(0.2, 0.12255) = 0.268$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.24003$$

$$\text{Now, } y(0.2) = y_0 + k = 0 + 0.24003 = 0.24003$$

Again, $x_1 = 0.4$, $y_1 = 0.24003$

$$k_1 = hf(x_1, y_1) = 0.2 * f(0.2, 0.24) = 0.292$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = 0.2 * f(0.3, 0.386) = 0.3471$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = 0.2 * f(0.3, 0.41355) = 0.3526$$

$$k_4 = hf(x_1 + h, y_1 + k_3) = 0.2 * f(0.4, 0.5926) = 0.4168$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.3513$$

$$\text{Now, } y(0.4) = y_1 + k = 0.24 + 0.3513 = 0.591$$

The angular displacement θ of a simple pendulum is given by the

equation $\frac{d^2\theta}{dt^2} + \frac{g}{l} \sin \theta = 0$ where $l = 98$ cm and $g = 980$ cm/sec² if

$\theta = 0$ and $\frac{d\theta}{dt} = 4.472$ at $t = 0$. Use Runge Kutta fourth order

method to find θ and $\frac{d\theta}{dt}$ when $t = 0.2$ sec. [2007 Spring]

Solution:

Here let us suppose that $\theta = Y$, $t = x$ then $y'' + \frac{g}{l} \sin y = 0$, $h = 0.2$

$$\text{Or } \frac{dy}{dx} + 10 \sin y = 0 \therefore y' = \frac{dy}{dx} = \frac{dy}{dx} = z = f(x, y, z)$$

$$\phi = y'' = -10 \sin y = \phi(x, y, z)$$

$$y(0) = 0, y'(0) = 4.472$$

Now using Runge-Kutta method,

$$k_1 = hf(x_0, y_0, z_0) = 0.2 * f(0, 0, 4.472) = 0.8944$$

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$$\begin{aligned}
 l_1 &= h\phi(x_0, y_0, z_0) = 0.2 * \phi(0, 0.4, 4.472) = 0 \\
 k_2 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) = 0.2 * f(0.1, 0.44, 4.472) \\
 &= 0.8944 \\
 l_2 &= h\phi\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) = 0.2 * \phi(0.1, 0.44, 4.472) \\
 &= -0.0156 \\
 k_3 &= hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) = 0.2 * f(0.1, 0.4472, 4.4642) \\
 &= 0.89284 \\
 l_3 &= h\phi\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) = 0.2 * \phi(0.1, 0.4472, 4.4642) \\
 &= -0.0156 \\
 k_4 &= hf(x_0 + h, y_0 + k_3, z_0 + l_3) = 0.2 * f(0.2, 0.89284, 4.456) \\
 &= 0.89127 \\
 l_4 &= h\phi(x_0 + h, y_0 + k_3, z_0 + l_3) = 0.2 * \phi(0.2, 0.89284, 4.456) \\
 &= -0.03116
 \end{aligned}$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.8933$$

$$l = \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4) = -0.01558$$

$$\text{Now, } y(0.2) = y_0 + k = 0.8933$$

$$z(0.2) = z_0 + l = 4.472 - 0.01558 = 4.456$$

$$\text{Hence, } \theta_{0.2} = 0.8933, \left(\frac{d\theta}{dt}\right)_{0.2} = 4.456$$

OR

$$\text{Here, } y' = y - \frac{2x}{y}, y(0) = 1, h = 0.2$$

$$F(x, y) = y - \frac{2x}{y}$$

Now using Rungee-Kutta method,

$$k_1 = hf(x_0, y_0) = 0.2 * f(0, 1) = 0.2$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.2 * f(0.1, 1.1) = 0.1836$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.2 * f(0.1, 1.0918) = 0.1817$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = 0.2 * f(0.2, 1.1817) = 0.1686$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.1832$$

$$\text{Now, } y(0.2) = y_0 + k = 1 + 0.1832 = 1.1832$$

Again,

$$x_1 = 0.4, y_1 = 1.1832$$

$$k_1 = hf(x_1, y_1) = 0.2 * f(0.2, 1.1832) = 0.169$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = 0.2 * f(0.3, 1.2677) = 0.1588$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = 0.2 * f(0.3, 1.2626) = 0.1574$$

$$k_4 = hf(x_1 + h, y_1 + k_3) = 0.2 * f(0.4, 1.3406) = 0.1487$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.15835$$

$$\text{Now, } y(0.4) = y_1 + k = 1.1832 + 0.15835 = 1.34155$$

Find the value of y at $x = 2.0$ for the following initial value problem by using RK-4 method. $y' = x + y^2$; $y(1) = -0.8$, step size $h = 0.5$

[2006, 2007, 2008 Spring]

Solution:

$$\text{Here } y' = x + y^2, y(0) = -0.8, x_0 = 1, h = 0.5$$

$$\text{Now } f(x, y) = x + y^2$$

Now using Rungee-Kutta method,

$$k_1 = hf(x_0, y_0) = 0.5 * f(1, -0.8) = 0.82$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.5 * f(1.25, -0.39) = 0.701$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.5 * f(1.25, -0.44) = 0.7260$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = 0.5 * f(1.5, -0.672) = 0.7527$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.7377$$

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$$\text{Now, } y(1.5) = y_0 + k = -0.8 + 0.7377 = -0.0622$$

$$\text{Again, } x_1 = 1.5, y_1 = -0.0622$$

$$k_1 = hf(x_1, y_1) = 0.5 * f(1.5, -0.0622) = 0.7519$$

$$k_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = 0.5 * f(1.75, 0.31375) = 0.9242$$

$$k_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = 0.5 * f(1.75, 0.399) = 0.9549$$

$$k_4 = hf(x_1 + h, y_1 + k_3) = 0.5 * f(2.0, 0.8927) = 1.3984$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.8719$$

$$\text{Now, } y(2) = y_1 + k = -0.0622 + 0.8719 = 0.80975$$

$y'' + 2y' + 6y = x$ using RK-4th Method, where $y(0) = 0, y'(0) = 1$ are the initial condition and find it for $y(0.2)$ taking $h = 0.2$

[2008 Fall]

Solution:

Here, $y'' + 2y' + 6y = x, y(0) = 0, y'(0) = 1, h = 0.2$

Let $y' = z = f(x, y, z), z(0) = 1$ then

$$z' = x - 2z - 6y = \phi(x, y, z)$$

Now using Rungee-Kutta method,

$$k_1 = hf(x_0, y_0, z_0) = 0.2 * f(0, 0, 1) = 0.2$$

$$l_1 = h\phi(x_0, y_0, z_0) = 0.2 * \phi(0, 0, 1) = -0.4$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) = 0.2 * f(0.1, 0.1, 0.8)$$

$$= 0.16$$

$$l_2 = h\phi\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}, z_0 + \frac{l_1}{2}\right) = 0.2 * \phi(0.1, 0.08, 0.8)$$

$$= -1.98$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) = 0.2 * f(0.1, 0.08, 0.01)$$

$$= 0.1796$$

$$l_3 = h\phi\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}, z_0 + \frac{l_2}{2}\right) = 0.2 * \phi(0.1, 0.0898, 0.01)$$

$$= -0.0279$$

$$k_4 = hf(x_0 + h, y_0 + k_3, z_0 + l_3)$$

$$= 0.2 * f(0.2, 0.1796, 0.9721) = 0.1944$$

$$l_4 = h\phi(x_0 + h, y_0 + k_3, z_0 + l_3)$$

$$= 0.2 * \phi(0.2, 0.1944, 0.9721) = 0.58212$$

$$\text{Then, } k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.178933$$

$$l = \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4) = -0.6389$$

$$\text{Now, } y(0.2) = y_0 + k = 0 + 0.178933 = 0.178933$$

$$z(0.2) = z_0 + l = 1 - 0.6389 = 1.638946$$

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CHAPTER 6

6.0 CLASSIFICATION OF PARTIAL DIFFERENTIAL EQUATIONS

We can classify partial differential equations of second order. The general linear partial equation of second order in two independent variables is in the form

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = 0$$

$$\text{Or, } Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0 \dots (i)$$

Where A, B, C, D, E and F are in general functions of x and y.

Equation (i) said to be

- (i) Elliptic at a point (x, y) in plane if $B^2 - 4AC < 0$
- (ii) Parabolic if $B^2 - 4AC = 0$
- (iii) Hyperbolic if $B^2 - 4AC > 0$

6.1 ELLIPTIC EQUATIONS

6.1.1 LAPLACE'S EQUATION - ITS SOLUTION

An important equation of the elliptic type is $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

$$\text{i.e. } u_{xx} + u_{yy} = 0 \dots (i)$$

this Equation is called Laplace's equation. Replacing the derivatives by the corresponding difference expressions in eqn(i), we get

$$\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} + \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{k^2} = 0$$

Taking a square mesh and putting $h = k$, we get from above

$$u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}] \dots (ii)$$

That is, the value of u at any interior mesh point is the arithmetic mean of its values at the four neighbouring mesh points to the left, right, below and above. This is called the *Standard five point formula (SFPF)*.

Instead of eqn(ii), we may also use the formula

$$u_{i,j} = \frac{1}{4} [u_{i-1,j-1} + u_{i+1,j-1} + u_{i-1,j+1} + u_{i+1,j+1}] \dots (iii)$$

Which shows, the value of $u_{i,j}$ is the arithmetic mean of its values at the four neighbouring diagonal mesh points. This is called the *Diagonal five point formula (DFPF)*.

The SFPF and DFPF are represented in figures (a) and (b) below.

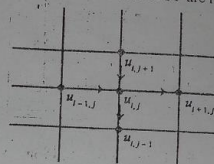


Figure a : SFPF

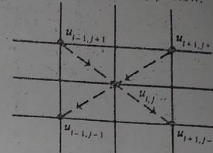


Figure b : DFPF

Note: The DFPF is valid since we know that Laplace equation remains invariant when the coordinate axes are rotated through 45° . But the error in DFPF is four times the error in SFPF. Therefore, we prefer SFPF.

6.1.2 POISSON'S EQUATION - ITS SOLUTION

The partial differential equation

$$\nabla^2 u = f(x, y) \quad \text{or,} \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

$$\text{or, } u_{xx} + u_{yy} = f(x, y) \dots (i)$$

where $f(x, y)$ is a given function of x and y is called the Poisson's equation.

To solve the Poisson equation numerically, the derivatives in eqn(i) are replaced by difference expressions at the point $x = ih, y = jk$ (here, $h = k$). Then we get

$$\frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} + \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{h^2} = f(ih, jh)$$

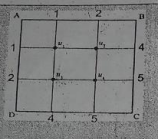
$$\text{or, } u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j} = h^2 f(ih, jh) \dots (ii)$$

Applying the above formula at each mesh point, we get similar equations in the pivotal values i, j . These equations can be solved by iteration techniques or direct techniques.

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University Question Solutions

Solve the elliptic equation $U_{xx} + U_{yy} = 0$ for the following square mesh with boundary values as shown in the figure. Iterate until the maximum difference between two successive values at any point is less than 0.001. [2009 Spring]



Solution:

Let u_1, u_2, u_3, u_4 be the values of x at the interior mesh points. From SFPP,

$$u_1 = \frac{1}{4}(1 + u_2 + u_3 + 1) \dots (i)$$

$$u_2 = \frac{1}{4}(u_1 + 2 + 4 + u_4) \dots (ii)$$

$$u_3 = \frac{1}{4}(u_1 + 2 + 4 + u_4) \dots (iii)$$

$$u_4 = \frac{1}{4}(u_2 + 5 + 5 + u_3) \dots (iv)$$

Here $u_2 = u_3$ then we can write above equations as,

$$4u_1 - 2u_2 = 2 \dots (a)$$

$$4u_2 - u_1 - u_4 = 6 \dots (b)$$

$$4u_4 - 2u_2 = 10 \dots (c)$$

Solving these equations by Direct Methods we get,

$$u_1 = 2 \quad u_2 = u_3 = 3 \quad u_4 = 4$$

NOTE: We can use the iterative method (Gauss-Seidal & Gauss Method) for solving the above equations. By iterative method we can't get the exact value.

A steel plate is of size 15 cm $\times$ 15 cm. If two of the sides are held at 100 °C and other two sides are held at 0 °C what is the steady state temperature at interior points assuming a grid size of 5 cm $\times$ 5 cm? [2006 Fall]

Solution: Let u_1, u_2, u_3, u_4 be the values of x at the interior mesh points. From SFPP,

$$u_1 = \frac{1}{4}(100 + u_2 + u_3 + 100) \dots (i)$$

$$u_2 = \frac{1}{4}(u_1 + 100 + u_4) \dots (ii)$$

$$u_3 = \frac{1}{4}(u_1 + 100 + u_4) \dots (iii)$$

$$u_4 = \frac{1}{4}(u_2 + u_3) \dots (iv)$$

Here $u_2 = u_3$ then we can write above equations as,

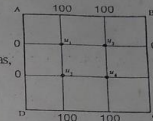
$$4u_1 - 2u_2 = 200 \dots (a)$$

$$4u_2 - u_1 - u_4 = 100 \dots (b)$$

$$4u_4 - 2u_2 = 0 \dots (c)$$

Solving these equations by Direct Methods we get,

$$u_1 = 75 \quad u_2 = u_3 = 50 \quad u_4 = 25$$



NOTE: We can use the iterative method (Gauss-Seidal & Gauss Method) for solving the above equations. By iterative method we can't get the exact value.

Solve the equation $\nabla^2 u = -10(x^2 + y^2 + 10)$ over the square mesh with sides $x = 0, y = 0, x = 3, y = 3$ with $u = 0$ on the boundary and mesh length $h = 1$. [2008 Fall]

Solution: The given differential equation is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -10(x^2 + y^2 + 10) \dots (i)$$

Let u_1, u_2, u_3, u_4 be the values of u at the four mesh points as shown in figure. Replacing LHS of Eqn (i) by finite difference expressions and putting $x = ih, y = jh$ on the RHS of it, we get

$$u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j} = -10(i^2 + j^2 + 10) \dots (ii)$$

For u_1 by putting $i=1, j=2$ in Eqn (ii), we get

$$0 + u_2 + u_3 + 0 - 4u_1 = -10(1 + 4 + 10)$$

$$\text{Or, } u_1 = \frac{1}{4}(u_2 + u_3 + 150)$$

$$\text{Or, } 4u_1 - u_2 - u_3 = 150 \dots (a)$$

For u_2 by putting $i=2, j=2$ in Eqn (ii), we get

$$\text{Or, } u_2 = \frac{1}{4}(u_1 + u_4 + 180)$$

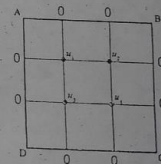
$$\text{Or, } 4u_2 - u_1 - u_4 = 180 \dots (b)$$

For u_3 by putting $i=1, j=1$ in Eqn (ii), we get

$$\text{Or, } u_3 = \frac{1}{4}(u_1 + u_4 + 120)$$

$$\text{Or, } 4u_3 - u_1 - u_4 = 120 \dots (c)$$

For u_4 by putting $i=2, j=1$ in Eqn (ii), we get



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$$\text{Or, } u_4 = \frac{1}{4}(u_2 + u_3 + 150)$$

$$\text{Or, } 4u_4 - u_2 - u_3 = 150 \dots\dots\dots (d)$$

Here we can see that $u_4 = u_1$. So with $u_4 = u_1$ Eqns (a)-(d) reduce to

$$4u_1 - u_2 - u_3 = 150$$

$$4u_2 - 2u_1 = 180$$

$$4u_3 - 2u_1 = 120$$

Now solve these equations by Direct Method. We get,

$$u_1 = 75 \quad u_2 = 82.5 \quad u_3 = 67.5 \quad u_4 = 75$$

Solve the Poisson's equation $\nabla^2 u = 2x^2 y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with Dirichlet boundary condition of $u(x, y) = 0$ and $h = k = 1$ using Gauss Seidel method. [2005 spring, 2007 spring, 2010 fall]

Solution: The given differential equation is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2x^2 y^2 \dots\dots\dots (i)$$

Let u_1, u_2, u_3, u_4 be the values of u at the four mesh points as shown in figure. Replacing LHS of Eqn (i) by finite difference expressions and putting $x = ih, y = jh$ on the RHS of it, we get

$$u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j} = 2i^2 j^2 \dots\dots\dots (ii)$$

For u_1 by putting $i=1, j=2$ in Eqn (ii), we get

$$\text{Or, } u_1 = \frac{1}{4}(u_2 + u_3 - 8)$$

$$\text{Or, } 4u_1 - u_2 - u_3 = -8 \dots\dots\dots (a)$$

For u_2 by putting $i=2, j=2$ in Eqn (ii), we get

$$\text{Or, } u_2 = \frac{1}{4}(u_1 + u_4 - 32)$$

$$\text{Or, } 4u_2 - u_1 - u_4 = -32 \dots\dots\dots (b)$$

For u_3 by putting $i=1, j=1$ in Eqn (ii), we get

$$\text{Or, } u_3 = \frac{1}{4}(u_1 + u_4 - 2)$$

$$\text{Or, } 4u_3 - u_1 - u_4 = -2 \dots\dots\dots (c)$$

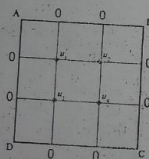
For u_4 by putting $i=2, j=1$ in Eqn (ii), we get

$$\text{Or, } u_4 = \frac{1}{4}(u_2 + u_3 - 8)$$

$$\text{Or, } 4u_4 - u_2 - u_3 = -8 \dots\dots\dots (d)$$

Here we can see that $u_4 = u_1$. So with $u_4 = u_1$ Eqns (a)-(d) reduce to

$$4u_1 - u_2 - u_3 = -8$$



$$4u_2 - 2u_1 = -22$$

$$4u_3 - 2u_1 = -8$$

Now solve these equations by Direct Method. We get,

$$u_1 = \frac{-22}{4} \quad u_2 = \frac{-43}{4} \quad u_3 = \frac{-13}{4} \quad u_4 = \frac{-22}{4}$$

NOTE: We can use the iterative method (Gauss-Seidel & Gauss Method) for solving the above equations. By iterative method we can't get the exact value.

In a square bar with dimension of 3 inch $\times$ 3 inch, torsion function,

$$\phi, \text{ can be obtained from the following P.D.E: } \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -2$$

where $\phi = 0$ on the outer boundary of the bar's cross section. Subdivide the region into nine equal squares to form a mesh and find the values of ϕ in the interior nodes. [2006 Spring]

The given differential equation is

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -2 \dots\dots\dots (i)$$

Let u_1, u_2, u_3, u_4 be the values of u at the four mesh points as shown in figure. Replacing LHS of Eqn (i) by finite difference expressions and putting $x = ih, y = jh$ on the RHS of it, we get

$$u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j} = -2 \dots\dots\dots (ii)$$

For u_1 by putting $i=1, j=2$ in Eqn (ii), we get

$$\text{Or, } u_1 = \frac{1}{4}(u_2 + u_3 + 2)$$

$$\text{Or, } 4u_1 - u_2 - u_3 = 2 \dots\dots\dots (a)$$

For u_2 by putting $i=2, j=2$ in Eqn (ii), we get

$$\text{Or, } u_2 = \frac{1}{4}(u_1 + u_4 + 2)$$

$$\text{Or, } 4u_2 - u_1 - u_4 = 2 \dots\dots\dots (b)$$

For u_3 by putting $i=1, j=1$ in Eqn (ii), we get

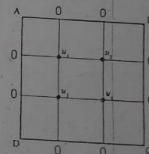
$$\text{Or, } u_3 = \frac{1}{4}(u_1 + u_4 + 2)$$

$$\text{Or, } 4u_3 - u_1 - u_4 = 2 \dots\dots\dots (c)$$

For u_4 by putting $i=2, j=1$ in Eqn (ii), we get

$$\text{Or, } u_4 = \frac{1}{4}(u_2 + u_3 + 2)$$

$$\text{Or, } 4u_4 - u_2 - u_3 = 2 \dots\dots\dots (d)$$



2013 05 14

I N D E X

Numerical Method

Syllabus

- 1) Solution of non-linear equation
- 2) Solution of system of linear equation
- 3) Interpolation and approximation
- 4) Differentiation and integration
- 5) Solution of ordinary differential equation
- 6) solution of partial differential equation

Text -Books

Higher Engg. Mathematics.

- 1) H.K. Das
- 2) B.S. Grewal
- 3) Erwin Kreszig.

CH#1

Solution of Non-Linear Equation

Solution of Non-Linear Equation

- 1) Hit and Trial Method
- 2) Graphical Method
- 3) Direct Method (or Analytical Method)
- 4) Iteration Method

#. Errors in numerical analysis :

- let, x_t = exact value
and x_c = Calculated value

- Absolute error = $|x_c - x_t|$

Relative error = $\frac{|x_c - x_t|}{x_t} \times 100\%$

When x_t is unknown, Then

let, x_k be the value in k^{th} iteration
and x_{k+1} be the value in $(k+1)^{\text{th}}$ iteration

Then,

Relative error = $\frac{|x_{k+1} - x_k|}{x_{k+1}} \times 100\%$

Types of numerical errors :

- (i) Round off error
- (ii) Truncation error

Notes:

(i) Roundoff Error:

Roundoff error occurs when a fixed number of digits are used to represent a exact number, since the numbers are stored at every stage of calculation. Roundoff error is introduced at the end of every arithmetics operations and due to its cumulative effect the exact answer is not obtained.

(ii) Truncation error:

Truncation error arise by using an approximate mathematical procedure in place of an exact mathematical formula.

In other word,

This type of error resulting from the termination of the mathematical process.

for eg: if we use $e^x = 1 + x + \frac{x^2}{2!} + \dots$ to calculate e^5 , then exact answer is not obtained.

→ Roundoff error and Truncation error are inherent error in any numerical process. These cannot be avoided completely, however we can minimize their effect.

Solution of Non-Linear Equation.

Iterative Methods:

- 1) Bisection Method
- 2) Regular falsi Method
- 3) Secant Method
- 4) Newton's Method
- 5) Fixed Point Method

(i) Bisection Method:

Choose the starting values (or initial guess) or x_1 & x_2 for the root of the eqn $f(x) = 0$.

If $f(x_1) \cdot f(x_2) < 0$

Then, x_1 and x_2 bracket a root of the given equation.

calculate $x_0 = \frac{x_1 + x_2}{2}$

And $f(x_0) = 0$

If $f(x_0) \neq 0$

then if $f(x_0) \cdot f(x_1) < 0$ then

x_0 & x_1 bracket a root.

* Algorithm: Bisection Method:

- step 0: Decide the initial guess x_1 & x_2 and stopping criteria ϵ .
- 1: compute $f(x_1) = f_1$ and $f(x_2) = f_2$
- 2: If $f_1 \cdot f_2 > 0$ then stop, otherwise continue
- 3: Calculate $x_0 = \frac{x_1 + x_2}{2}$ and $f_0 = f(x_0)$
- 4: If $f_0 = 0$, then x_0 is the exact root & stop otherwise continue
- 5: If $f_1 \cdot f_0 < 0$, then x_0 and x_1 bracket a root and set $x_2 = x_0$, go to step 1.

Q. Ex 1 Using bisection method find a root of the eqn $x^2 - 3x - 2 = 0$

→ Here,

$$\text{Let, } f(x) = x^2 - 3x - 2$$

$$\text{Take, } x_1 = 0, \quad x_2 = -1$$

$$f(x_1) = f(0) = -2$$

$$f(x_2) = f(-1) = 2$$

By Bisection Method, the new root is obtained

$$\text{by } x_0 = \frac{x_1 + x_2}{2}$$

Iterative Table

It No.	x_1	x_2	x_0	$f(x_0)$
1	0	-1	-0.5	-0.25
2	-0.5	-1	-0.75	0.8125
3	-0.5	-0.75	-0.625	0.26
4	-0.5	-0.625	-0.5625	0.003
5	-0.5	-0.5625	-0.5313	-0.12
6	-0.5313	-0.5625	-0.5469	

$$\therefore \text{approx. root} = -0.5481$$

Q. No. 2 Using Bisection Method, find a root of the eqn

$$a) \quad e^x - x - 2 = 0$$

→ Here,

$$\text{let, } f(x) = e^x - x - 2$$

Take initial guess as-

$$x_1 = 1 \quad \& \quad x_2 = 2$$

$$f(x_1) = f(1) = -0.281$$

$$f(x_2) = f(2) = 3.389$$

By Bisection Method, the new root is obtained by

$$\therefore x_0 = \frac{x_1 + x_2}{2}$$

Iteration Table				
It No.	(-ve) x_1	(+ve) x_2	x_0	$f(x_0)$
1	1	2	1.5	0.981
2	1	1.5	1.25	0.24
3	1	1.25	1.125	-0.044
4	1.125	1.25	1.1875	0.0913
5	1.125	1.1875	1.15625	0.0217
6	1.125	1.156	1.140	0.0132
7	1.125	1.140	1.132	-0.301
8	1.132	1.140	1.136	-0.0021

$\therefore$ Approximation root is 1.136

Q. Using bisection Method, find a root of $\sin x - 2x + 1 = 0$. Correct to three decimal places.

$\rightarrow$ Solution,

Let, $f(x) = \sin x - 2x + 1$

Take initial guess as -

$x_1 = 0$ & $x_2 = 1$

$\therefore f(x_1) = f(0) = 1$

$\therefore f(x_2) = f(1) = -0.1585$

By section Method

$\therefore x_0 = \frac{x_1 + x_2}{2}$

Iteration Table				
It No.	(+ve) x_1	(-ve) x_2	x_0	$f(x_0)$
1	0	1	0.5	1.479
2	0.5	1	0.75	+0.182
3	0.75	1	0.875	0.268
4	0.875	1	0.938	-0.0696
5	0.875	0.938	0.9065	-0.026
6	0.875	0.9065	0.890	-0.0029
7	0.875	0.890	0.882	0.0067
8	0.883	0.890	0.887	0.0012
9	0.887	0.89	0.8885	-0.00087
10	0.887	0.8885	0.8887	-0.000188
11	0.887	0.8887	0.8888	

$\therefore$ Approx. root is 0.8888 //

(a) $\log x - \cos x = 0$

Here,

let, $f(x) = \log x - \cos x$

Take, initial guess as $x_1 = 1$ & $x_2 = 2$

$\therefore f(x_1) = f(1) = -0.54$

$\therefore f(x_2) = f(2) = 1.109$

By using Bisection method is

$x_0 = \frac{x_1 + x_2}{2}$

Iteration Table

It. NO.	x_1	x_2	x_0	$f(x_0)$
1	1	2	1.5	0.335
2	1	1.5	1.25	-0.092
3	1.25	1.5	1.375	0.106
4	1.25	1.375	1.313	0.017
5	1.25	1.313	1.282	-0.036
6	1.282	1.313	1.298	-0.0086
7	1.298	1.313	1.306	0.0053
8	1.298	1.306	1.302	-0.0017
9	1.302	1.306	1.304	0.0018
10	1.302	1.304	1.303	0.00006
11	1.302	1.303	1.303	0.00006

$\therefore$ Approx. Root is 1.303

(b) $x \log x - 1 = 0$

Solⁿ:

let $f(x) = x \log x - 1$

Take, initial guess as $x_1 = 1$ & $x_2 = 2$

$\therefore f(x_1) = f(1) = -1$

$\therefore f(x_2) = f(2) = 0.386$

By using Bisection method is

$x_0 = \frac{x_1 + x_2}{2}$

Iteration Method

It. NO.	x_1	x_2	x_0	$f(x_0)$
1	1	2	1.5	-0.392
2	1.5	2	1.25	-0.721
3	1.25	2	1.125	-1.537
4	1.125	2	1.063	-0.935
5	1.063	2	1.031	-0.986
6	1.031	2		
7	1.5	2	1.75	-0.0206
8	1.75	2	1.875	0.179
9	1.75	1.875	1.812	0.078
10	1.75	1.812	1.781	0.028
11	1.75	1.781	1.766	0.004
12	1.75	1.766	1.758	-0.0081
13	1.758	1.766	1.762	-0.0019
14	1.762	1.766	1.764	0.0012